# Identification and Prioritization of Underground Hydrogen ($H_2$) Storage Sites in Nigeria.

**Abstract**

Underground hydrogen storage (UHS) is recognised internationally as a technically viable solution for large-scale, long-term energy storage. However, no systematic assessment of Nigeria's UHS potential has been conducted. This study addresses that gap by identifying, evaluating, and prioritising suitable geological formations for underground hydrogen storage in Nigeria using a GIS-based Multi-Criteria Decision Analysis framework integrated with the Analytical Hierarchy Process.

Nine criteria were defined across geological (geology, lithology, hydrogeology), structural (fault density), infrastructure (pipelines, energy hubs, roads, slope), and environmental (land use/land cover) categories. Criterion weights were derived using the Analytical Hierarchy Process, yielding a consistency ratio of 0.012, well below the 0.10 threshold for acceptable consistency. All spatial data were harmonised to WGS 84 UTM Zone 32N at 100-metre resolution and processed in ArcGIS Pro 3.5. Protected areas were applied as a binary exclusion mask. A weighted linear combination was used to generate suitability surfaces, classified into five classes from Unsuitable (0) to Very High (5).

The constrained suitability map shows that after excluding protected areas (12.7% of Nigeria's land area), High suitability (Class 4) occupies 206,307.17 km² (23.1%) and Very High suitability (Class 5) occupies 70,885.24 km² (8.0%). The combined area being a total of 277,192.41 km² or 31.1% of Nigeria. Prioritised zones are concentrated in eleven States across four geopolitical regions. The Niger Delta basin (Rivers, Delta, Bayelsa, Akwa Ibom, Edo) contains the largest contiguous High and Very High suitability areas, supported by favourable geology and existing infrastructure. The Anambra Basin (Enugu) and Upper Benue Trough (Gombe, Adamawa, Taraba) represent secondary prospects requiring additional characterisation.

This study provides the first systematic UHS site assessment for Nigeria, quantifies storage potential at State and Local Government Area resolution, confirms the Niger Delta as the highest priority region, and validates literature-derived AHP weights for data-limited settings. These findings establish the foundational evidence base for Nigerian policymakers and industry stakeholders to prioritise storage investments, target field investigations, and integrate underground hydrogen storage into national energy planning.

**Keywords:** Underground hydrogen storage, GIS, Multi-Criteria Decision Analysis, Analytical Hierarchy Process, Nigeria, site prioritisation, energy transition.

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| | | |
|---|---|---|
| AHP | – | Analytical Hierarchy Process |
| BGS | – | British Geological Survey |
| GIS | – | Geographic Information System |
| GLiM | – | Global Lithological Map |
| $H_2$ | – | Hydrogen |
| IUCN | – | International Union for Conservation of Nature |
| LULC | – | Land Use / Land Cover |
| MCDA | – | Multi-Criteria Decision Analysis |
| NIEP-SIP | – | National Integrated Electricity Policy and Strategic Implementation Plan |
| OGIM | – | Oil and Gas Infrastructure Mapping |
| UHS | – | Underground Hydrogen Storage |
| UNEP-WCMC | – | United Nations Environment Programme – World Conservation Monitoring Centre |
| UTM | – | Universal Transverse Mercator |
| WDPA | – | World Database on Protected Areas |
| WFP | – | World Food Programme |
| WGS 84 | – | World Geodetic System 1984 |

## 1.0 Introduction

## 1.1 Background of the Study

The global energy system is undergoing a fundamental transformation driven by the imperative to decarbonise while maintaining energy security and supporting economic development. Hydrogen has emerged as a critical vector in this transition due to its unique properties: it is energy-dense, storable, transportable, and emits only water when utilised. According to the International Energy Agency, global hydrogen demand continues to grow, concentrated primarily in refining, chemicals production, and iron and steel manufacturing. However, this consumption remains almost completely satisfied by hydrogen produced from unabated fossil fuels. Low-emissions hydrogen produced via water electrolysis using renewable electricity, from bioenergy, or from fossil fuels combined with carbon capture and storage has attracted significant policy attention given its potential to reduce greenhouse gas emissions and diversify energy supply, particularly in import-dependent economies. These challenges show that hydrogen's contribution to the energy transition depends not only on production capacity but equally on enabling infrastructure, particularly storage.

The intermittency of renewable energy sources; solar and wind creates an inherent mismatch between supply and demand that must be addressed through storage. While batteries provide effective short-duration storage (hours to days), they cannot economically address seasonal fluctuations where summer surplus must offset winter deficits. Underground hydrogen storage (UHS) offers the only technically and economically viable solution for large-scale, long-duration energy storage capable of balancing renewable supply across seasons. Zeng *et al*. (2024), in their comprehensive review published in *Engineering*, demonstrate UHS's unique position among storage technologies, noting that salt caverns provide gigawatt-scale storage capacity, depleted hydrocarbon reservoirs offer terawatt-scale potential, while surface storage tanks are limited to kilowatt-scale applications. This hierarchical scaling means that only underground storage can accommodate the volumes required for grid-scale energy management and industrial decarbonisation. The authors systematically identify that storage performance in porous media depends critically on three interconnected processes: hydrogen delivery (injection, flow, and withdrawal efficiency), hydrogen retention (trapping and containment within the pore space), and accurate equation-of-state modelling to describe hydrogen behaviour under reservoir conditions. High permeability and porosity enhance injection and withdrawal rates, while capillary sealing capacity determines containment security. Economic analysis of depleted gas reservoirs reveals significant advantages, with levelised costs for hydrogen storage in converted gas fields ranging from USD 0.55–0.90 per kilogram, and costs declining substantially as annual cycling frequency increases (Zeng *et al*., 2024).

Three principal geological settings are suitable for UHS. Salt caverns, created through solution mining in salt formations, offer the most secure storage due to salt's self-sealing properties and chemical inertness with hydrogen. Depleted hydrocarbon reservoirs leverage existing porous rock formations with proven containment, utilising available well infrastructure and geological characterisation data. Deep saline aquifers, while less characterised, provide additional capacity potential in regions lacking salt formations or depleted fields. Iranfar *et al*. (2026), applying seven multi-criteria decision-making methods to evaluate five storage options across 34 parameters in *Environmental Science-Advances*, confirm that salt caverns emerge as the most suitable sites across all methods, followed by lined rock caverns, while porous reservoirs and saline aquifers rank consistently lower. Their comparative assessment provides a transparent basis for risk evaluation and highlights the complexity inherent in UHS site selection decisions.

Nigeria stands at a critical juncture in its energy development. Despite decades of reform, the national electricity grid continues to underperform in reliability, reach, and resilience (The Electricity Hub, 2026). The National Integrated Electricity Policy and Strategic Implementation Plan (NIEP-SIP), formally ratified by President Bola Tinubu in May 2025, explicitly acknowledges this reality by shifting Nigeria's electricity strategy away from exclusive focus on centralised grid expansion towards a diversified, decentralised, and market-driven power system (OGTV, 2025). The policy aims to unlock $122.2 billion in investments to overhaul Nigeria's power sector between 2024 and 2045, incorporating hydrogen, solar photovoltaic technology, biomass, wind, and gas projects combined with carbon capture and storage technologies (OGTV, 2025). This policy framework creates natural space for hydrogen technologies, including storage.

Nigeria's Energy Transition Plan and associated climate commitments drive the country to decarbonise energy supply while sustaining economic growth (The Electricity Hub, 2026). These dual imperatives; energy access and energy transition demand solutions that are flexible, scalable, and locally adaptable. Off-grid wind–solar hydrogen systems align closely with these priorities, offering a practical mechanism to stabilise renewable power, displace diesel generation, and support productive energy use (The Electricity Hub, 2026). The NIEP-SIP explicitly recognises the economic burden imposed by self-generation using diesel and petrol, which accounts for a substantial share of electricity consumed by businesses in Nigerian cities. Off-grid hydrogen systems should therefore be evaluated not against international hydrogen cost benchmarks but against diesel-based energy systems, which remain Nigeria's de facto alternative. By converting renewable electricity into hydrogen, businesses can reduce fuel costs, stabilise long-term energy expenditures, and improve environmental performance.

Nigeria's industrial clusters; cement production, ammonia and fertiliser manufacturing, steel and metal processing, and refined petrochemicals represent concentrated hydrogen demand centres. A techno-economic assessment of green hydrogen for Nigerian industrial applications published on Zenodo (2025) demonstrates that hybrid solar–hydro configurations could achieve electrolyser capacity factors of 40–60%, meeting up to 70% of cluster hydrogen demand during initial deployment phases. The study develops an integrated modelling framework co-optimising renewable supply dispatch, electrolyser sizing, hydrogen storage, short-distance pipeline distribution, and demand aggregation across industrial clusters. Results show that the hybrid solar–hydro configuration improves electrolyser capacity factor substantially (40–60% versus 15–35% for solar-only), reduces levelised cost of hydrogen by 10–25% compared with solar-only baselines, and enables dispatchable hydrogen supply (Zenodo, 2025). This industrial demand provides the offtake certainty necessary to justify storage infrastructure investment.

Recent Nigerian scholarship has begun to examine subsurface hydrogen storage potential specifically. Agbaje *et al*. (2025), in the *NIPES Journal of Science and Technology Research*, conduct a comparative analysis of key geological storage solutions; salt caverns, deep saline aquifers, depleted oil and gas reservoirs, disused coal mines, and excavated rock caverns drawing upon international best practices from Europe, North America, and Scandinavia. Their study identifies substantial hydrogen storage opportunities across several Nigerian sedimentary formations, including the Sokoto Basin, Benue Trough, and the prolific Niger Delta, while also highlighting the potential advantages of repurposing the country's extensive oil and gas infrastructure for energy storage purposes (Agbaje *et al*., 2025). The analysis explores key parameters including storage capacity, safety, cost efficiency, and geographical constraints, concluding that leveraging international best practices while investing in localized research and pilot projects can position Nigeria as a leader in hydrogen storage across Sub-Saharan Africa. Innovative approaches to utilising depleted oil fields for hydrogen storage are also emerging internationally. Research from KAUST demonstrates that combining hydrogen with liquid organic molecules known as Liquid Organic Hydrogen Carriers (LOHCs) could cut costs, simplify storage, and even help to recover residual oil from depleted reservoirs (KAUST Discovery, 2026). Tariq *et al*. (2026) simulated how LOHC systems would perform in a depleted sandstone reservoir at approximately 2,200 metres depth, finding that about three-quarters of the carrier could be recovered after each cycle while also recovering more than half of the residual oil trapped in the field. This additional oil recovery would offset storage costs, with estimates suggesting that such projects could generate $70 million more in value than they consume (KAUST Discovery, 2026). Such approaches may have relevance for Nigeria's mature oil fields.

Despite progress in hydrogen production discussions and emerging geological characterisation, systematic assessment of underground storage potential in Nigeria using rigorous, criteria-based multi-criteria decision analysis remains absent. While international experience demonstrates that depleted hydrocarbon fields offer the most immediate and cost-effective storage opportunities (Zeng *et al*., 2024; Wei *et al*., 2026), Nigeria's extensive oil and gas infrastructure has not been systematically evaluated for this purpose. Salt formations, present in the Benue Trough and other basins, similarly lack characterisation for cavern storage. The convergence of global UHS technology maturation, European demonstration progress, Nigerian industrial hydrogen demand, and supportive policy frameworks creates both the imperative and the opportunity for this investigation. Establishing Nigeria's underground hydrogen storage potential is not merely an academic exercise but a foundational input to energy planning that will shape the country's participation in the emerging hydrogen economy.

### 1.2 Problem Statement.

The global transition towards renewable energy sources is fundamentally constrained by their intermittent nature. Solar and wind generation fluctuate with weather patterns, time of day, and seasonal cycles, creating an inherent mismatch between supply and demand that must be addressed through energy storage. While batteries provide effective short-duration storage, they cannot economically address seasonal fluctuations where summer surplus must offset winter deficits. As Marvin and Sarkinbaka (2024) observe in their comprehensive review of power-to-X technologies for Nigeria, the effectiveness of renewable energy resources is often hampered by seasonal variations, which limit the amount of energy that can be produced to meet growing demand. Long-term energy storage, particularly during periods of low demand, represents a critical enabling infrastructure for renewable integration. This storage challenge is particularly acute for hydrogen, which requires large-scale, long-duration storage solutions that only underground formations can provide (Zeng *et al*., 2024).

Nigeria's energy policy framework has begun to acknowledge these storage imperatives. The Federal Government has initiated plans to deploy battery energy storage systems to enable the integration of 4,200 MWp of solar photovoltaic power into the national grid by 2030 (Vanguard, 2025; Guardian, 2025). Speaking at the inaugural workshop on the Nigeria Battery Energy Storage System Feasibility Study, the Minister of Power explicitly identified "the issues of power intermittency, limited dispatchability of renewables, grid instability and underutilised energy generation" as persistent challenges despite major sector reforms (ThisDay, 2025). The African Development Bank has committed $1.2 million to support battery storage feasibility studies, recognising that while Africa holds nearly 60 per cent of the world's best solar resources, it accounts for only 2 per cent of global energy storage capacity (The Electricity Hub, 2025; AllAfrica, 2025). However, these initiatives focus exclusively on battery-based, short-duration storage. They do not address the long-duration, seasonal storage requirements that only underground hydrogen storage can satisfy.

Despite Nigeria's significant potential for implementing power-to-X initiatives including green hydrogen, the country currently falls behind in technology deployment and viable production pathways for sustainable implementation (Marvin & Sarkinbaka, 2024). This shortfall is primarily due to lack of policies, frameworks, and financing schemes to support infrastructural development, especially for long-term energy storage. Although green hydrogen has substantial potential as an energy carrier in Nigeria, its immediate use is limited by high production costs (Marvin & Sarkinbaka, 2024). Critically, even if production costs decline, storage infrastructure remains absent. Systematic assessment of underground hydrogen storage potential in Nigeria has not been conducted. While international experience demonstrates that depleted hydrocarbon fields offer the most immediate and cost-effective storage opportunities (Zeng *et al*., 2024), Nigeria's extensive oil and gas infrastructure has not been evaluated for this purpose. Salt formations, present in the Benue Trough and other basins, similarly lack characterisation for cavern storage.

The absence extends beyond mere identification to prioritisation. Even where potential storage formations may exist, no framework exists to rank them according to technical suitability, economic viability, safety considerations, and infrastructure access. Multi-criteria decision analysis has been applied internationally to evaluate and rank geological storage options (Iranfar *et al*., 2026; Wei *et al*., 2026), but such frameworks have not been developed or adapted for Nigeria's geological context. The convergence of these factors; renewable intermittency demanding long-duration storage, policy recognition of storage imperatives limited to short-duration batteries, complete absence of systematically identified UHS sites, and lack of prioritisation frameworks establishes the central problem this thesis addresses: Nigeria lacks a systematic, criteria-based identification and prioritisation of underground hydrogen storage sites, leaving a critical gap in the infrastructure needed to enable its energy transition.

### 1.3 Research Questions.

i. What geological, technical, and infrastructural factors most significantly influence underground hydrogen storage suitability in Nigeria?
ii. How can the Analytical Hierarchy Process be applied to systematically weight and prioritise these factors within a transparent decision-making framework?
iii. Which regions within Nigeria demonstrate the highest spatial suitability for underground hydrogen storage when evaluated using GIS-based MCDA?
iv. How robust and consistent are the prioritised storage sites when subjected to AHP consistency testing and sensitivity analysis?

### 1.4 Aim and Objectives.

The aim of this study is to identify, evaluate, and prioritise suitable geological formations for underground hydrogen storage (UHS) in Nigeria using a GIS-based Multi-Criteria Decision Analysis framework integrated with the Analytical Hierarchy Process.

The aim will be achieved through the following objectives:

i. To identify and define the key geological, technical, infrastructural, and environmental criteria relevant to underground hydrogen storage suitability in Nigeria.
ii. To develop a structured Multi-Criteria Decision Analysis (MCDA) framework using the Analytical Hierarchy Process (AHP) to assign relative weights to the identified criteria.
iii. To generate spatial suitability maps for underground hydrogen storage across selected Nigerian sedimentary basins using GIS-based weighted sum analysis.
iv. To prioritise and rank potential underground hydrogen storage sites based on suitability scores and consistency-validated decision weights.

## 1.5 Research Hypothesis.

*Null Hypothesis ($H_0$):* There are no significant differences in suitability for underground hydrogen storage between Nigeria's sedimentary basins when evaluated using GIS-based multi-criteria decision analysis and the Analytical Hierarchy Process.

*Alternative Hypothesis ($H_1$):* Nigeria's sedimentary basins exhibit varying degrees of suitability for underground hydrogen storage, with the Niger Delta basin containing the highest concentration of High and Very High suitability zones when evaluated using GIS-based MCDA-AHP.

## 1.6 Scope and Limitations.

### *1.6.1 Scope of the Study.*

This study focuses on the identification and prioritisation of suitable geological formations for underground hydrogen storage (UHS) in Nigeria using a Geographic Information System (GIS)–based Multi-Criteria Decision Analysis (MCDA) framework integrated with the Analytical Hierarchy Process (AHP).

i. Identification and evaluation of geological criteria relevant to UHS, including reservoir depth, lithology, porosity, permeability, and caprock integrity.
ii. Consideration of technical and infrastructural factors, such as proximity to energy infrastructure and accessibility.
iii. Integration of expert-informed and literature-based weighting through the Analytical Hierarchy Process.
iv. Spatial analysis and generation of UHS suitability maps using GIS-based weighted sum techniques.
v. Ranking and prioritisation of potential underground hydrogen storage zones at a regional screening level.

### *1.6.2 Limitations of the Study.*

i. **Regional-Scale Focus**: The assessment does not include site-specific reservoir simulation, geomechanical stability modelling, or operational risk analysis required for final storage design.
ii. **Data Constraints:** The analysis relies on secondary geospatial and geological datasets, which may vary in resolution, temporal coverage, and completeness across different regions of Nigeria.
iii. **Subjectivity in Criteria Weighting:** While AHP includes consistency checks, the weighting process incorporates expert judgment, which introduces controlled subjectivity (Saaty, 1980).

iv. **Subsurface Uncertainty:** Limited borehole and seismic data in some sedimentary basins restrict detailed characterisation of subsurface heterogeneity.

v. **Evolving Technological Context:** Advances in hydrogen storage technologies and changes in regulatory frameworks may alter future suitability assessments.

vi. **Data Accessibility Constraints:** Access to high-resolution geological, seismic, and reservoir datasets was limited during the course of the study. Some relevant institutions and organisations either did not possess the required datasets in accessible formats or were unable to grant access due to confidentiality and proprietary restrictions.

### 1.7 Structure of the Thesis

This thesis is structured into six chapters as follows:

**Chapter 1 – Introduction:** Introduces the research background, problem statement, aim and objectives, research questions, scope and limitations, and overall thesis structure.

**Chapter 2 – Literature Review:** Examines existing research on underground hydrogen storage, geological storage options, hydrogen behaviour in subsurface environments, and the application of GIS-based MCDA and AHP in energy infrastructure siting.

**Chapter 3 – Study Area and Methodology:** Presents the geological and tectonic setting of the selected Nigerian sedimentary basins, followed by data sources, criteria selection, MCDA framework, AHP weighting procedure, consistency testing, and GIS-based spatial analysis workflow.

**Chapter 4 – Results:** Presents criteria weights, spatial suitability maps, and prioritised underground hydrogen storage zones.

**Chapter 5 – Discussion:** Interprets results in relation to geological feasibility, hydrogen storage requirements, uncertainty considerations, and comparison with previous studies.

**Chapter 6 – Conclusions and Recommendations:** Summarises key findings, highlights contributions to knowledge, discusses limitations, and provides recommendations for future research and policy applications.

## 2.0 Literature Review

### 2.1 Overview of Nigeria

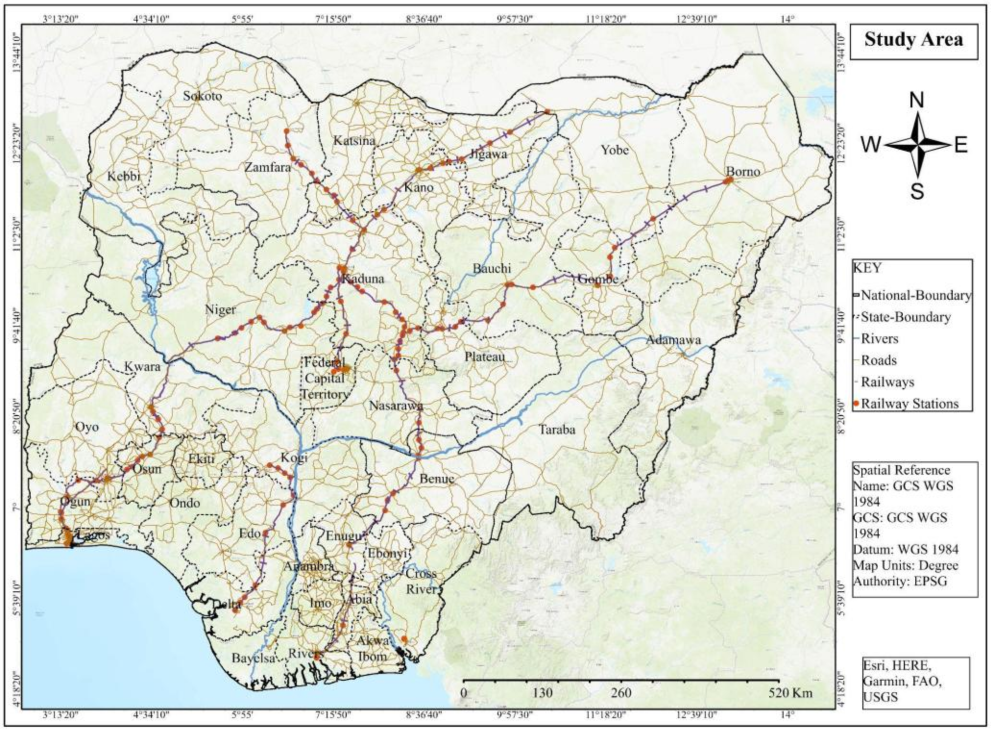


*Figure 1: Map of the Nigeria (source: World Bank, 2025)*

Nigeria is located in West Africa, situated between latitudes 4° and 14°N and longitudes 2° and 15°E (Obaje, 2009). The country shares borders with Niger to the north, Chad and Cameroon to the east, Benin to the west, and the Gulf of Guinea to the south (Obaje, 2009). Nigeria covers a total land area of approximately 923,768 square kilometres, making it one of the largest countries in West Africa (Obaje, 2009). The country's topography varies from mangrove swamps and tropical rainforests along the southern coast to savannah and semi-arid regions in the north (Obaje, 2009).

As of 2024, Nigeria's population was estimated at approximately 227 million people, making it the most populous country in Africa (World Bank, 2024). The official language is English, inherited from its colonial history, although over 500 indigenous languages are spoken across the country's numerous ethnic groups (Obaje, 2009). Nigeria is divided into 36 states and one Federal Capital Territory (Abuja), with states further grouped into six geopolitical zones: South-West, South-South, South-East, North-West, North-Central (Middle Belt), and North-East (Obaje, 2009). Each state is further subdivided into 774 Local Government Areas (LGAs), which serve as the basic administrative units for local governance (Obaje, 2009).

Nigeria is Africa's largest oil producer and holds the continent's largest natural gas reserves (Omejeh, 2025). The country has proven oil reserves estimated at 35-37 billion barrels, ranking ninth globally, with daily production averaging approximately 1.4 million barrels (Omejeh, 2025). Natural gas reserves are estimated at approximately 186 trillion cubic feet, ranking seventh globally (Omejeh, 2025). The petroleum sector dominates Nigeria's economy, with oil revenues accounting for a significant portion of government revenue and foreign exchange earnings (Omejeh, 2025). Despite its resource wealth, Nigeria faces significant energy access challenges, with approximately 80 million people lacking access to electricity, and many households and businesses relying on private diesel and petrol generators due to an unreliable national grid (Omejeh, 2025).

Nigeria contains several sedimentary basins that are of significant interest for hydrocarbon exploration and, more recently, for potential underground hydrogen storage (Obaje, 2009). The Niger Delta Basin is the most prolific sedimentary basin, located in the southern coastal region, covering approximately 75,000 km² onshore with sedimentary thickness exceeding 10,000 metres (Obaje, 2009). The basin contains over 500 proven hydrocarbon fields and accounts for the vast majority of Nigeria's oil and gas production (Obaje, 2009). The Benue Trough is a major Cretaceous rift basin extending northeast-southwest across central Nigeria for approximately 800 kilometres, divided into Lower, Middle, and Upper Benue segments with distinct structural characteristics (Obaje, 2009). The Anambra Basin, located in southeastern Nigeria adjacent to the Lower Benue Trough, has yielded several gas discoveries and contains the Ajali Sandstone, a potential reservoir unit (Obaje, 2009). The Chad Basin (Bornu Basin) in northeastern Nigeria forms part of the larger Lake Chad Basin and has been explored for hydrocarbons with hydrocarbon shows encountered, though no commercial discovery has been made (Obaje, 2009). Other sedimentary basins include the Sokoto Basin (Nigerian sector of the Iullemmeden Basin), the Mid-Niger (Bida) Basin, and the Dahomey Basin (Obaje, 2009).

Petroleum exploration in Nigeria began in 1908 when a German company, Nigerian Bitumen Corporation, conducted exploratory activities in the Araromi area of present-day Ondo State (Obaje, 2009). Systematic exploration resumed after the Second World War, and in 1956, Shell-BP made the first commercially significant oil discovery at Oloibiri village in present-day Bayelsa State (Obaje, 2009). Crude oil production began in 1957, and the first export occurred in 1960, the same year Nigeria gained independence (Obaje, 2009). Following the Oloibiri discovery, other major international oil companies, including Mobil, Gulf, Agip, Elf, and Chevron, commenced exploration and production activities (Obaje, 2009). Approximately 1,500 exploration wells have been drilled across Nigeria's sedimentary basins (Obaje, 2009). The enactment of the Petroleum Industry Act (PIA) in 2021 introduced a new legal and regulatory framework for the sector, aiming to increase transparency and attract investment (Omejeh, 2025). Nigeria's extensive exploration history and well-characterised sedimentary basins provide a valuable knowledge base for assessing potential for underground hydrogen storage, particularly in depleted hydrocarbon reservoirs where containment has been proven over geological timescales.

### *2.1.1 Geopolitical Structure of Nigeria*

Nigeria operates as a federal republic with three levels of government: federal, state, and local. The country is administratively divided into 36 states and the Federal Capital Territory (FCT), Abuja. These 36 states and the FCT are further grouped into six geopolitical zones, a classification introduced in 1995 during the military administration of General Sani Abacha (Punch Newspapers, 2025).

The six geopolitical zones and their constituent states are as follows:

| **Geopolitical Zone** | **Constituent States** |
|---|---|
| **North-Central** | Benue, Kogi, Kwara, Nasarawa, Niger, Plateau, and the Federal Capital Territory (Abuja) |
| **North-East** | Adamawa, Bauchi, Borno, Gombe, Taraba, Yobe |
| **North-West** | Jigawa, Kaduna, Kano, Katsina, Kebbi, Sokoto, Zamfara |
| **South-East** | Abia, Anambra, Ebonyi, Enugu, Imo |
| **South-South** | Akwa Ibom, Bayelsa, Cross River, Delta, Edo, Rivers |
| **South-West** | Ekiti, Lagos, Ogun, Ondo, Osun, Oyo |

The geopolitical zones were created to reflect the cultural, economic, and political realities of the country and have since been adopted by political parties in their constitutions as part of their party structures (Punch Newspapers, 2025). Federal government projects and political appointments are often carried out on the basis of these zones (Punch Newspapers, 2025).

### *2.1.2 Major Geological and Petroleum Regions*

Nigeria contains several sedimentary basins that are of significant interest for hydrocarbon exploration and more recently, for potential underground hydrogen storage. These basins formed in response to tectonic events associated with the breakup of Gondwana and the opening of the South Atlantic Ocean during the Mesozoic era (Edegbai, Schwark and Oboh-Ikuenobe, 2019). The sedimentary basins collectively cover approximately 40% of Nigeria's total land area and contain sedimentary thicknesses ranging from 2,000 metres in inland basins to over 10,000 metres in the Niger Delta (Obaje, 2009).

**Niger Delta Basin**

The Niger Delta Basin is the most prolific sedimentary basin in Nigeria, located in the southern coastal region and extending offshore into the Gulf of Guinea (Obaje, 2009). The basin covers approximately 75,000 km² onshore and over 100,000 km² offshore, with sedimentary thickness exceeding 10,000 metres in the depocentre (Obi, Okonkwo and Ezeagwuna, 2025). The stratigraphy consists of three diachronous formations: the basal Akata Formation (marine shales, source rock), the overlying Agbada Formation (paralic sandstones and shales, reservoir), and the Benin Formation (continental sands, aquifer) (Obaje, 2009). The basin contains over 500 proven hydrocarbon fields and accounts for the vast majority of Nigeria's oil and gas production (Obi, Okonkwo and Ezeagwuna, 2025).

**Benue Trough**

The Benue Trough is a major Cretaceous rift basin extending northeast-southwest across central Nigeria for approximately 800 to 1,000 kilometres, divided into Lower, Middle, and Upper Benue segments (Obaje, 2009). The trough originated from the breakup of Gondwana during the Cretaceous, which caused the divergence of the African and South American plates (Edegbai, Schwark and Oboh-Ikuenobe, 2019). The Upper Benue Trough is further subdivided into the Gongola and Yola sub-basins, with the Gongola Sub-basin containing the Kolmani oil field, where hydrocarbon discoveries have been made in the Bima, Yolde, and Gombe formations (Discover Geoscience, 2025). The Benue Trough contains fifteen sub-basins, including the Bornu, Damaturu, Lau, Numan, Yola, and Gongola sub-basins (Discover Geoscience, 2025).

**Chad Basin (Bornu Basin)**

The Chad Basin, also known as the Bornu Basin, is located in northeastern Nigeria and forms part of the larger Lake Chad Basin shared with Niger, Chad, and Cameroon (Obaje, 2009). The Nigerian portion of the basin has been explored for hydrocarbons, with several wells drilled and hydrocarbon shows encountered, though no commercial discovery has been made (Obaje, 2009). A feature known as the "Dumbulwa-Bage High or Zambuk Ridge" is an anticlinal ridge that divides the Benue Trough from the Chad Basin (Discover Geoscience, 2025).

**Sokoto Basin**

The Sokoto Basin is the Nigerian sector of the Iullemmeden Basin, located in northwestern Nigeria (Obaje, 2009). The basin contains Cretaceous to Tertiary sedimentary sequences and has been explored for hydrocarbons, though with limited success compared to the Niger Delta (Obaje, 2009). During the Maastrichtian transgressive episode, the Sokoto Basin experienced widespread marginally marine conditions, resulting in the deposition of coal, clay, ironstone, and aquiferous units (Edegbai, Schwark and Oboh-Ikuenobe, 2019).

**Dahomey Basin**

The Dahomey Basin is an amalgamation of inland, coastal, and offshore basins extending from southeastern Ghana through Togo and the Republic of Benin to southwestern Nigeria (Falufosi and Osinowo, 2021). The basin is part of the string of basins formed during the opening of the Gulf of Guinea in the Cretaceous. There is lithological continuity between the eastern end of the Dahomey Basin and the western end of the Niger Delta, which has contributed to complexity in formation discrimination (Falufosi and Osinowo, 2021). The basin has proven offshore hydrocarbon fields and enormous onshore deposits of heavy oil and tar sands, though suitable hydrocarbon trapping mechanisms are most prevalent in the deep offshore (Falufosi and Osinowo, 2021).

**Bida Basin (Mid-Niger Basin)**

The Bida Basin, also known as the Mid-Niger Basin, is located in west-central Nigeria and is bounded to the north and south by Precambrian basement rocks (Obaje, 2009). During the Campanian to Maastrichtian period, a second transgression reestablished the Trans-Saharan seaway through a westward Bida Basin route, bringing marginally marine conditions to the basin (Edegbai, Schwark and Oboh-Ikuenobe, 2019). The basin contains sedimentary sequences of Cretaceous age and has been explored for hydrocarbons, though commercial discoveries remain limited (Obaje, 2009).

**Anambra Basin**

The Anambra Basin is a Cretaceous sedimentary basin located in southeastern Nigeria, adjacent to the western flank of the Lower Benue Trough (Obaje, 2009). The basin has yielded several gas discoveries and contains the Ajali Sandstone, a potential reservoir unit with favourable porosity characteristics (Obaje, 2009). During the Maastrichtian transgression, the Anambra Basin experienced widespread marginally marine conditions, resulting in deposition of coal, clay, and ironstone units (Edegbai, Schwark and Oboh-Ikuenobe, 2019). The basin's complex structural and depositional settings have historically affected exploration success, though it remains prospective for further hydrocarbon exploration (Discover Geoscience, 2025).

## 2.2 Justification for Selecting Nigeria as a Case Study for Underground Hydrogen Storage

### *2.2.1 Geological Suitability*

Nigeria's geological framework presents several characteristics that make it a suitable candidate for underground hydrogen storage assessment. These characteristics include the presence of extensive sedimentary basins, deep saline aquifers, numerous depleted hydrocarbon reservoirs, competent caprock formations, and favourable reservoir depth and trapping conditions.

**Sedimentary Basins**

Nigeria contains several sedimentary basins that collectively cover approximately 40% of the country's total land area (Obaje, 2009). These basins include the Niger Delta Basin, Benue Trough, Anambra Basin, Chad Basin (Bornu Basin), Sokoto Basin, Dahomey Basin, and Bida Basin (Obaje, 2009). Sedimentary thicknesses range from approximately 2,000 metres in inland basins to over 10,000 metres in the Niger Delta depocentre (Obi, Okonkwo and Ezeagwuna, 2025). The presence of thick sedimentary sequences is a primary requirement for underground hydrogen storage, as adequate reservoir thickness is necessary to accommodate sufficient storage volumes (Heinemann *et al*., 2021). The extensive areal coverage of these basins provides multiple potential storage sites across different geopolitical regions of the country.

**Deep Saline Aquifers**

Potential deep saline aquifer systems occur within several Nigerian sedimentary basins, particularly the Niger Delta Basin, Benue Trough, and Chad Basin. These formations commonly comprise porous sandstone units containing saline formation water at depths generally exceeding 800 m (Obaje, 2009). Deep saline aquifers are considered promising candidates for underground hydrogen storage due to their large theoretical storage capacity, broad spatial distribution, and limited interaction with potable groundwater systems (Heinemann *et al*., 2021). Unlike depleted hydrocarbon reservoirs, however, deep saline aquifers in Nigeria have not yet been extensively characterised for hydrogen containment performance, injectivity behaviour, or long-term storage integrity. Nevertheless, their geological characteristics suggest significant potential for future regional-scale underground hydrogen storage assessments (Tarkowski, 2019).

**Depleted Oil and Gas Reservoirs**

Nigeria has over 500 proven hydrocarbon fields, primarily located in the Niger Delta Basin (Obi, Okonkwo and Ezeagwuna, 2025). These fields contain depleted reservoirs that have produced oil and gas over several decades. Depleted hydrocarbon reservoirs are considered prime candidates for underground hydrogen storage because they have demonstrated containment of buoyant fluids over geological timescales (Zeng *et al*., 2024). The extensive well data, pressure measurements, and production histories available for these reservoirs provide valuable characterisation information that reduces exploration risk (Heinemann *et al*., 2021). The presence of numerous depleted reservoirs across different basins offers multiple candidate sites for hydrogen storage assessment.

**Caprock Formations**

The sedimentary basins of Nigeria contain competent caprock formations that have effectively trapped hydrocarbons over geological timescales (Obaje, 2009). In the Niger Delta, the Akata Formation shales provide regional seal for the Agbada reservoir sandstones (Obi, Okonkwo and

Ezeagwuna, 2025). The Anambra Basin contains shale units within the Nkporo and Mamu formations that serve as seals for gas accumulations (Obaje, 2009). Caprock integrity is a critical safety criterion for hydrogen storage, as hydrogen's small molecular size and low density increase leakage potential compared to natural gas (Miocic *et al*., 2023). The proven sealing capacity of Nigerian caprock formations provides confidence in their suitability for hydrogen containment.

**Reservoir Depth and Trapping Conditions**

Hydrocarbon reservoirs in Nigeria occur at depths ranging from approximately 1,500 metres to over 4,000 metres (Obi, Okonkwo and Ezeagwuna, 2025). Optimal depth ranges for hydrogen storage are between 500 metres and 2,000 metres, where sufficient pressure exists to maintain hydrogen in dense phase without excessive compression costs (Heinemann *et al*., 2021). Many Nigerian reservoirs fall within this optimal depth window. The trapping conditions in Nigerian basins include structural traps (fault-assisted anticlinal closures), stratigraphic traps, and combination traps (Obi, Okonkwo and Ezeagwuna, 2025). These trapping styles have proven effective for hydrocarbon retention and are transferable to hydrogen storage applications, though hydrogen's unique properties require additional assessment (Miocic *et al*., 2023).

*2.2.2 Existing Oil and Gas Infrastructure*

Nigeria's long history of oil and gas production has resulted in the development of extensive infrastructure that is directly relevant to underground hydrogen storage assessment and potential deployment.

**Oil and Gas Production History**

Petroleum exploration in Nigeria began in 1908, with the first commercially significant oil discovery made at Oloibiri in 1956 (Obaje, 2009). Commercial production commenced in 1957, and Nigeria has since produced over 30 billion barrels of crude oil (Obi, Okonkwo and Ezeagwuna, 2025). Approximately 1,500 exploration and production wells have been drilled across Nigeria's sedimentary basins (Obaje, 2009). This long production history has generated extensive geological, petrophysical, and engineering data that are essential for assessing depleted reservoirs for hydrogen storage suitability (Heinemann *et al*., 2021). The knowledge gained from decades of hydrocarbon production provides a foundation for evaluating reservoir behaviour under gas injection and withdrawal cycles.

**Pipelines and Processing Facilities**

Nigeria operates an extensive network of crude oil and natural gas pipelines connecting production fields to processing facilities, refineries, and export terminals. The Niger Delta region contains the highest density of pipeline infrastructure, with major transmission corridors including the Trans-Niger Pipeline and the Escravos-Lagos Pipeline System (Omejeh, 2025). Existing gas processing facilities, compression stations, and flow stations are distributed across the producing basins. Proximity to existing pipeline infrastructure is a critical factor in hydrogen storage economic viability, as pipeline transport is the most cost-effective option for large-scale gas movement over land (IEA, 2023). The presence of existing pipelines that could potentially be repurposed for hydrogen transport reduces the capital requirements for new storage development.

**Existing Reservoir Data**

Decades of exploration and production have generated extensive reservoir data for Nigerian sedimentary basins. These datasets include well logs, core analyses, pressure measurements, fluid samples, and seismic surveys (Obaje, 2009). The availability of such data reduces the characterisation effort required for assessing depleted reservoirs for hydrogen storage compared to greenfield sites (Heinemann *et al*., 2021). Key reservoir parameters including porosity, permeability, depth, thickness, pressure, temperature, and caprock integrity have been measured across numerous fields (Obi, Okonkwo and Ezeagwuna, 2025). This existing knowledge base enables more reliable assessment of reservoir suitability for hydrogen containment.

**Petroleum Industry Expertise**

Nigeria has developed substantial petroleum industry expertise over more than five decades of oil and gas production. Technical personnel including geologists, geophysicists, petroleum engineers, and drilling specialists are available locally (Omejeh, 2025). The Nigerian National Petroleum Corporation (NNPC) and various international oil companies operating in Nigeria maintain technical teams experienced in reservoir characterisation, gas storage operations, and subsurface monitoring. This local expertise is a valuable resource for any future hydrogen storage development, reducing the need for extensive technology transfer from other regions.

*2.2.3 Energy Resources and Hydrogen Production Potential*

**Natural Gas Reserves (Blue Hydrogen Potential)**

Nigeria possesses proven natural gas reserves of approximately 206 trillion standard cubic feet, ranking seventh globally and first in Africa (Idris, Taiwo, Odeleye, Ahmed and Sani, 2025). These reserves provide a substantial resource base for blue hydrogen production through steam methane reforming or autothermal reforming combined with carbon capture and storage (Idris *et al*., 2025). Autothermal reforming in conjunction with carbon capture and storage represents the most effective and low-emission route for blue hydrogen production from natural gas (Idris *et al*., 2025). The Minister of State for Petroleum Resources has confirmed Nigeria's intention to promote blue hydrogen through the utilization of existing gas resources (Channels Television, 2024). The availability of extensive natural gas infrastructure, including pipelines and processing facilities, further supports blue hydrogen development (Omejeh, 2025).

**Solar Energy Potential**

Nigeria receives significant solar irradiation across its territory, with average annual sunshine hours of approximately 2,000 to 2,500 hours (Olarewaju, Ogunjuyigbe, Ayodele, Ayanlade, Feng and Liu, 2025). The northern regions of the country receive the highest solar radiation levels, with Sokoto, Bauchi, and Kano recording the greatest solar potential (Olarewaju, Ogunjuyigbe, Ayodele, Ayanlade, Feng and Liu, 2025). Mean annual hydrogen production from solar photovoltaic modules has been estimated at 64.33 tons per annum in Lagos to 140.28 tons per annum in Delta State (Olarewaju *et al*., 2025). The cost of hydrogen production from solar photovoltaic systems ranges from 5.67 to 6.12 USD per kilogram across different Nigerian sites (Olarewaju *et al*., 2025). The federal government has identified the country's significant solar potential, with approximately 2,000 hours of sunshine annually, as an ideal foundation for green hydrogen production (Channels Television, 2024).

**Biomass Resources**

Nigeria possesses substantial biomass resources derived from agricultural residues, municipal solid waste, livestock waste, and forest residues (Ebisi, 2024). Popular commodity crops in Nigeria can yield an estimated 2,918.50 million metric tonnes of biomass annually, translating to approximately 3.391 million gigawatt-hours of energy potential (Ebisi, 2024). Livestock waste offers additional energy potential of 950.53 thousand metric tonnes, equivalent to 263,889 megawatt-hours (Ebisi, 2024). Major Nigerian cities could potentially generate between 0.4 and 15.36 billion gigawatt-hours of energy annually from municipal waste, with total city waste capable of generating approximately 20,378 metric tonnes daily (Ebisi, 2024). A multicriteria GIS-based assessment of biomass energy potentials found that crop residues theoretical energy potential is highest in the north-east region at 1,163.32 petajoules per year, while forest residues theoretical energy potential is highest in the north-west at 260.18 petajoules per year (Ukoba, Diemuodeke, Briggs, Ojapah, Okedu, Owebor, Akhtar and Ilhami, 2024).

**Future Hydrogen Demand Potential**

Nigeria's hydrogen demand potential is expected to grow across multiple sectors, including transportation, power generation, industrial applications, and export markets (6Wresearch, 2025). Key end-user segments include energy companies, industrial sectors (particularly ammonia and fertiliser production), transportation, chemical industries, and power plants (6Wresearch, 2025). The Nigeria4H2 project, a collaborative initiative involving WASCAL and Nigerian universities, has estimated that Nigeria has the potential to produce up to 4 million tonnes of green ammonia annually (News Agency of Nigeria, 2025). The federal government estimates the global hydrogen economy could reach approximately $200 billion by 2030, with Nigeria positioning itself to capture a share of this market (Channels Television, 2024).

## 2.3 Hydrogen as an Energy Carrier.

### *2.3.1 Production, Storage, and Transport*

Hydrogen is the most abundant element in the universe, yet it does not occur naturally in its pure form on Earth and must be produced from other compounds containing hydrogen, such as water, hydrocarbons, or biomass (US Department of Energy, 2017). This distinction between hydrogen as an element and hydrogen as an energy carrier is fundamental: hydrogen is not an energy source but a vector for storing and delivering energy produced from other primary sources. Its value lies in its unique properties: it has very high energy content by weight (approximately 120 MJ/kg, nearly three times that of gasoline), is storable in multiple forms, and emits only water when utilised (Bhandari & Adhikari, 2024).

Production pathways are conventionally categorised by colour, though this nomenclature reflects the production method and associated emissions rather than physical properties. Grey hydrogen, produced from natural gas via steam methane reforming without carbon capture, dominates current supply. Blue hydrogen applies the same process but incorporates carbon capture and storage to reduce emissions. Green hydrogen, produced via water electrolysis powered by renewable electricity, represents the only pathway fully aligned with decarbonisation objectives (Du *et al.*, 2024). Electrolysis methods have advanced significantly, with proton exchange membrane and alkaline electrolysers emerging as the primary technologies, achieving improved efficiency and cost-effectiveness through recent innovations (Bhandari & Adhikari, 2024). High-temperature electrolysis and photocatalytic water splitting represent next-generation approaches under active development (El-Emam & Özcan, 2024).

Global hydrogen production reached nearly 100 million tonnes in 2024, with demand expected to surpass that milestone in 2025 (IEA, 2025a). This demand, however, remains almost exclusively from established industrial sectors—refining, ammonia production, methanol synthesis, and fossil-based direct reduced iron, with new applications such as biofuels upgrading, mobility, power generation, and synthetic fuels accounting for less than one percent of total demand (IEA, 2025a). Low-emissions hydrogen use increased by nearly ten percent in 2024 but remains below one percent of total demand due to persistent cost challenges and insufficient policy support (IEA, 2025a). The cost structure of hydrogen production varies significantly by method. Levelised costs currently range from $1–3 per kilogram for fossil-based sources to $3.4–7.5 per kilogram for electrolysis using low-emission electricity (Nyangon & Darekar, 2024). These costs are projected to decrease, particularly for electrolytic hydrogen in regions with abundant solar resources, though high production costs remain a substantial barrier to widespread adoption (Nyangon & Darekar, 2024). The global life-cycle cost of hydrogen production declines as production scales up, showing the importance of infrastructure investment and market development (Nyangon & Darekar, 2024).

Storage methods encompass physical, chemical, and advanced porous materials approaches, each with distinct advantages and limitations (Bhandari & Adhikari, 2024). Physical storage includes compressed gaseous hydrogen in pressurised vessels, liquid hydrogen at cryogenic temperatures (-253°C), and underground storage in geological formations. Chemical storage involves binding hydrogen in other compounds such as metal hydrides, liquid organic hydrogen carriers, or ammonia, from which it can be released through controlled reactions. The selection of storage method depends on the intended application, duration of storage, and required discharge rates. For large-scale, long-duration storage required to balance seasonal renewable energy fluctuations, underground geological storage offers the only technically and economically viable solution (Zeng *et al*., 2024).

Transport infrastructure remains a critical constraint on hydrogen deployment. As of 2025, approximately 37,000 kilometres of hydrogen pipelines have been announced for development by 2035, yet less than six percent have reached final investment decision (IEA, 2025b). Most announced projects are concentrated in Europe and China, where the world's longest, largest-diameter hydrogen pipelines are currently under construction. In Germany, work began in 2025 on repurposing a 400-kilometre section of natural gas pipeline, a world-first repurposing project at this scale (IEA, 2025b). Trade considerations substantially influence project development: nearly 45 percent of low-emissions hydrogen from announced production projects is intended for export, exceeding 16 million tonnes per year hydrogen-equivalent by 2030 if all projects materialise (IEA, 2025b).

### *2.3.2 Role in Decarbonisation*

Hydrogen's role in decarbonisation derives from its capacity to address sectors where direct electrification is impractical, technically challenging, or economically prohibitive (El-Emam & Özcan, 2024). These "hard-to-abate" sectors include heavy industry (steel, cement, chemicals), long-haul transport (shipping, aviation, heavy trucking), and high-temperature industrial heat. In these applications, hydrogen serves either as a feedstock (replacing fossil-derived hydrogen in ammonia and methanol production), as a reducing agent (replacing coke in steel production), or as a fuel (providing high-temperature heat or power for combustion engines and fuel cells). Industrial decarbonisation represents the most immediate and substantial opportunity for hydrogen deployment. Refining and chemical sectors currently lead the adoption of low-emissions hydrogen, with projects that are operational, under construction, or having reached final investment decision expected to consume more than two million tonnes per year by 2030 (IEA, 2025a). In steel production, hydrogen-based direct reduction offers a pathway to near-zero emissions, replacing coal as both reducing agent and energy source. Bhandari and Adhikari (2024) emphasise hydrogen's crucial role in reducing carbon dioxide emissions in industrial processes, noting that progress in this sector is essential for achieving global climate objectives.

Power sector applications focus on grid balancing and long-duration storage. Hydrogen can absorb excess renewable electricity during periods of high generation through electrolysis, store that energy for extended periods, and regenerate electricity through fuel cells or combustion turbines when renewable output is low (El-Emam & Özcan, 2024). This capability addresses the fundamental challenge of renewable intermittency: while batteries provide effective short-duration storage (hours to days), they cannot economically address seasonal fluctuations where summer surplus must offset winter deficits (Zeng *et al*., 2024). However, progress in power sector hydrogen deployment remains slow and mostly concentrated in Japan and Korea, where policy support is helping first movers progress with projects despite setbacks under early support programmes (IEA, 2025a). By 2030, hydrogen could reduce carbon dioxide emissions in electricity and power generation by 50–100 million tonnes annually, requiring 10–20 million tonnes of hydrogen and investment of $50–100 billion (Nyangon & Darekar, 2024).

Transport sector applications are advancing unevenly across modes. In road transport, heavy trucks remain the only fast-growing market for fuel cell electric vehicles, despite their higher total cost of ownership compared to battery electric or diesel alternatives (IEA, 2025a). China maintains its leadership position, accounting for almost 95 percent of the world's fuel cell commercial vehicle stock. Fuel cell vehicles offer advantages including driving ranges exceeding 300 miles on a single tank, refuelling times of just a few minutes, and fuel cell efficiency more than twice that of internal combustion engines (US Department of Energy, 2017). In shipping, the fleet of vessels capable of using hydrogen-based fuels is growing and offtake agreement activity has increased, but concerns about fuel supply availability are moderating momentum (IEA, 2025a). The recent approval of the International Maritime Organization's Net-Zero Framework could provide a boost, though its impact remains uncertain as it may favour first-generation biofuels or liquefied natural gas in the short term (IEA, 2025a).

Residential and commercial applications include combined heat and power systems, where fuel cells simultaneously generate electricity and useful heat at efficiencies exceeding 80 percent. Japan's ENE-FARM programme exemplifies this approach, deploying over 400,000 residential fuel cell units since its inception (Nyangon & Darekar, 2024). Such distributed generation applications reduce grid demand, enhance energy security, and utilise hydrogen's versatility across the energy system.

Greenhouse gas reduction potential varies significantly with production pathway and application. Compared to conventional gasoline vehicles, fuel cell vehicles can reduce carbon dioxide emissions by up to half if hydrogen is produced from natural gas, and by up to 90 percent if hydrogen is produced from renewable energy (US Department of Energy, 2017). Du *et al*. (2024) systematically reviewed green hydrogen's abatement potential across the value chain, finding that applications including carbon avoidance and embedding to realise carbon recycling have considerable carbon reduction potential. However, uncertainty limits the influence of carbon reduction cost assessment indicators based on financial analysis methods for policy guidance.

Abatement costs vary widely across value chains, electricity sources, baseline scenarios, technology mixes, and temporal contexts (Du *et al*., 2024). The synthesis of this literature reveals a consistent finding: hydrogen's contribution to decarbonisation depends not only on production capacity but equally on enabling infrastructure, particularly storage. Without adequate storage, hydrogen cannot fulfil its grid-balancing function, industrial applications cannot secure reliable supply, and the seasonal mismatches between renewable generation and demand cannot be reconciled. This infrastructure gap and its complete absence of systematic assessment in Nigeria establish the foundation for the present investigation.

### 2.4 Applications of Hydrogen

Hydrogen has a wide range of promising applications across multiple sectors beyond its role as an energy carrier. Okolie *et al.,* (2021), in a comprehensive review published in the *International Journal of Hydrogen Energy*, identified and categorised the key applications of hydrogen across various industries. In the energy sector, hydrogen is used as a direct fuel for heating and power generation, as well as in power-to-gas systems where excess electricity from renewable sources is converted into hydrogen for storage or injection into natural gas networks. Hydrogen also plays a critical role in fuel cells for stationary and mobile power generation, offering a clean alternative to internal combustion engines. In the transportation sector, hydrogen is applied in aerospace and maritime applications, serving as a clean fuel for aircraft and ships. Liquid hydrogen, due to its high energy content per unit mass, is particularly suitable for long-haul aviation and space missions where weight is a critical factor. In the industrial sector, hydrogen is essential for fuel upgrading processes in petroleum refineries, including hydrodesulfurisation, hydrodeoxygenation, and hydrocracking. These processes remove harmful impurities such as sulphur, nitrogen, and oxygen from crude oil fractions while improving fuel quality. Hydrogen is also used in ammonia production via the Haber process, which is fundamental for fertiliser manufacturing. In metallurgy, hydrogen is utilised in oxy-hydrogen flames for cutting and welding metals, as well as a reducing agent for metal extraction from ores. Hydrogen reduction offers a cleaner alternative to carbon-based reduction methods, producing water instead of carbon dioxide as a by-product. In the pharmaceutical industry, hydrogen is used in drug manufacturing, medical imaging, and hydrogen-deuterium exchange mass spectrometry (HDX-MS) for protein structure and drug discovery studies. Other applications include the synthesis of liquid hydrocarbons through the Fischer-Tropsch process, syngas fermentation for alcohol production, and the production of platform chemicals. The diversity of hydrogen applications across these sectors underscores its strategic importance as a clean energy carrier and industrial feedstock for sustainable development.


*Figure 2: Promising applications of hydrogen (source: Okolie et al., 2021)*

## 2.5 Underground Hydrogen Storage Technologies

Underground hydrogen storage (UHS) utilises natural geological formations to enable large-scale, long-term storage with a small footprint and low operating costs, while providing seasonal or long-cycle energy balancing capabilities (Huang *et al*., 2025; Ali *et al*., 2025). Compared to surface storage options such as high-pressure tanks or liquid hydrogen systems, geological storage offers volumetric capacities orders of magnitude greater, making it the only viable solution for seasonal renewable energy balancing, backup power, and industrial buffering (Krevor *et al*., 2023). The principal geological structures considered for UHS include salt caverns, depleted hydrocarbon reservoirs, and deep saline aquifers, each with distinct characteristics, advantages, and limitations.

### *2.5.1 Salt Caverns*

Salt caverns are artificially constructed cavities within salt deposit layers, formed by dissolving and extracting salt from underground salt formations through solution mining, leaving behind empty chambers (Bai *et al*., 2025). Among all UHS options, salt caverns represent the most mature and widely demonstrated technology, with operational experience dating back decades for natural gas storage and more recently for hydrogen (Zheng *et al*., 2025). The Advanced Clean Energy Storage project in Delta, Utah, exemplifies current commercial-scale development, combining 220 megawatts of alkaline electrolysis with two massive 4.5 million barrel salt caverns to store clean hydrogen for seasonal deployment (US Department of Energy, 2022).

The fundamental advantages of salt caverns derive from the unique properties of rock salt. First, salt exhibits viscoplasticity and extremely low permeability, undergoing slow deformation under external forces that seals any microfractures and enhances containment over time (Bai *et al*., 2025). Second, rock salt is chemically inert to hydrogen, eliminating concerns about geochemical reactions that could consume stored hydrogen or produce contaminants (Zheng *et al*., 2025). Third, salt formations are practically impermeable to gases, providing secure long-term storage with minimal leakage risk. These characteristics enable salt caverns to achieve the highest hydrogen recovery rates and purity maintenance among all storage options. Operational experience confirms these advantages. Salt caverns are typically located at depths ranging from 500 to 1500 metres and have demonstrated excellent sealing performance with hydrogen loss rates approaching zero (Zheng *et al*., 2025). The internal pressure in salt caverns can be actively managed over a wide range, providing high flexibility for frequent injection-withdrawal cycles. This makes salt caverns particularly suitable for grid-scale applications requiring rapid response to electricity demand fluctuations.

However, salt caverns also present limitations. Their construction requires thick, high-purity salt formations, which are geographically restricted. The solution mining process involves significant capital investment, with construction costs estimated at approximately €0.6–1.5 per kilogram of hydrogen storage capacity (Zheng *et al*., 2025). Additionally, cavern creation generates large volumes of brine requiring responsible disposal. Storage capacity is inherently limited by cavern dimensions, typically ranging from 500,000 to 1 million cubic metres per cavern, though multiple caverns can be developed in a single salt formation to aggregate capacity. Despite these constraints, salt caverns remain the preferred option where suitable salt formations exist due to their demonstrated reliability, hydrogen compatibility, and operational flexibility. The SHASTA project (Subsurface Hydrogen Assessment, Storage, and Technology Acceleration) led by US National Laboratories continues to investigate salt cavern performance, though their research increasingly focuses on expanding storage options to porous media where salt formations are absent (US Department of Energy, 2021).

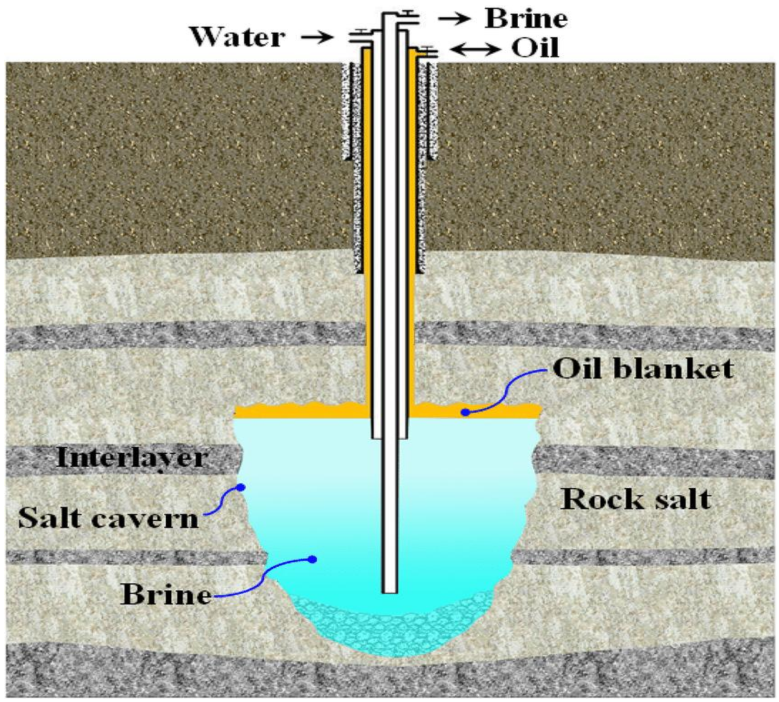


*Figure 3: Sketch of a salt cavern construction (Shi et al. 2017)*

### *2.5.2 Depleted Oil and Gas Reservoirs*

Depleted hydrocarbon reservoirs are porous rock formations that previously contained oil or natural gas and have been brought to the end of their economic production life. These formations have received increasing attention for UHS due to their substantial storage capacities, well-characterised geological conditions, and availability of existing infrastructure (Kanaani *et al*., 2022; Sarı & Çiftçi, 2024). The primary advantage of depleted reservoirs lies in their proven containment. These formations have trapped hydrocarbons over geological timescales, demonstrating the integrity of their caprock seals and structural traps. This proven containment record significantly reduces uncertainty compared to greenfield storage developments (Khajooie *et al*., 2025). Additionally, depleted reservoirs benefit from extensive characterisation data acquired during exploration and production, including well logs, seismic surveys, core analyses, and production histories. This existing knowledge base reduces the need for new data acquisition and enables more reliable performance prediction.

Infrastructure re-use represents a further economic benefit. Existing wells, pipelines, compression facilities, and monitoring systems can often be repurposed for hydrogen storage, substantially reducing capital requirements compared to developing new facilities from scratch (Yang *et al*., 2025). Studies indicate that utilising existing infrastructure can achieve significant cost reductions, with storage costs in depleted reservoirs estimated at approximately €0.6–1.2 per kilogram of hydrogen (Zheng *et al*., 2025). The International Journal of Hydrogen Energy (2025) confirms that repurposing depleted reservoirs is significantly more economical and scalable, especially given the demonstrated containment provided by pre-existing infrastructure. However, depleted reservoirs present technical challenges unique to hydrogen storage. Unlike salt caverns, porous media involve complex interactions between hydrogen, residual hydrocarbons, formation water, and rock minerals. Physical adsorption or chemical reactions may occur between hydrogen and residual phases, potentially affecting hydrogen purity and recovery efficiency (Zamehrian & Sedaee, 2022b; Wang *et al*., 2024c). Microbial activity present in some depleted reservoirs can consume hydrogen through methanogenesis or sulfate reduction, leading to hydrogen loss and potential reservoir plugging (Dopffel *et al*., 2021).

The requirement for cushion gas represents another significant consideration. Depleted reservoirs typically require 50–60% of total gas volume as cushion gas to maintain reservoir pressure and prevent excessive water influx (Navaid *et al*., 2023). This cushion gas, which remains permanently in the reservoir, represents a substantial upfront investment. However, residual hydrocarbons already present can serve as partial cushion gas, reducing this requirement compared to aquifers. Recent research has extended the depleted reservoir concept to unconventional formations. Khajooie *et al*., (2025) provides the first systematic review of UHS in depleted shale gas reservoirs, highlighting that shale formations offer unique advantages including organic matter that adsorbs hydrogen (reducing reactivity with minerals) and extremely low matrix permeability that minimises leakage risk. The complex fracture networks created during production also facilitate hydrogen transport and provide additional storage capacity. However, technical challenges including hydrogen-brine-shale interactions and multi-field coupling processes require further investigation.

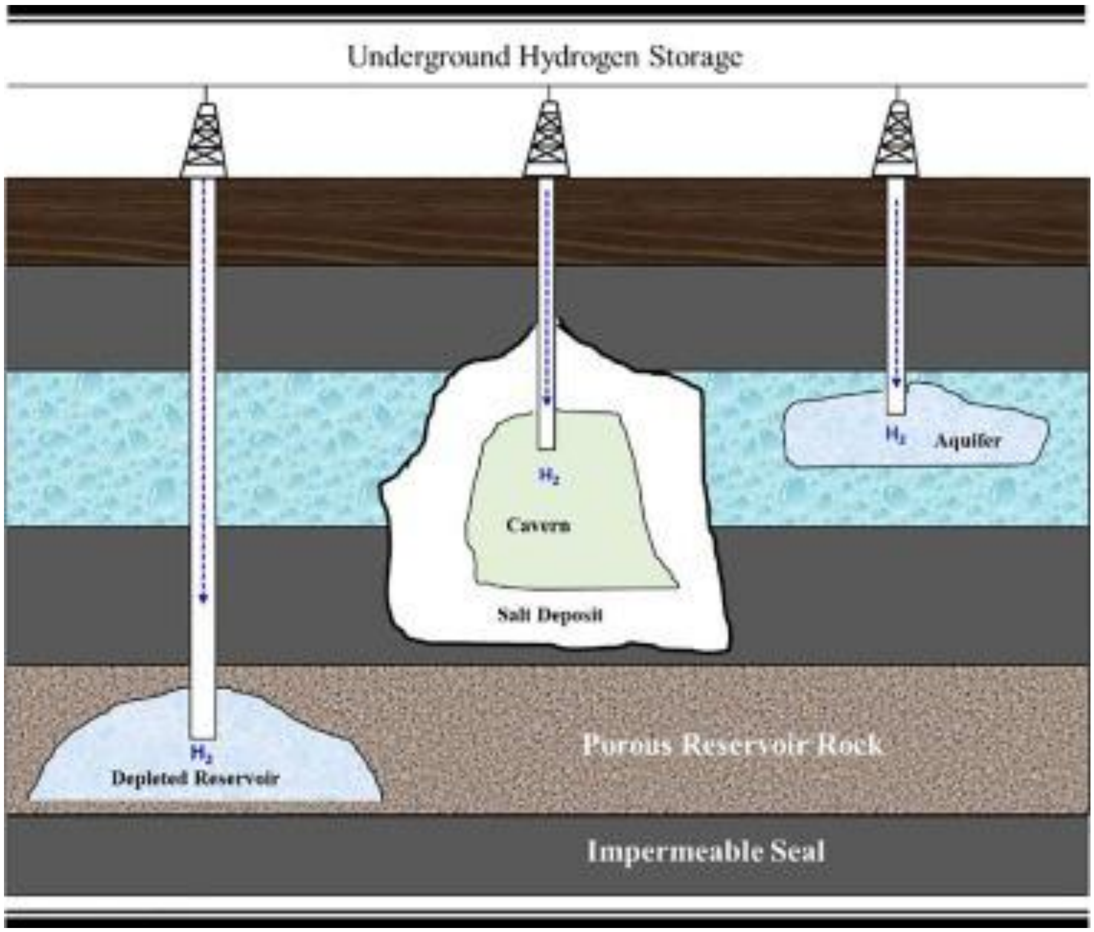


*Figure 4: Sketch of an underground hydrogen storage showing depleted oil and gas reservoir (ScienceDirect, 2025)*

### *2.5.3 Deep Saline Aquifers*

Deep saline aquifers are porous rock formations saturated with brine (high-salinity water) that are not suitable for potable water supply or other beneficial uses. Due to their widespread geographic distribution, aquifers offer significant flexibility and availability in site selection for UHS, with immense theoretical storage capacities far exceeding those of depleted reservoirs or salt caverns (Baru *et al*., 2025). The primary attraction of saline aquifers is their capacity potential. Unlike depleted reservoirs, which are limited to areas where hydrocarbon accumulations existed, aquifers are present in most sedimentary basins worldwide. This geographic availability means that regions lacking salt formations or depleted fields may still access UHS through aquifers. Studies indicate that aquifers could provide the storage volumes necessary for grid-scale seasonal hydrogen balancing at national and continental scales (Alafnan, 2024).

However, saline aquifers present the greatest technical uncertainty among UHS options. Several factors contribute to this complexity. First, aquifer diagenetic characteristics, while generally providing structural stability, require careful evaluation to ensure adequate seal integrity. Compared to salt caverns, aquifers typically have weaker caprock sealing properties, posing risks of gas leakage that must be addressed through rigorous site selection and operational protocols (Braid *et al*., 2024). Second, hydrogen injection into aquifers involves complex multiphase flow dynamics. Hydrogen's low density and viscosity relative to brine can lead to viscous fingering, capillary trapping, and buoyancy-driven migration toward the caprock (Baru *et al*., 2025). These phenomena affect both storage efficiency and recovery rates. During injection, hydrogen displaces formation water; during withdrawal, water may encroach and trap hydrogen in pore spaces, reducing recoverable volumes. Reservoir heterogeneity significantly influences these processes, with layered or fractured formations presenting particular challenges. Third, geochemical reactions between hydrogen, brine, and reservoir minerals can consume hydrogen and produce secondary gases. Hydrogen may participate in redox reactions with minerals containing iron, sulfur, or other oxidised species, potentially generating hydrogen sulfide or other contaminants that affect gas quality and safety (Baru *et al*., 2025). The extent of these reactions depends on mineralogy, brine chemistry, temperature, and pressure, requiring site-specific evaluation. Fourth, microbial activity in aquifers represents a potentially significant hydrogen loss mechanism. Indigenous microbial populations can consume hydrogen through methanogenesis (producing methane), sulfate reduction (producing hydrogen sulfide), or acetogenesis (Baru *et al*., 2025). Historical experience from town gas storage in aquifers at Lobodice and Beynes documented substantial hydrogen consumption by microorganisms (Dopffel *et al*., 2021). Mitigation strategies may include biocides, nutrient limitation, or operational approaches to minimise microbial activity. Fifth, cushion gas requirements for aquifers are substantially higher than for depleted reservoirs. Since aquifers initially contain no gas, up to 80% of the total gas volume may need to be injected as cushion gas to maintain reservoir pressure and control water movement (Thiyagarajan *et al*., 2022). This represents a significant capital investment that affects project economics. Economic analysis emphasises the significant influence of cushion gas volumes and hydrogen recovery efficiency on the levelized cost of storage (Baru *et al*., 2025).

Despite these challenges, aquifers remain widely recognised as strong candidates for large-scale UHS due to their enormous capacity potential and geographic availability. Baru *et al*. (2025) provide a comprehensive roadmap for developing scalable, safe, and economically viable hydrogen storage in deep saline aquifers, emphasising the importance of integrated, coupled-process modelling and comprehensive site characterisation to optimise storage design and operation. The SHASTA project specifically investigates storage opportunities in porous media to expand UHS beyond salt-dominated regions (US Department of Energy, 2021).

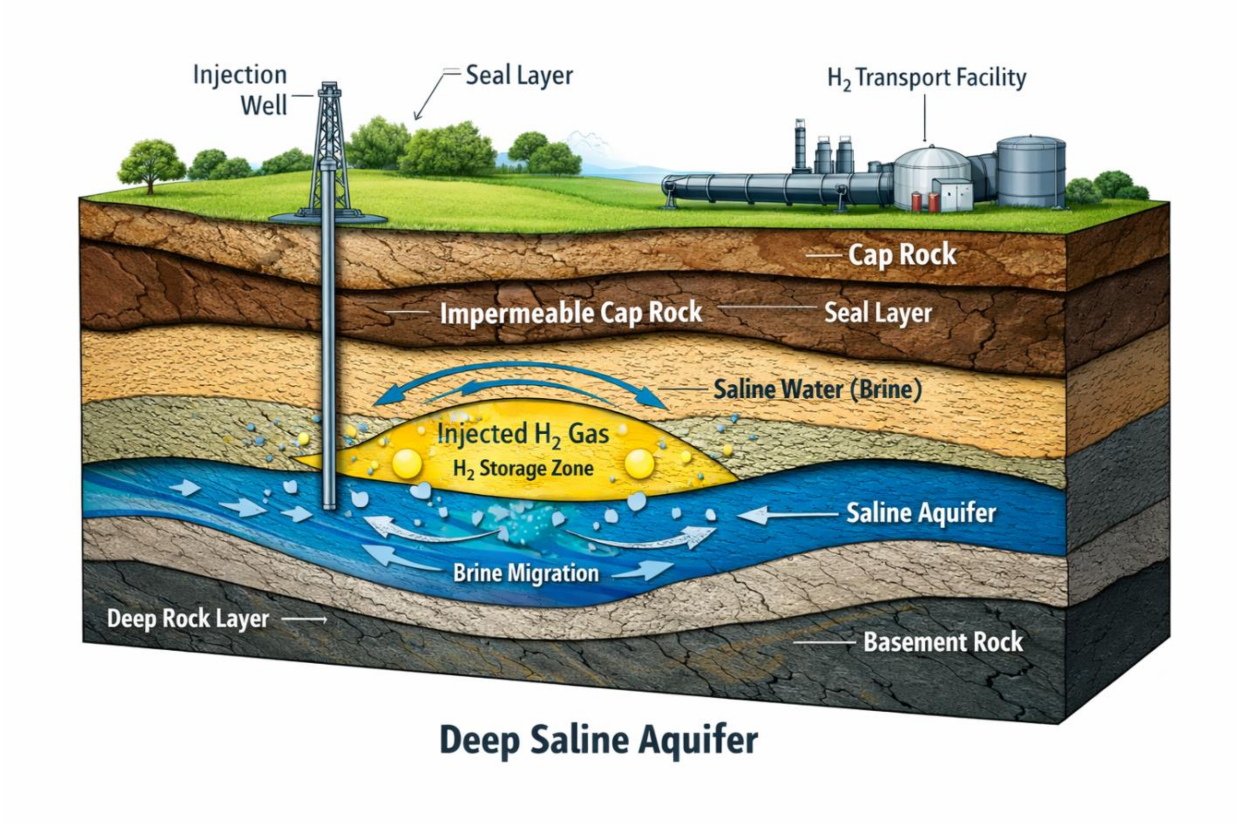


*Figure 5: Sketch of a deep saline aquifer (Author, 2026)*

*2.5.4 Comparative Assessment of Storage Types*

A growing body of literature provides comparative assessments of UHS technologies to guide site selection and technology prioritisation. Mu *et al*., (2025) systematically reviews five UHS methods; salt caverns, hydro-reservoirs (aquifers), depleted oil and gas reservoirs, abandoned mines, and hard rock caverns comparing them across multiple performance metrics including storage capacity, economy, construction difficulty, hydrogen underground migration, and hydrogen purity.

The comparative analysis reveals clear patterns. Salt caverns consistently emerge as the most mature and widely used option due to their excellent sealing properties and extremely low hydrogen loss. They offer the highest operational flexibility and maintain hydrogen purity most effectively. However, their construction is geographically constrained and capital-intensive. Depleted reservoirs provide the best balance of capacity, cost, and existing infrastructure, making them key targets for current large-scale UHS research and applications. Their proven containment record and characterisation data reduce technical uncertainty, while infrastructure re-use improves economics. However, concerns about hydrogen interactions with residual hydrocarbons and microbial activity require careful management. Aquifers offer the largest theoretical capacity and widest geographic distribution but face the greatest technical uncertainty and highest cushion gas requirements. They represent a longer-term option requiring substantial research and demonstration before commercial deployment can be considered reliable.

Iranfar *et al*. (2026), applying seven multi-criteria decision-making methods to evaluate five storage options across 34 parameters, confirm that salt caverns emerge as the most suitable sites across all methods. Their comparative assessment provides a transparent basis for risk evaluation and highlights the complexity inherent in UHS site selection decisions. For Nigeria, this comparative analysis has direct implications. The Niger Delta's extensive depleted hydrocarbon fields represent immediate candidates for UHS assessment, offering proven containment, existing infrastructure, and substantial characterisation data. The Benue Trough's salt formations merit investigation for cavern storage potential, which would provide higher-performance storage for applications requiring rapid cycling and maximum hydrogen purity. Deep saline aquifers, present across multiple Nigerian basins, represent longer-term potential requiring more extensive characterisation but offering the largest ultimate capacity.

## 2.6 Geological and Technical Requirements for UHS

The suitability of a geological formation for underground hydrogen storage is determined by a combination of intrinsic rock properties and ambient reservoir conditions that together govern storage capacity, injectivity, containment security, and recovery efficiency. This section critically reviews the literature on porosity and permeability, caprock integrity, and depth-pressure-temperature constraints that collectively define the technical envelope for viable UHS operations.

### *2.6.1 Porosity and Permeability*

Porosity and permeability constitute the fundamental storage and flow characteristics of any potential UHS reservoir. Porosity, defined as the fraction of void space within the rock, determines the total volume available for hydrogen storage, while permeability quantifies the rock's capacity to transmit fluids under an applied pressure gradient, directly controlling injection and withdrawal rates (Nasiru *et al*., 2024). For hydrogen, whose low density and viscosity create unique flow behaviour, these parameters assume heightened significance.

Quantitative thresholds for viable UHS have been established through systematic review of operational projects and experimental studies. Adequate reservoir porosity must exceed 10 percent, with values greater than 15 percent considered favourable for commercial-scale storage (Nasiru *et al*., 2024). This requirement reflects the need to accommodate sufficient hydrogen volumes within economically viable footprint areas while maintaining acceptable working gas ratios. Permeability requirements are similarly defined, with minimum thresholds of 100 millidarcies necessary for adequate injectivity and deliverability, though values exceeding 500 millidarcies enable higher cycling rates and improved operational flexibility (Nasiru *et al*., 2024).

The pore-scale behaviour of hydrogen in porous media differs fundamentally from that of natural gas or carbon dioxide due to hydrogen's distinctive physical properties. Dorhjie and Cheremisin (2024) conducted pore-scale network simulations to investigate hydrogen-brine relative permeability, revealing that capillary pressure and relative permeability are highly sensitive to the distribution of surface contact angles. Their results demonstrate that the relative permeability of the hydrogen phase decreases as the frequency of pores with stronger water-wet contact angle values increases, with relative permeability endpoints significantly influenced by pore and throat geometry, size distribution, and pore connectivity. This finding has direct implications for site selection: formations with uniform pore networks and favourable wettability characteristics will exhibit superior hydrogen flow performance.

The relative permeability endpoint (residual saturation) determines how much hydrogen becomes trapped in the pore space during withdrawal cycles and is therefore unrecoverable. Al homoud *et al*. (2024) numerically evaluated the influence of wettability and relative permeability hysteresis on hydrogen recovery in saline aquifers, demonstrating that neglecting hysteresis leads to substantial overestimation of gas production. Their base model, which ignored hysteresis effects, predicted a recovery factor of 78 percent by the fourth operational cycle. However, when hysteresis was incorporated using the Carlson model with laboratory-measured trapped gas saturation of approximately 17 percent, the hydrogen recovery factor fell to 45 percent by the fourth cycle. This 33 percent reduction in recoverable volume has profound implications for project economics and underscores the necessity of incorporating hysteretic behaviour in storage performance assessments.The study further revealed that gas hysteresis significantly impacts water production, with a 12.5 percent increase in water volume observed in models incorporating gas hysteresis (Al homoud *et al*., 2024). Water hysteresis also proved substantial, contributing to additional reductions in hydrogen recovery. Critically, the authors concluded that relative permeability hysteresis exerts a more significant influence on hydrogen production than other petrophysical phenomena such as wettability, which demonstrated limited impact on operational feasibility and poses little threat to hydrogen storage in sandstone formations. These finding redirects attention from wettability characterisation, which has received disproportionate research emphasis, towards comprehensive hysteresis quantification.

Dorhjie and Cheremisin (2024) also examined the effect of different cushion gases on hydrogen-brine relative permeability, finding that the relative permeabilities of methane and nitrogen are similar to that of hydrogen. This observation supports the use of these gases as cushion gas options without fundamentally altering flow behaviour, though their different densities and chemical properties affect other aspects of storage performance. The pore-scale processes governing hydrogen flow operate within broader petrophysical frameworks that must be characterised at multiple scales. Nasiru *et al*. (2024) emphasise that comprehensive evaluation of geological reservoirs for UHS projects requires detailed assessment of key reservoir properties, with adequate porosity and permeability identified as critical prerequisites for ensuring injectivity and deliverability. The interaction between pore-scale heterogeneity and larger-scale reservoir architecture determines whether favourable local properties translate into viable storage performance at field scale.

*2.6.2 Caprock Integrity*

The caprock, or seal, is the geological barrier that prevents upward migration of stored hydrogen from the reservoir to overlying formations or the surface. For hydrogen, whose small molecular diameter, low density, and high diffusivity create exceptional leakage potential, caprock integrity represents perhaps the most critical safety and containment parameter. Evaluation of caprock sealing capacity encompasses multiple mechanisms: capillary entry pressure, geochemical stability, and mechanical integrity under cyclic loading.

Recent experimental advances have enabled high-resolution quantification of caprock sealing performance under UHS conditions. Li *et al*. (2025) developed a novel core-flooding apparatus integrated with a micro-capillary flow meter capable of measuring ultra-low permeabilities down to 10 nano-Darcy, flow rates as low as 10 nano-litres per hour, and both threshold and breakthrough pressures. Benchmark tests with nitrogen and methane were followed by hydrogen experiments across caprocks spanning a wide range of permeability and porosity. Their results demonstrate clear trends between caprock properties and sealing performance: threshold and breakthrough pressures rise with higher gas-brine interfacial tension, and caprock permeability and porosity inversely correlate with sealing capacity. Importantly, hydrogen exhibited slightly lower threshold and breakthrough pressures compared to nitrogen and methane, reinforcing the need for accurate site-specific caprock evaluation rather than reliance on natural gas storage analogues.

The capillary sealing mechanism is governed by the Young-Laplace equation, which relates the capillary pressure required for gas to enter a water-saturated pore throat to interfacial tension, contact angle, and pore throat radius. Thicker caprock layers substantially reduce gas leakage rates. Capillary bundle models estimate higher leakage rates, while pore network models accounting for the shielding effect of small channels predict lower rates, demonstrating the importance of appropriate modelling approaches for accurate leakage estimation. Geochemical interactions between hydrogen, brine, and caprock minerals represent an additional dimension of seal integrity that has received increasing research attention. Zeng *et al*. (2023) developed kinetic batch models to characterise time-dependent redox reactions unique to underground hydrogen storage, combined with analytical estimates for hydrogen penetration into caprock. Their results show that dissolution degrees of tested minerals in three types of shales remain below one percent over 30 years, indicating strong caprock integrity and containment ability during UHS from a geochemical perspective. Reactive transport calculations indicate that hydrogen affects only a few metres of caprock above the reservoir, leaving storage integrity of thick caprocks unaffected. The overall amount of hydrogen penetrating into caprock is typically much less than one percent of stored volumes. These findings suggest that hydrogen-brine-shale geochemical interactions may not compromise caprock integrity during UHS, providing reassurance regarding long-term seal performance.

The relationship between caprock permeability and porosity and sealing capacity established by Li *et al*. (2025) provides a quantitative framework for evaluating candidate storage sites. By measuring threshold and breakthrough pressures across caprocks with varying properties, their work enables screening-level assessments based on routine core analysis data rather than requiring specialised experimental measurements for every candidate site. This practical contribution supports systematic evaluation of depleted reservoirs, which the authors identify as growing in interest due to cost advantages and geographic availability compared to salt caverns.

*2.6.3 Depth, Pressure, and Temperature Constraints*

Depth exerts primary control over pressure and temperature conditions in potential storage formations, which together determine hydrogen phase behaviour, storage density, containment security, and operational safety. The relationship between depth and these parameters creates a window of viable storage conditions that must be respected in site selection.

Minimum depth requirements are driven by the need for sufficient pressure to maintain hydrogen in dense phase and ensure effective containment. Hydrogen exists as a gas under surface conditions but compresses with increasing pressure. At depths exceeding approximately 800 metres, pressure and temperature conditions reach the supercritical state for hydrogen, where it exhibits gas-like diffusivity and liquid-like density, maximising storage efficiency (Qureshi *et al*., 2022). Maximum depth constraints arise from economic and technical considerations. As depth increases, drilling and completion costs escalate non-linearly, and reservoir pressures approach fracture gradients that limit achievable injection pressures. Nasiru *et al*. (2024) indicate that reservoir depth must range from greater than 200 metres to 2000 metres to maintain sufficient pressure without significantly increasing capital and operational costs. The upper bound reflects both economic limits and the practical challenges of operating at high pressures and temperatures.

Temperature increases with depth according to local geothermal gradients, which vary across sedimentary basins typically ranging from 20 to 35 degrees Celsius per kilometre. Temperature influences hydrogen properties including density, viscosity, and diffusivity, as well as the kinetics of geochemical reactions and microbial activity. Experimental investigations conducted at two representative conditions—55 bar and 20 degrees Celsius representing shallower storage, and 100 bar and 45 degrees Celsius representing deeper storage—revealed that relative permeability curves remain very close across these conditions (Yekta *et al*., 2018). This stability is attributed to the nearly constant viscosity of hydrogen under the investigated pressure and temperature range, in contrast to other fluid pairs such as the carbon dioxide-water system where capillary numbers vary strongly with pressure and temperature. Similarly, capillary pressure data varied little between experimental conditions, suggesting that relative permeability and capillary pressure results may be applicable across a wide range of pressure and temperature conditions relevant to UHS (Yekta *et al*., 2018).

The relationship between depth and storage capacity follows from gas compressibility. Qureshi *et al*. (2022) present normalised volume of hydrogen as a function of depth over typical geothermal gradients, demonstrating that hydrogen volume decreases with depth as pressure increases, enabling greater mass storage within fixed reservoir pore volume. This compressibility effect means that deeper storage within the viable window achieves higher volumetric efficiency, though this advantage must be weighed against increased costs. Pressure conditions also determine the required cushion gas volume; the gas that must remain permanently in the reservoir to maintain pressure and prevent water influx. Depleted reservoirs typically require 50 to 60 percent of total gas volume as cushion gas, while aquifers may require up to 80 percent (Nasiru *et al*., 2024). The specific pressure regime influences these requirements, with overpressured reservoirs potentially requiring less cushion gas than normally pressured formations. The integrated consideration of depth, pressure, and temperature establishes the operational envelope for UHS. Formations shallower than 500 metres generally lack sufficient pressure for efficient storage and present elevated leakage risks due to inadequate containment forces. Formations deeper than 2500 metres encounter elevated costs, high pressures approaching fracture limits, and potential geochemical reaction rates that could affect hydrogen quality and reservoir properties. Within the optimal window of approximately 800 to 2000 metres, hydrogen exists in supercritical state with favourable density, flow behaviour remains consistent across typical temperature variations, and containment can be assured with appropriate caprock.

## 2.7 Site Selection and Prioritization Methods

The identification of suitable locations for underground hydrogen storage is inherently a spatial decision problem requiring the simultaneous evaluation of multiple, often conflicting, criteria. This section critically reviews the methodological literature on site selection and prioritization, focusing on Geographic Information Systems (GIS), Multi-Criteria Decision Analysis (MCDA), and the Analytical Hierarchy Process (AHP), which form the analytical foundation of this study.

### *2.7.1 GIS-Based Suitability Mapping*

Geographic Information Systems provide the technological platform for acquiring, storing, managing, analyzing, and visualizing spatial data relevant to site selection. In the context of energy infrastructure siting, GIS enables the integration of diverse spatial datasets; geological maps, infrastructure networks, environmental constraints, and topographic data within a common coordinate framework, facilitating systematic evaluation across large geographical areas.

The application of GIS to underground hydrogen storage site selection has been demonstrated in several recent studies. A seminal work by researchers at the Mineral and Energy Economy Research Institute of the Polish Academy of Sciences developed a methodology for selecting optimal locations for hydrogen storage in salt caverns using GIS-based multi-criteria decision analysis coupled with spatial data analysis (Tarkowski & Czapowski, 2023). Their approach, published in the *International Journal of Hydrogen Energy*, enables the creation of rock salt deposit suitability maps that help identify favorable locations for hydrogen storage facilities, taking into account the storage capacity of geological structures and various surface and underground factors that may limit or preclude storage potential (Tarkowski & Czapowski, 2023). The methodology's value extends to governmental institutions, geological services, power plants producing electricity from renewable sources, and chemical and petrochemical plants.

The effectiveness of GIS-MCDA integration for energy storage site selection has been systematically reviewed by Lopez *et al*. (2024), who assessed studies published between 2000 and June 2024 utilizing methodologies that combine GIS and MCDA for pumped hydroelectric energy storage site planning. Following PRISMA guidelines, their systematic review confirmed that the combined use of GIS and MCDA is generally effective in identifying potential sites for energy storage installations, considering a range of environmental, economic, and technical criteria (Lopez *et al*., 2024). However, they also highlighted challenges, including limited availability of geospatial data and the complexity of decision-making models, noting that while these approaches offer advantages in accuracy and efficiency, limitations exist regarding the need for advanced technical skills and high implementation costs.

GIS-based suitability assessment has also been applied to green hydrogen production facility siting. A Master's thesis from Aalto University generated site suitability maps using GIS-based Multi-Criteria Analysis for green hydrogen projects in Norrbotten County, Sweden, considering factors including renewable energy potential, proximity to key resources and demand points, existing grid support, underground geology suitability for storage, and proximity to restricted and protected zones (Ramsing Thakur & Adolfsson, 2024). The study demonstrated that GIS-based MCA can provide reasonably good pointers for making viable green hydrogen business cases, particularly when integrated with demand analysis and resource sufficiency assessment.

The fundamental value of GIS in site selection lies in its capacity to handle the spatial dimensions of decision criteria. Unlike tabular data or non-spatial analytical methods, GIS explicitly represents geographical variation in factors such as geological suitability, infrastructure proximity, and environmental constraints. This spatial explicitness is essential for UHS site selection because storage potential varies continuously across sedimentary basins according to depth, lithology, fault density, and distance to pipelines and demand centres. By enabling weighted overlay analysis and suitability surface generation, GIS transforms raw spatial data into actionable intelligence for decision-makers.

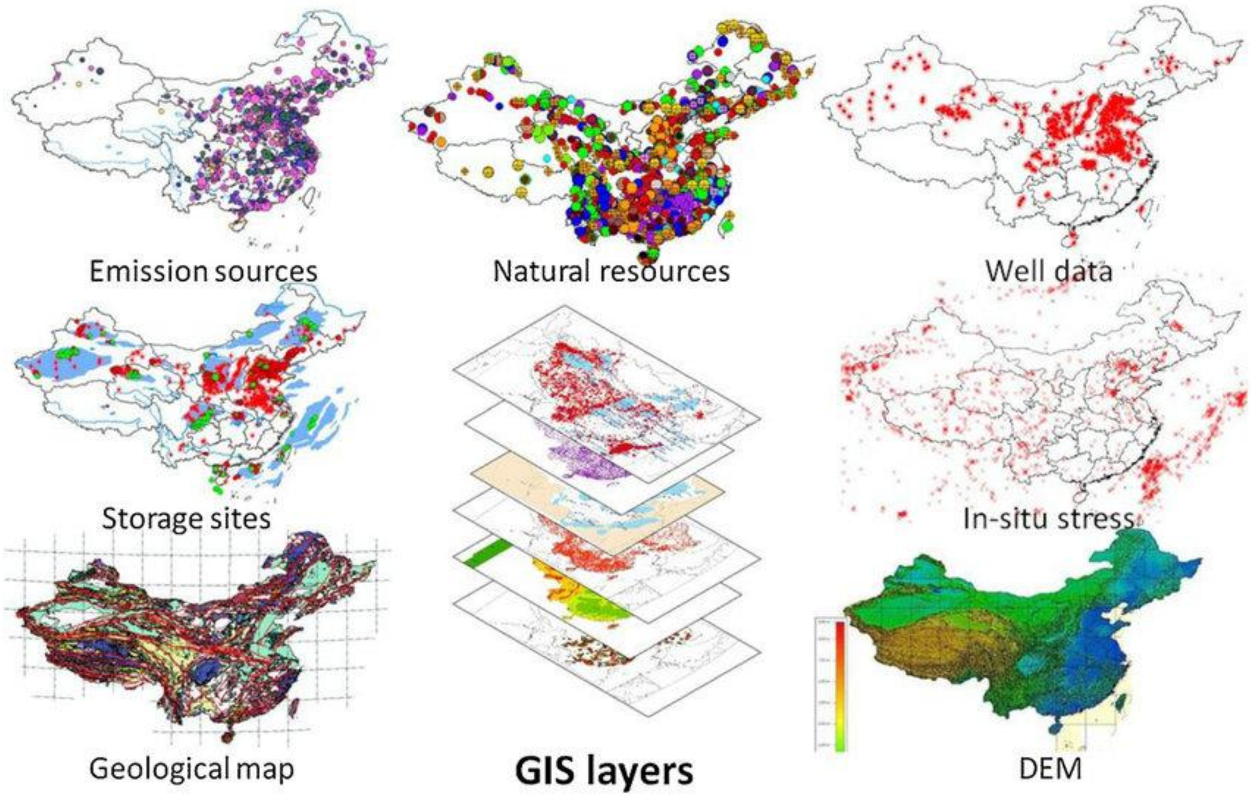


*Figure 6: Datasets for GIS site suitability evaluation (Ning et al., 2013)*

### *2.7.2 Multi-Criteria Decision Analysis (MCDA)*

Multi-Criteria Decision Analysis encompasses a family of methods designed to support decision-making when multiple, often conflicting, criteria must be considered simultaneously. In the context of energy infrastructure siting, MCDA provides structured, transparent, and replicable frameworks for handling the trade-offs inherent in evaluating locations against diverse criteria measured in different units.

The application of MCDA to energy storage site selection has grown substantially in recent years. A systematic review by Lopez *et al*. (2024) identified numerous studies employing various MCDA methods for pumped hydroelectric energy storage site planning, confirming that MCDA approaches are generally effective in identifying potential sites while considering environmental, economic, and technical criteria. The review highlighted that MCDA methods enable the incorporation of both quantitative measurements and qualitative judgments within consistent analytical frameworks, making them particularly suitable for complex spatial decisions where data completeness varies and stakeholder perspectives differ.

For underground hydrogen storage specifically, Tarkowski and Czapowski (2023) demonstrated that MCDA coupled with spatial data analysis enables objective site selection and assessment of the impact of individual criteria. Their methodology, developed for assessing rock salt deposit suitability, explicitly frames site selection as a multi-criteria decision problem requiring systematic evaluation of geological, technical, environmental, and social factors. The approach recognizes that optimal locations for underground storage facilities are determined by multiple interacting criteria, and that transparent handling of these criteria is essential for defensible decision-making.

The MCDA literature distinguishes between multiple objective decision-making (MODM) and multiple attribute decision-making (MADM) methods. MODM approaches are typically used for design problems with continuous solution spaces, while MADM methods are applied to evaluation problems with discrete alternatives. For site selection problems, where the decision space consists of discrete locations or areas, MADM methods are most appropriate. Within MADM, various schools have developed, including:

i. Outranking methods such as ELECTRE (Elimination et Choix Traduisant la Réalité) and PROMETHEE (Preference Ranking Organization Method for Enrichment Evaluation)
ii. Distance-to-ideal methods such as TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution)
iii. Additive methods such as the Analytical Hierarchy Process and weighted linear combination (Tarkowski & Czapowski, 2023)

The choice among these methods depends on problem characteristics, including the nature of criteria (quantitative vs. qualitative), the availability of preference information, and the desired properties of the decision rule. For energy infrastructure siting, AHP has emerged as the most widely applied method due to its intuitive structure, transparency, and ability to integrate with GIS.

Emerging research has extended MCDA applications to broader energy system contexts. The RESTOR (Renewable Energy STORage) project, initiated in January 2025, aims to develop and apply a multi-criteria planning model for islands to support decision-making for energy system transformation (IRUSE, 2025). The project incorporates multiple sustainability criteria; economic, environmental, technical, social—in the evaluation, selection, and ranking of possible energy storage projects, taking into account regional needs and end-user perspectives. While not specific to hydrogen storage, the RESTOR framework demonstrates the growing recognition that multi-criteria approaches are essential for navigating the complexity of energy storage decisions in real-world contexts.

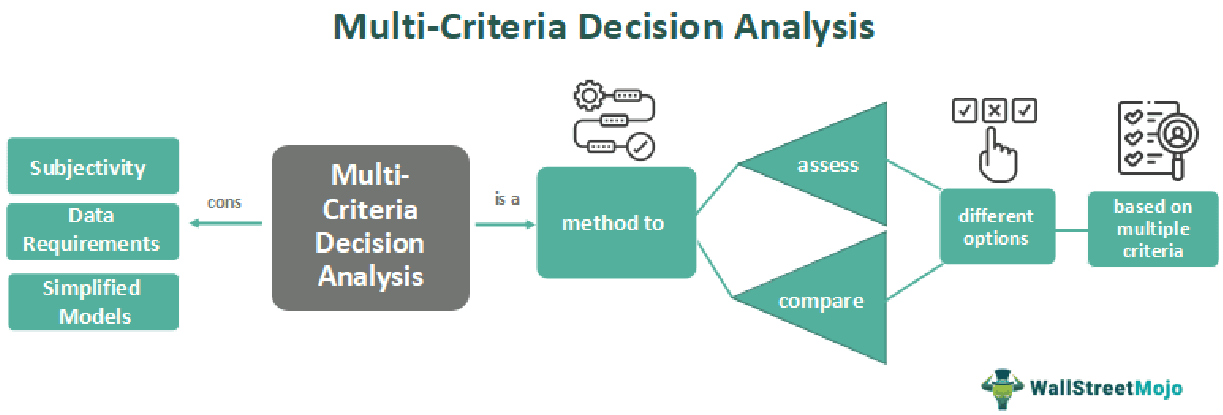


*Figure 7: Sketch of an MCDA (ScienceDirect, 2021)*

### *2.7.3 Analytical Hierarchy Process (AHP)*

The Analytical Hierarchy Process, developed by Saaty (1980), is a structured technique for organizing and analyzing complex decisions based on mathematics and psychology. It has become the most widely used MCDA method in energy infrastructure siting due to its intuitive pairwise comparison structure, built-in consistency checking, and seamless integration with GIS.

Theoretical foundations: AHP decomposes complex decisions into a hierarchical structure of criteria, sub-criteria, and alternatives. At each level, pairwise comparisons establish the relative importance of elements using Saaty's 9-point fundamental scale (1 = equal importance, 3 = moderate importance, 5 = strong importance, 7 = very strong importance, 9 = extreme importance). These comparisons are organized into matrices, from which priority weights are derived through eigenvector calculation. The method's mathematical foundation ensures that subjective judgments are converted into ratio-scale measurements that can be combined across criteria.

Consistency verification: A distinctive strength of AHP is its built-in mechanism for assessing the logical coherence of judgments. The Consistency Ratio (CR) compares the consistency of the decision-maker's judgments against that of randomly generated matrices. Saaty (1980) established that CR values below 0.10 indicate acceptable consistency; values exceeding this threshold suggest that judgments should be revised. This verification provides statistical confidence that derived weights are not random and that pairwise comparisons are logically coherent.

Integration with GIS: The combination of AHP with GIS has become a standard approach for spatial decision problems. Başeğmez (2025) demonstrated this integration for nuclear power plant siting, using AHP to determine criterion weights based on expert opinions, then implementing Weighted Linear Combination in a GIS environment to produce comprehensive suitability maps. The study identified fault lines, main roads, and protected areas as the three most significant factors, demonstrating how AHP can reveal priority ordering among diverse criteria while GIS provides spatial expression of the resulting suitability surface. For renewable energy applications, the integration of fuzzy set theory with AHP has addressed uncertainty and imprecision in subjective judgments. Çebi *et al*. (2025) developed an integrated model combining fuzzy set theory, AHP, GIS, and information axiom methods for solar power plant site selection in Türkiye. Their approach used fuzzy AHP to determine importance degrees of decision criteria, GIS to analyze geographically referenced information, and the information axiom to select the best alternative under qualitative and quantitative criteria. The study's sensitivity analysis further examined the impact of criterion weights on results, demonstrating the robustness of the AHP-GIS framework.

Applications in energy storage siting: For underground hydrogen storage specifically, Tarkowski and Czapowski (2023) employed AHP as the core weighting method within their GIS-based multi-criteria decision analysis framework. Their methodology for assessing rock salt deposit suitability explicitly uses AHP to generate weights for criteria including depth, thickness, environmental constraints, and infrastructure access. The resulting suitability maps integrate these weights through weighted linear combination, enabling identification of optimal locations for hydrogen storage facilities. The approach recognizes that AHP's transparency and replicability are essential for decisions with significant economic and safety implications. Koo *et al*. (2012) applied AHP to electric facility siting, engaging stakeholders including resident representatives, academic experts, local assembly members, government officers, and journalists in the weighting process. Their study found that distance from residential areas (35.06%), cultural assets (16.68%), and landscape conservation (13.11%) emerged as the most important factors, demonstrating how AHP can capture diverse stakeholder perspectives and produce weights that reflect societal values. This participatory dimension of AHP is particularly relevant for energy infrastructure decisions that may face community opposition.

Weighted Linear Combination: Within GIS environments, AHP-derived weights are typically applied through Weighted Linear Combination (WLC), where each standardized criterion layer is multiplied by its weight and the results are summed to produce a composite suitability index. Başeğmez (2025) implemented this approach for nuclear plant siting, producing suitability maps that revealed 49.48% of the study area meeting suitability thresholds, with 19.82% classified as perfectly suitable and 29.66% as very suitable. The WLC method's mathematical simplicity and intuitive interpretation make it the preferred integration technique for GIS-MCDA applications.

The synthesis of this literature establishes that GIS-based MCDA with AHP weighting represents the current state of practice for regional-scale energy infrastructure siting, including underground hydrogen storage. The method's transparency, replicability, and ability to incorporate both quantitative measurements and qualitative judgments align directly with the requirements of this study. Furthermore, the existence of validated applications to salt cavern assessment (Tarkowski & Czapowski, 2023) provides methodological precedent and confidence in the approach's suitability for the Nigerian context.

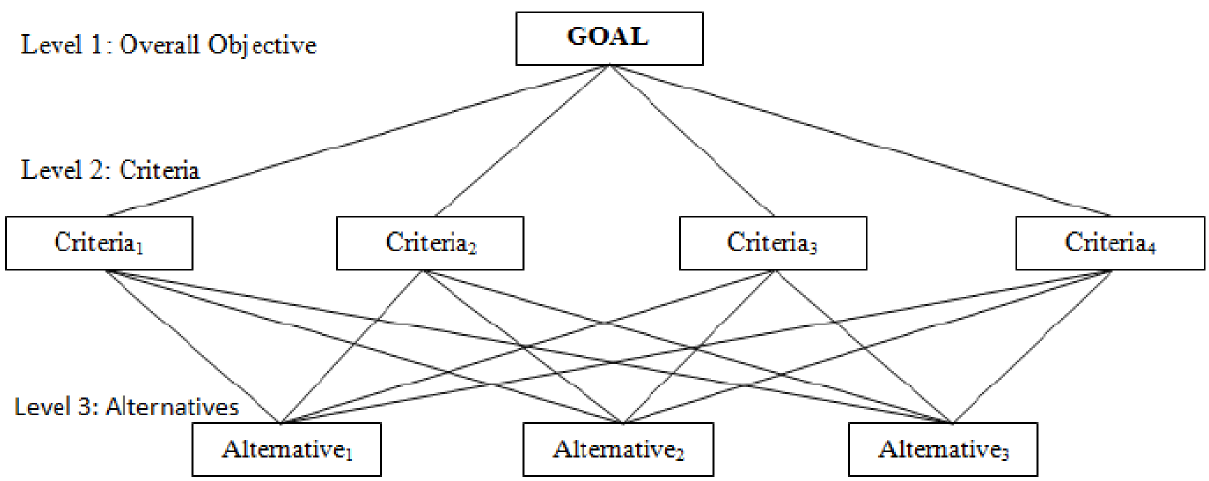


*Figure 8: General hierarchy structure of AHP (Monark et al., 2014)*

## 2.8 International Case Studies

### *2.8.1 Germany: Operational Salt Cavern Experience*

Germany offers the world's most extensive operational experience with underground hydrogen storage, providing critical insights into technical feasibility, safety performance, and infrastructure requirements. The German energy transition (Energiewende) has driven investment in storage technologies to balance increasing shares of variable renewable generation, with hydrogen identified as a key long-duration storage vector. The primary German hydrogen storage assets are salt caverns located in northern Germany. The Kiel cavern has stored town gas containing approximately 60% hydrogen since 1971 without evidence of leakage, microbiological degradation, or geochemical reactions, demonstrating long-term compatibility (Caglayan *et al*., 2020). The Etzel cavern complex, with a total capacity exceeding one billion cubic metres for natural gas, is actively being evaluated for conversion to 100% hydrogen service (Kruck *et al*., 2013). German salt formations typically occur at depths of 500-1500 metres, with individual cavern volumes ranging from 500,000 to one million cubic metres, achieving working gas capacities of approximately 30 million cubic metres per cavern.

The methodological approach to German UHS development combines extensive site characterisation, experimental research, and pilot-scale demonstration. The H2STORE project conducted laboratory experiments on hydrogen-rock interactions under reservoir conditions, while HyCAVmobil established a mobile test facility for cavern integrity assessment (Bauer *et al*., 2017). Site selection employs multi-criteria frameworks considering depth, salt thickness, purity, and proximity to hydrogen transport infrastructure. Key findings demonstrate that salt caverns maintain hydrogen purity above 98% through multiple withdrawal-injection cycles, achieve high cycling rates suitable for grid balancing, and benefit from existing surface infrastructure. Germany's regulatory framework, developed through decades of natural gas storage operation, provides transferable precedents for permitting, safety zoning, and operational monitoring.

### *2.8.2 Netherlands: Pilots and Infrastructure Integration*

The Netherlands has emerged as a European leader in integrated hydrogen storage development, explicitly linking cavern storage to renewable generation, pipeline networks, and industrial demand centres. Dutch hydrogen strategy targets 3-5 GW of electrolysis capacity by 2030, with storage essential for system balancing. The HyStock project at Zuidwending represents the most advanced integrated hydrogen storage initiative. Three salt caverns are under development, with the first operational since 2021 (Gasunie, 2021). Geological characteristics include Zechstein salt formations at depths of 500-1500 metres, providing excellent sealing properties and chemical inertness. Each cavern offers working gas capacity of approximately 30 million cubic metres, with injection/withdrawal rates designed to match offshore wind output profiles.

The methodological approach distinguishes the Dutch experience: storage development is embedded within Gasunie's hydrogen backbone planning, connecting caverns directly to pipeline networks serving industrial clusters (TNO, 2022). Public-private partnership structures share investment risk, with Gasunie collaborating with Statoil and Vattenfall on cavern development and operational testing. Key outcomes confirm technical viability while highlighting integration requirements. Zuidwending operational data demonstrate cavern suitability for hydrogen with minimal purity degradation. Critically, Dutch experience shows that storage cannot be developed in isolation—it requires simultaneous development of transport infrastructure, demand aggregation, and renewable generation. The Netherlands provides the clearest model for integrated hydrogen storage planning applicable to Nigeria's industrial clusters.

### *2.8.3 United Kingdom: Depleted Reservoir Analogues*

The United Kingdom offers essential lessons from depleted reservoir operations, particularly relevant for Nigeria given the Niger Delta's extensive hydrocarbon fields. UK experience addresses the question of converting existing gas storage and production infrastructure to hydrogen service. The Rough gas storage facility, operated in the Southern North Sea since 1985 as the UK's largest storage asset, provides the primary analogue. Rough utilised a depleted gas reservoir at depths of approximately 2700 metres, with a caprock proven over geological timescales and 30 years of operational cycling data (Heinemann *et al*., 2021). Current assessments evaluate conversion potential for hydrogen, investigating well integrity, caprock response to hydrogen exposure, and cushion gas requirements.

The methodological approach combines reservoir simulation, laboratory experiments on core samples, and risk assessment frameworks adapted from natural gas storage and carbon capture experience. The HyNet and Net Zero Teesside projects incorporate hydrogen storage in depleted offshore reservoirs as core components of regional decarbonisation strategies, using dynamic modelling to predict hydrogen flow, mixing, and recovery efficiency (Miocic *et al*., 2023). Key findings highlight both opportunities and challenges. Depleted reservoirs offer enormous capacity-orders of magnitude greater than salt caverns—but face greater uncertainty. Hydrogen interactions with residual methane can affect delivered gas composition. Microbial activity in reservoirs may consume hydrogen or produce hydrogen sulfide. Cushion gas requirements are substantial (50-60% of total volume). However, existing wells, platforms, and pipelines can be repurposed, significantly reducing capital requirements. For Nigeria, the UK experience underscores that depleted reservoir storage is technically feasible but requires detailed site-specific characterisation addressing hydrogen's unique properties.

#### *2.8.4 Synthesis: Lessons Learned and Applicability to Nigeria*

International case studies yield five transferable insights for Nigeria. First, salt caverns offer the highest operational performance and maintain hydrogen purity, justifying investigation of Benue Trough salt formations following German and Dutch exploration methodologies. Second, depleted reservoirs provide immediate opportunities due to existing infrastructure and characterisation data, directly applicable to the Niger Delta using UK assessment frameworks (Heinemann *et al*., 2021). Third, integrated planning linking storage to pipeline networks and demand centres is essential, demonstrated by Dutch experience—storage cannot be developed in isolation. Fourth, public-private partnership models reduce investment risk and accelerate deployment, with Gasunie's collaborations offering transferable structures. Fifth, regulatory frameworks must evolve alongside technical development, with German permitting precedents providing guidance for Nigerian policy development. These lessons directly inform the criteria selection, weighting, and prioritisation framework developed in subsequent chapters, ensuring Nigeria benefits from international experience rather than repeating foundational research.

### 2.9 Research Gaps and Justification

Despite significant international progress in underground hydrogen storage research and demonstration, critical gaps remain that this thesis addresses. These gaps span geographical, methodological, and technical dimensions.

Geographical gap: No systematic assessment of underground hydrogen storage potential exists for Nigeria. While international studies have evaluated storage potential across Europe (Caglayan *et al*., 2020), North America (Lord *et al*., 2014), and selected developing regions, Nigeria's sedimentary basins remain uncharacterized for UHS. The Niger Delta's extensive depleted fields, Benue Trough's salt formations, and inland basins' aquifers have not been evaluated against established suitability criteria. This absence leaves Nigerian energy planning without the foundational knowledge needed to integrate hydrogen storage into emerging policy frameworks (The Electricity Hub, 2026).

Methodological gap: Existing UHS screening studies employ varying criteria with limited transparency or replicability. Wei *et al*. (2026) recently highlighted that previous screening systems lacked evidence-based thresholds derived from experimental data. While their work establishes rigorous criteria for depleted reservoirs, no comparable framework has been developed or applied to Nigeria's diverse geological settings. Furthermore, multi-criteria decision analysis has been applied to UHS site selection internationally (Iranfar *et al*., 2026; Tarkowski & Czapowski, 2023), but these methods have not been adapted to Nigeria's data availability constraints and geological context.

Technical gap: Key uncertainties regarding hydrogen behaviour in subsurface formations remain, particularly for porous media. Questions persist about hydrogen interactions with residual hydrocarbons, microbial consumption rates, and caprock integrity under cyclic loading (Dopffel *et al*., 2021; Zeng *et al*., 2023).

Infrastructure gap: Nigeria's extensive oil and gas infrastructure; pipelines, wells, processing facilities has not been evaluated for hydrogen service suitability. International experience demonstrates that infrastructure repurposing significantly reduces storage costs (Heinemann *et al*., 2021), but this potential remains unquantified for Nigeria.

Policy gap: Nigeria's Energy Transition Plan and National Integrated Electricity Policy recognise hydrogen's potential but lack the site-specific evidence needed to prioritise investments or guide regulatory development (OGTV, 2025). Without identified and ranked storage sites, policy remains aspirational rather than actionable.

Justification: This thesis addresses these gaps through the first systematic, criteria-based identification and prioritisation of underground hydrogen storage sites in Nigeria. By adapting international screening frameworks (Wei *et al*., 2026), applying validated MCDA methodologies (Iranfar *et al*., 2026), and integrating Nigerian geological and infrastructure data, the study provides the foundational evidence base for policy development, investment prioritisation, and future research. The timing aligns with Nigeria's emerging hydrogen policy framework and international momentum toward storage demonstration, ensuring results can inform decisions as the sector develops.

# 3. Study Area and Methodology

## 3.1 Geological Framework of Nigeria

The study was carried out for Nigeria, in West Africa. Software used for spatial analysis is the ESRI's ArcGIS Pro v3.5.

### *3.1.1 Basement Complex*

The Basement Complex of Nigeria comprises Precambrian to early Paleozoic crystalline rocks exposed across approximately 50% of the country's landmass (Obaje, 2009). These rocks include migmatites, gneisses, schists, quartzites, and older granites that have undergone multiple phases of deformation and metamorphism (Nwachukwu, 2015). The Basement Complex is divided into three broad units: the Migmatite-Gneiss Complex (dominant in the west and northwest), the Schist Belts (occurring as linear north-south trending belts), and the Older Granites (intruding both units) (Obaje, 2009). Structurally, the basement is characterized by fault systems that have influenced the development and geometry of overlying sedimentary basins, particularly the Benue Trough (Benkhelil, 1989). For underground hydrogen storage, the Basement Complex is unsuitable due to negligible primary porosity and permeability, and is therefore excluded from further consideration (Miocic *et al.,* 2023).

### *3.1.2 Sedimentary Basins*

Nigeria's sedimentary basins cover approximately 40% of the country's total area and are the focus of this study (Obaje, 2009). These basins formed in response to tectonic events associated with the breakup of Gondwana and the opening of the South Atlantic Ocean during the Mesozoic era (Whiteman, 2010). The basins are classified into inland basins (Benue Trough, Anambra Basin, Chad Basin, Sokoto Basin, Nupe Basin, Dahomey Basin) and the coastal Niger Delta Basin (Nwajide, 2013). Each basin contains sedimentary thicknesses ranging from 2,000 metres in inland basins to over 10,000 metres in the Niger Delta (Reijers, 2011). These basins host Nigeria's proven hydrocarbon reserves and contain the porous reservoir rocks and sealing caprocks necessary for underground storage applications (Heinemann *et al.,* 2021). For this study, four basins were selected for detailed assessment based on data availability, sedimentary thickness, presence of reservoir-seal systems, and existing infrastructure.

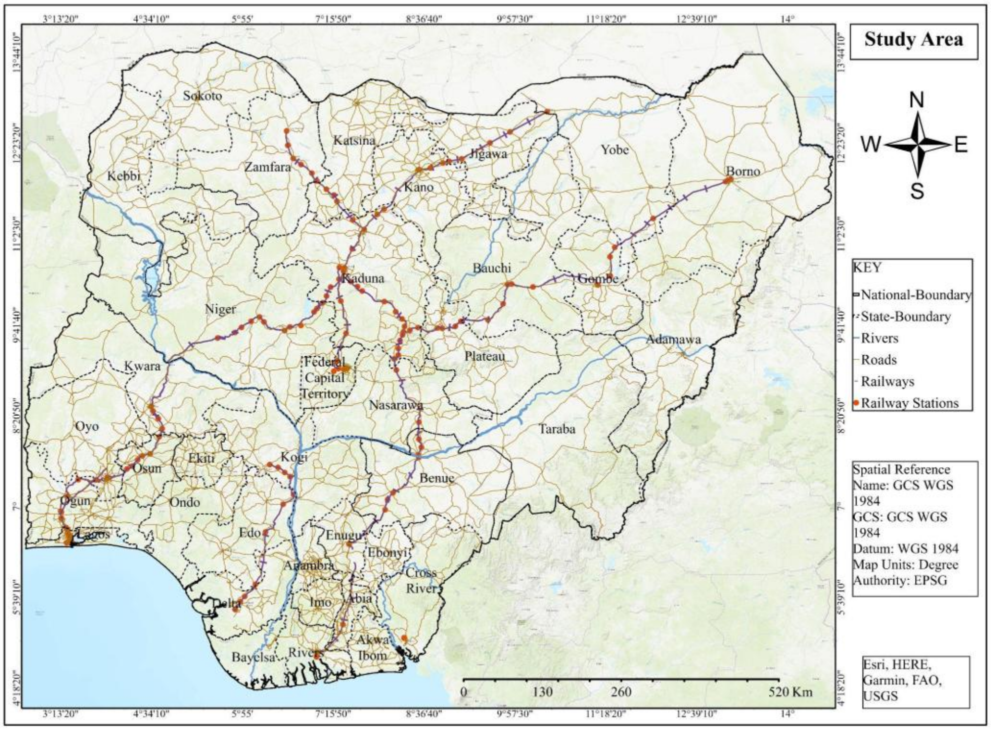


*Figure 9: Map of the study area (source: World Bank, 2025)*

## 3.2 Candidate Basins for Underground Hydrogen Storage

### *3.2.1 Niger Delta Basin*

The Niger Delta Basin is Nigeria's largest and most prolific sedimentary basin, located in the southern coastal region and extending offshore into the Gulf of Guinea (Doust & Omatsola, 1990). The basin covers approximately 75,000 km² onshore and over 100,000 km² offshore, with sedimentary thickness exceeding 10,000 metres in the depocentre (Reijers, 2011). The stratigraphy consists of three diachronous formations: the basal Akata Formation (marine shales, source rock), the overlying Agbada Formation (paralic sandstones and shales, reservoir), and the Benin Formation (continental sands, aquifer) (Short & Stäuble, 1967). The Agbada Formation contains numerous sandstone reservoirs with porosity ranging from 15-30% and permeability from 100–1,000 mD, making them suitable for porous media storage (Ekwe *et al.,* 2022). The basin contains over 500 proven hydrocarbon fields with extensive well control, seismic coverage, and existing pipeline infrastructure (NNPC, 2019). Faulting is dominated by growth faults and rollover anticlines that provide structural traps (Boboye & Adewusi, 2019). For UHS, the Niger Delta Basin offers the highest immediate potential due to data availability, proven containment, and existing infrastructure (Adebayo & Salami, 2023).

### *3.2.2 Anambra Basin*

The Anambra Basin is a Cretaceous sedimentary basin located in southeastern Nigeria, adjacent to the western flank of the lower Benue Trough (Obaje *et al.,* 2004). The basin covers approximately 15,000 km² and contains sedimentary thicknesses of 2,000-4,000 metres (Nwajide, 2013). The stratigraphic succession includes the Nkporo Shale (campanian), Mamu Formation (coal measures), Ajali Sandstone (potential reservoir), and Nsukka Formation (Obaje, 2009). The Ajali Sandstone is a quartz-rich, poorly consolidated sandstone with reported porosity of 20–25% and permeability of 200-500 mD (Adebayo & Salami, 2023). The basin has been explored for hydrocarbons with several gas discoveries, though it remains less developed than the Niger Delta (Ekwe *et al.,* 2022). Infrastructure is limited, with few pipelines and no refineries. For UHS, the Anambra Basin represents a secondary prospect requiring additional characterisation but offering potential in its Ajali Sandstone reservoir.

### *3.2.3 Benue Trough*

The Benue Trough is a major Cretaceous rift basin extending northeast-southwest across central Nigeria for approximately 800 km (Benkhelil, 1989). The trough is divided into the Lower, Middle, and Upper Benue segments, each with distinct stratigraphic and structural characteristics (Obaje, 2009). Sedimentary thickness ranges from 3,000–6,000 metres, comprising marine and fluvial sandstones, shales, limestones, and volcaniclastics (Nwajide, 2013). The trough contains salt deposits (albite-rich syenite and related evaporites) in the Lower Benue segment, which have undergone diapiric movements creating structural traps (Benkhelil, 1989). Sandstone units within the Asu River Group, Eze-Aku Formation, and Awgu Formation exhibit porosity ranging from 10–20% and permeability of 50–200 mD (Adebayo & Salami, 2023). Hydrocarbon shows have been documented, but no commercial production exists. The Benue Trough's salt structures may offer cavern storage potential, while its sandstones offer porous media potential. For UHS, the Benue Trough is a frontier area requiring significant characterisation but offering geological diversity.

### *3.2.4 Chad Basin*

The Chad Basin (also known as the Bornu Basin) is a Cretaceous to Quaternary intracratonic basin located in northeastern Nigeria, forming part of the larger Lake Chad Basin shared with Niger, Chad, and Cameroon (Obaje, 2009). The Nigerian portion covers approximately 40,000 km² with sedimentary thicknesses of 2,000–3,500 metres (Nwajide, 2013). The stratigraphy includes the Bima Sandstone (basal), Gongila Formation (shales and limestones), Fika Shale (marine), and Kerri-Kerri Formation (continental sands) (Obaje *et al.,* 2004). The Bima Sandstone is a potential reservoir with reported porosity of 10–18% and permeability of 50–150 mD (Ekwe *et al.,* 2022). Hydrocarbon shows have been encountered in exploration wells, but no commercial discovery has been made (NNPC, 2019). Infrastructure is minimal, with no pipelines or refineries in the basin. For UHS, the Chad Basin is a distant prospect requiring significant infrastructure development before storage becomes viable, though its deep aquifers may offer long-term capacity potential.

## 3.3 Research Design and Methodological Framework

### *3.3.1 Nature of the Study*

This study is quantitative, applied, and spatially explicit. It is quantitative, relying on numerical data and computational techniques for measurement and analysis (Liu *et al.,* 2020). It is applied, seeking to directly inform decision-making by Nigerian policymakers, energy planners, and industry stakeholders, bridging academic geological knowledge with practical site selection needs (Derakhshani *et al.,* 2024). It is spatially explicit, recognizing that UHS potential varies geographically according to geological conditions, infrastructure, and environmental constraints. This spatial nature necessitates Geographic Information Systems (GIS) as the computational backbone, integrating location and value-based preferences, the central characteristic of this study (Malczewski & Rinner, 2015).

### *3.3.2 Rationale for Using GIS and Decision-Support Methods*

UHS site selection is inherently a spatial multi-criteria problem: evaluating locations across Nigeria's sedimentary basins against multiple, often conflicting criteria with both spatial and non-spatial dimensions. GIS-MCDA provides a coherent framework combining GIS data management, spatial analysis, and visualization with MCDA structured decision support (Malczewski & Rinner, 2015).

First, GIS is essential because UHS suitability is location-dependent. Geological properties (depth, lithology, porosity, permeability, caprock integrity, fault density) and infrastructure factors (pipeline proximity, demand centre distance) vary continuously across space. GIS provides the platform to integrate multiple spatial factors into a unified analytical environment (Tian *et al.,* 2024).

Second, MCDA is necessary because multiple criteria must be balanced. Site selection requires trading off geological suitability, technical feasibility, economic viability, and environmental protection. MCDA provides structured, transparent methods for handling these trade-offs when criteria are measured in different units (Hayat, 2025). The Analytic Hierarchy Process (AHP), developed by Saaty (1980) and widely applied in energy siting (Awasthi *et al.,* 2024), decomposes complex decisions into pairwise comparisons with built-in consistency verification.

Third, GIS-MCDA integration creates analytical synergy. GIS enhances MCDA with buffer operations, overlay analysis, and suitability mapping; MCDA enhances GIS with criteria weighting and sensitivity analysis (Malczewski & Rinner, 2015). This integration has been applied to wind farm siting (Arca & Keskin Citiroglu, 2019), geothermal exploration (Hayat, 2025), solar power planning (Paschoalotto, 2023), and green hydrogen siting (Virtanen, 2024).

Fourth, GIS-MCDA aligns with UHS best practices. Derakhshani *et al*. (2024) employed it for salt cavern storage, and Abdulhaq *et al*. (2024) used it to assess abandoned hydrocarbon fields, confirming its suitability for regional-scale UHS assessment. While this study does not involve extensive stakeholder engagement, the framework is designed to be extensible to group decision contexts (Feick & Hall, 2001).

### *3.3.3 Methodological Workflow*

**Phase 1: Data Collection**

All spatial and attribute data required for UHS suitability assessment were compiled from available global and regional datasets. Administrative boundaries were obtained from the World Bank (2025). Protected areas data were acquired from UNEP-WCMC and IUCN (2026). Land use and land cover data were derived from ESRI Sentinel-2 classification (Karra *et al.,* 2021). Oil and gas infrastructure data, including pipelines, refineries, and petroleum terminals, were obtained from the Oil and Gas Infrastructure Mapping (OGIM) database (O'Brien *et al.,* 2025). Digital elevation data were sourced from the World Food Programme (2012). Road networks were compiled from Meijer *et al*. (2018) and the World Bank (2023). Lithological data were obtained from the Global Lithological Map (GLiM) database (Hartmann & Moosdorf, 2012). Groundwater hydrogeology data were sourced from the British Geological Survey Africa Groundwater Atlas (BGS, 2019-2021). Geological environment data were also obtained from the British Geological Survey (BGS, 2019-2021).

**Phase 2: Data Preprocessing and Harmonization**

All spatial data were reprojected to WGS 84 UTM Zone 32N to ensure uniformity and minimize distortion. The analysis environment was configured with a cell size of 100 metres, with the snap raster set to the digital elevation model and the extent set to Nigeria's administrative boundary. This approach ensured alignment of pixels and avoided misalignment or bias in subsequent multi-criteria analysis. Euclidean distance tools were applied to calculate proximity to pipelines, roads, and energy hubs. Lineaments were extracted from the digital elevation model, and fault density was calculated using the line density tool. Slope was derived from the digital elevation model using the surface slope tool. All datasets were converted to raster format at 100-metre resolution.

**Phase 3: Reclassification and Standardization**

All raster data and processing results were reclassified and assigned suitability values ranging from 1 to 5, where 1 represents the least suitable class and 5 the most suitable. The rescaling to 5 classes and standardization of datasets into uniform rasters, coordinate reference system, cell sizes and scales is to ensure consistency in the analysis and avoid distortion. Weighted overlay forces and compresses results into discrete classes and can affect variability, so weighted sum was adopted in this analysis for a continuous raster and no forced integer scaling. Lithological classes were reclassified according to their inferred relevance to underground hydrogen storage in porous media. Siliciclastic sedimentary rocks were assigned the highest suitability, unconsolidated sediments high suitability, metamorphic and volcanic rocks low to moderate suitability, and acid plutonic rocks very low suitability. Geological environment classes were reclassified with sedimentary basin units assigned the highest scores, unconsolidated sedimentary units moderate to high scores, and volcanic, granitic, and basement terrains low to very low scores. ESRI land cover data were reclassified into five suitability classes based on land-use compatibility, with bare ground and rangeland assigned highest suitability, built-up areas and agricultural lands lower suitability. Water bodies and flooded vegetation were treated as exclusion constraints. Protected areas were processed as a binary mask using the set null analysis tool.

**Phase 4: Criteria Weighting Using AHP**

The Analytical Hierarchy Process (AHP) was adopted to assign weights to nine evaluation criteria: geology, lithology, groundwater hydrogeology, lineaments (fault density), pipelines, energy hubs (refineries and petroleum terminals), roads, slope, and land use/land cover. A pairwise comparison matrix was constructed following Saaty's fundamental scale. The matrix was informed by literature-derived values and professional contributions. The pairwise comparison matrix yielded a consistency ratio of 0.012, which is below the acceptable threshold of 0.10, confirming that the judgments were logically consistent.

**Phase 5: Suitability Analysis**

The weighted sum tool was used to combine the nine reclassified criteria layers according to their AHP-derived weights. Following this initial weighted sum, the protected areas binary mask was applied to produce the final suitability map. Suitability levels were classified into five ranges: unsuitable, very low, low, moderate, high, and very high suitability.

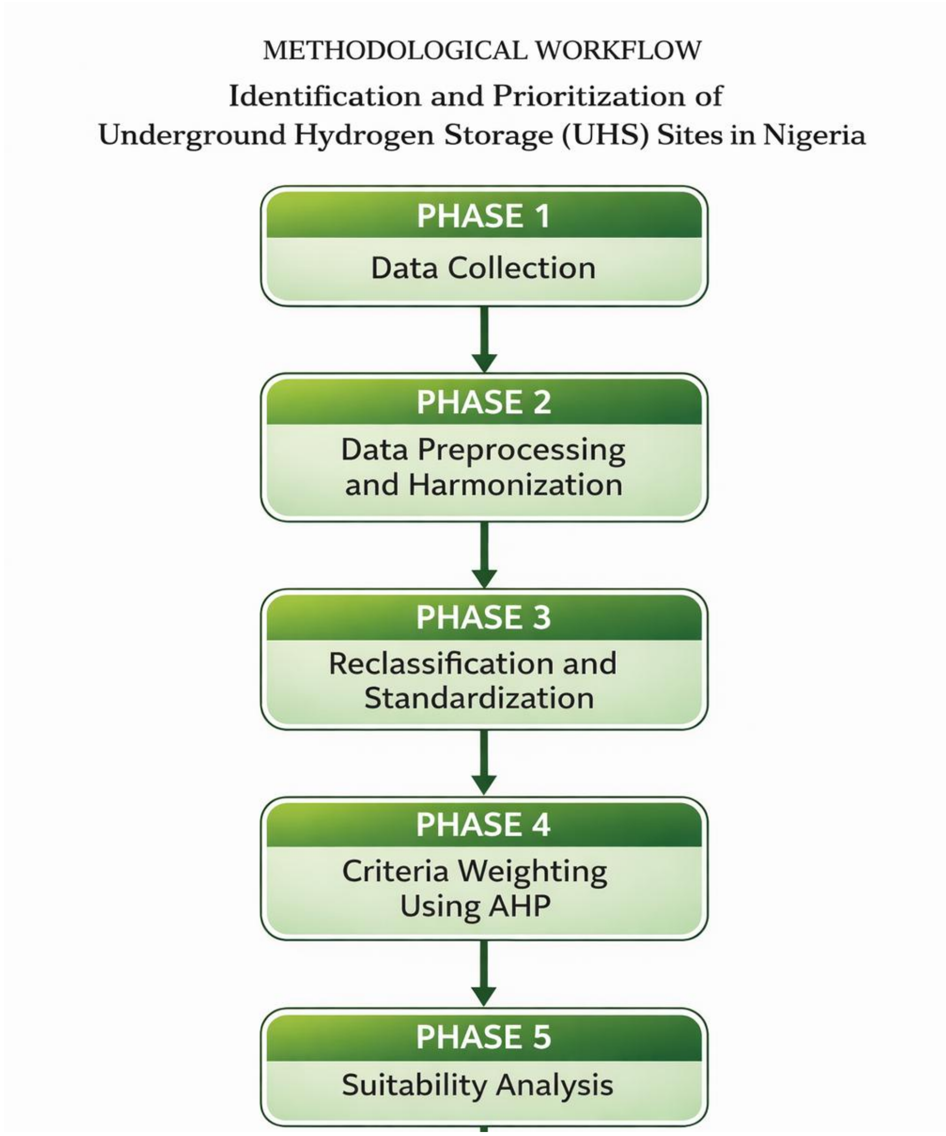


*Figure 10: UHS methodological flowchart (Source; Authour)*

## 3.4 Data Collection and Sources

### Administrative and Environmental Data

Administrative boundaries for Nigeria at state and local government levels were obtained from the World Bank Official Boundaries dataset (World Bank, 2025). Protected area boundaries were sourced from the World Database on Protected Areas (WDPA) maintained by UNEP-WCMC and IUCN (2026), which provides comprehensive coverage of nationally designated protected areas, forest reserves, and wetlands.

**Land Use and Land Cover Data**

Land use and land cover classification was derived from Sentinel-2 satellite imagery using deep learning classification methods (Karra *et al.,* 2021). This dataset provides 10-metre resolution classification of land cover types including water, trees, flooded vegetation, crops, built area, bare ground, snow/ice, clouds, and rangeland.

**Infrastructure Data**

Oil and gas infrastructure data were obtained from the Oil and Gas Infrastructure Mapping (OGIM) database version 2.7 (O'Brien *et al.,* 2025). This dataset includes gas pipelines, oil and gas fields, refineries, and petroleum terminals. Pipeline routes and energy facility locations were extracted for proximity analysis. Road network data were compiled from multiple sources including the Global Road Inventory Project (Meijer *et al.,* 2018) and World Bank Nigeria Roads dataset (World Bank, 2023).

**Elevation and Terrain Data**

Digital elevation data were obtained from the World Food Programme Common Operational Dataset for Nigeria (WFP, 2012). Slope was derived from this elevation model using surface slope tools. Lineaments for fault density analysis were extracted from the same elevation dataset.

**Geological and Hydrogeological Data**

Lithological data were obtained from the Global Lithological Map (GLiM) database (Hartmann & Moosdorf, 2012), which provides a representation of rock properties at the Earth's surface at 0.5-degree spatial resolution. Geological environment and groundwater hydrogeology data were sourced from the British Geological Survey Africa Groundwater Atlas (BGS, 2019-2021), which provides regional-scale mapping of aquifer productivity and geological units.

## 3.5 Data Pre-processing and Harmonization

### *3.5.1 Coordinate Reference System*

All the data were reprojected to WGS 84 UTM Zone 32N for uniformity and to avoid distortion of the dataset. This projection provides optimal coverage for Nigeria with minimal areal distortion and is consistent with established practice for Nigerian energy resource assessments (Olatunji *et al.,* 2024). The selection of UTM Zone 32N ensures that distance calculations (e.g., pipeline proximity, road proximity) and area calculations (e.g., site size) are accurate, which is critical for weighted sum analysis in multi-criteria decision-making (Malczewski & Rinner, 2015).

### *3.5.2 Data Harmonization*

A geodatabase was created and all obtained data were imported into the ArcGIS Pro environment. The analysis environment was configured with a cell size of **100 metres**, with the snap raster set to the digital elevation model and the extent set to Nigeria's administrative boundary. This ensured alignment of pixels and avoided misalignment or bias in subsequent multi-criteria analysis (Derakhshani *et al.,* 2024). Vector data were converted to raster format, and all raster layers were standardized to consistent bit depth (32-bit float for continuous data, 8-bit for classified data). 9 of the 10 datasets are criteria, while protected layers were adopted as an environmental constraint.

**Processing operations applied included:**

i. **Geology, lithology, and groundwater hydrogeology data** were converted directly to raster format.
ii. **Powerplants, refineries, and petroleum terminals** were merged together as a single "energy hubs" layer (Figure 9).
iii. Euclidean distance tool was used to obtain distance buffers for pipelines, roads, and energy hubs (Figures 9, 10 and 11).
iv. **Lineaments** were extracted from the digital elevation model and **fault density** was derived from these lineaments using the line density tool in ArcGIS Pro (Figure 12).
v. **Slope** was extracted from the digital elevation model using the surface slope tool (Figure 13).

These resampling and harmonization procedures ensured spatial consistency across all datasets prior to reclassification and weighted sum analysis. The use of Euclidean distance for proximity criteria follows standard practice in energy infrastructure siting, providing an objective measure of accessibility to existing infrastructure (IEA, 2023). Fault density analysis using line density tools is widely employed for assessing structural integrity, with higher fault densities indicating increased leakage risk for stored hydrogen (Miocic *et al.,* 2023). Slope extraction from digital elevation models enables assessment of terrain suitability for surface facilities and access roads (Goodman *et al.,* 2020). Of the ten datasets compiled, nine were used as weighted criteria, while protected areas were adopted as an exclusionary environmental constraint.

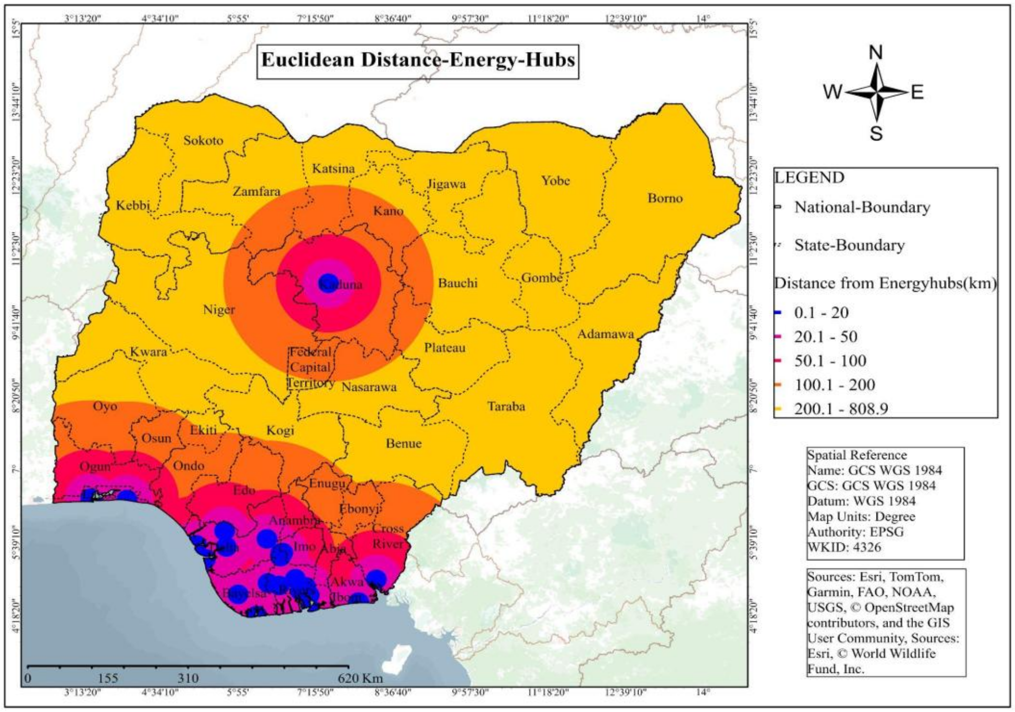


*Figure 11: Energy hubs*

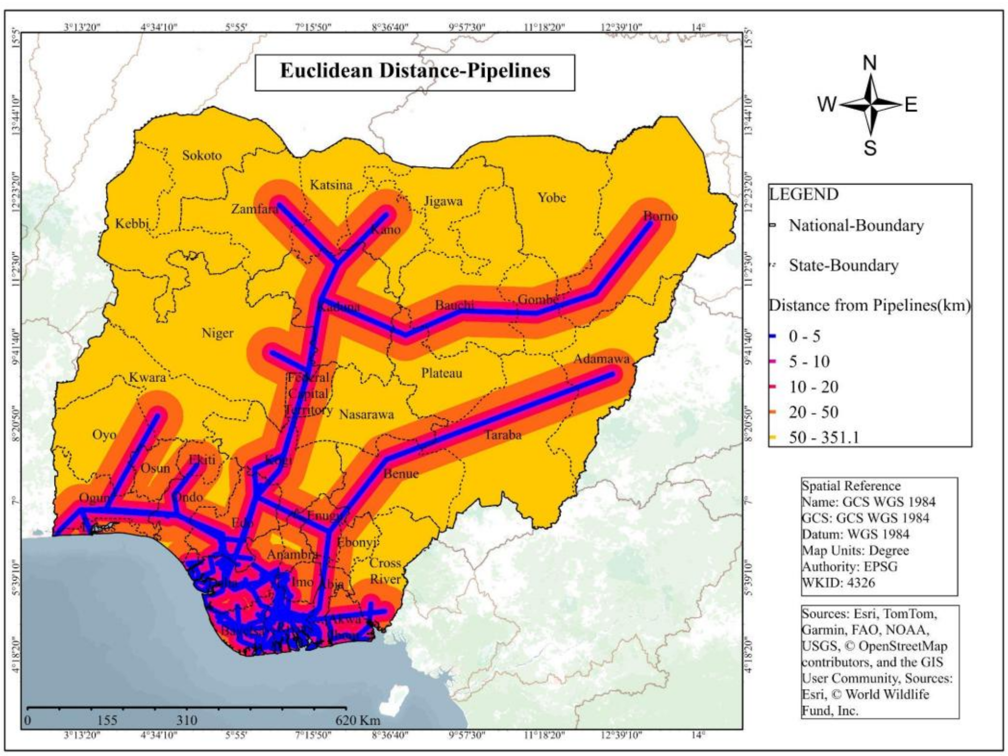


*Figure 12: Pipelines*

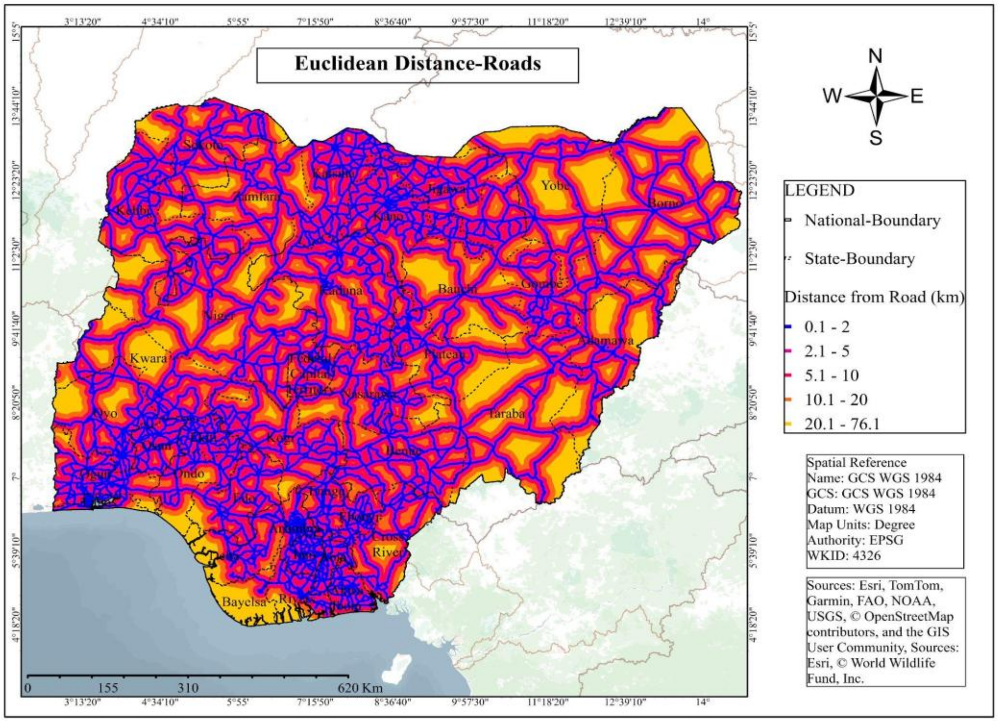


*Figure 13: Roads*

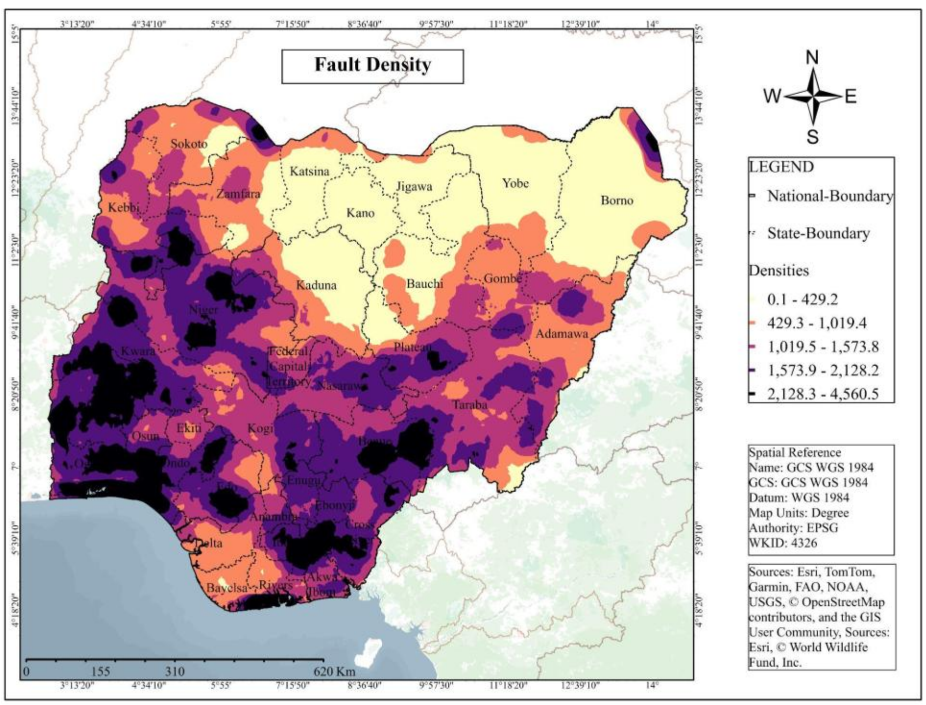


*Figure 14: Fault Density*

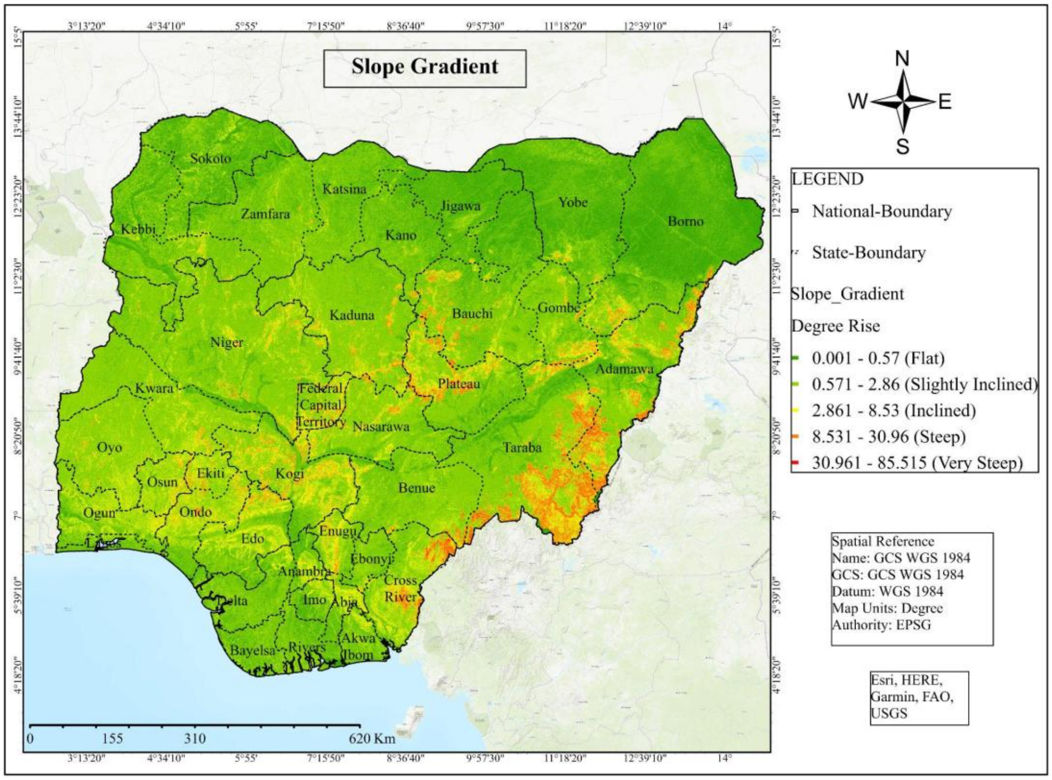


*Figure 15: Slope gradient*

### *3.3.3 Resolution and Scale Considerations*

A unified raster resolution of **100 metres × 100 metres** was selected for all continuous surfaces. This resolution balances analytical detail with computational feasibility while supporting the input data scales (ranging from 10 metres for land cover to 1 kilometre for geological data) (Malczewski & Rinner, 2015). The selection of 100-metre resolution is appropriate for regional-scale assessment of sedimentary basins, where the objective is to identify prospective rather than site-specific engineering locations (Heinemann *et al.,* 2021).

The **minimum mapping unit** for site identification was set at **10 km²** (approximately 1,000 cells at 100-metre resolution), eliminating isolated high-suitability cells too small for practical storage development (Heinemann *et al.,* 2021). This threshold aligns with international UHS screening studies that consider minimum storage volumes for economic viability (Caglayan *et al.,* 2020).

## 3.6 Criteria for Underground Hydrogen Storage Site Identification and Reclassification

Based on data availability and relevance to underground hydrogen storage, ten criteria were selected for this study. These criteria were drawn from international UHS literature and adapted to the Nigerian context. Each criterion was reclassified to a 1-5 suitability scale, with 1 representing the least suitable and 5 the most suitable. Protected areas were treated as an exclusion constraint rather than a weighted criterion.

**Reclassification Approach**

To enable systematic comparison and combination of criteria measured in different units, all criterion layers were reclassified to a common suitability scale. Reclassification involved converting raw data values (e.g., distance in kilometres, slope in degrees, categorical rock types) into standardized scores based on their relevance to underground hydrogen storage. A 1–5 scale was adopted, where 1 represents the least suitable condition and 5 the most suitable condition for UHS development. This approach follows established practice in GIS-based multi-criteria decision analysis for energy infrastructure siting (Malczewski & Rinner, 2015).

*3.6.1 Geological Criteria*

**Lithology**

Lithology determines the fundamental suitability of a formation for hydrogen storage. For porous media storage, quartz-rich sandstones with minimal reactive minerals are preferred, as clay minerals may catalyse hydrogen reactions or cause swelling that reduces permeability (Yekta *et al.,* 2018). Carbonate reservoirs present greater uncertainty due to heterogeneous porosity and potential for reactive dissolution (Edlmann *et al.,* 2021). In this study, lithological classes were reclassified according to their inferred relevance to underground hydrogen storage in porous media based on the Global Lithological Map (GLiM) database (Hartmann & Moosdorf, 2012).

Siliciclastic sedimentary rocks were assigned the highest suitability (score 5) because they are the most likely to include sandstone-dominated reservoir units with favourable porosity and permeability. Unconsolidated sediments were assigned high suitability (score 4) as indicative of sedimentary environments, though with lower confidence for deep storage application. Metamorphic and basic to intermediate volcanic rocks were assigned low to moderate suitability (scores 2–3) due to their generally poor primary porosity and dependence on fracture-controlled flow. Acid plutonic rocks were assigned very low suitability (score 1) because of their crystalline nature and negligible primary porosity. Water bodies were excluded from analysis. See, Figure 14 and Table 1.

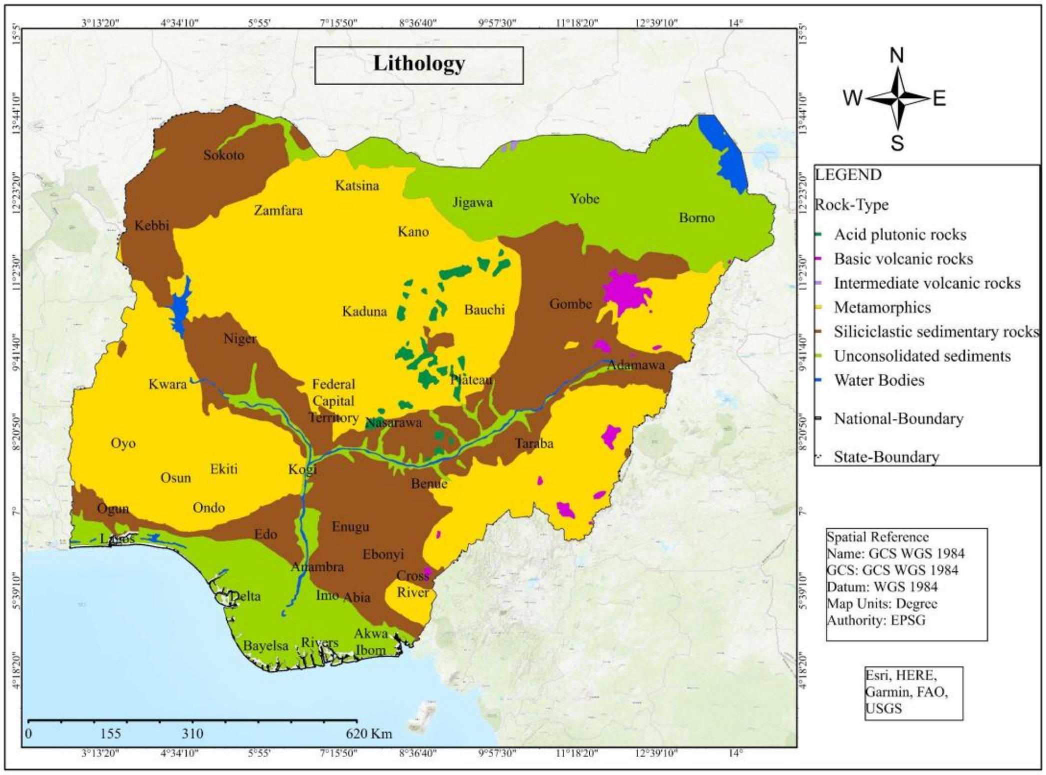


*Figure 16: Lithology (source: GliM; Hartmann & Moosdorf, 2012)*

**Geology**

Geological classes were reclassified according to their regional suitability for underground hydrogen storage. Sedimentary basin units were assigned the highest scores because they are more likely to contain reservoir-seal systems, adequate sediment thickness, and favourable subsurface conditions for porous storage (BGS, 2019–2021). The Niger Delta Basin (Tertiary), Lower Benue Basin (Cretaceous), Upper Benue Basin (Cretaceous and Tertiary), Sokoto Basin (Cretaceous and Tertiary), and Nupe Basin (Cretaceous-Tertiary) were assigned the highest suitability scores (Figure 15 and Table 1).

Unconsolidated sedimentary units were assigned moderate to high scores (3-4) depending on basin context. Volcanic, granitic, and basement terrains were assigned low to very low scores (1-2) due to their generally poor primary porosity and limited storage potential. Surface water was excluded from the analysis.

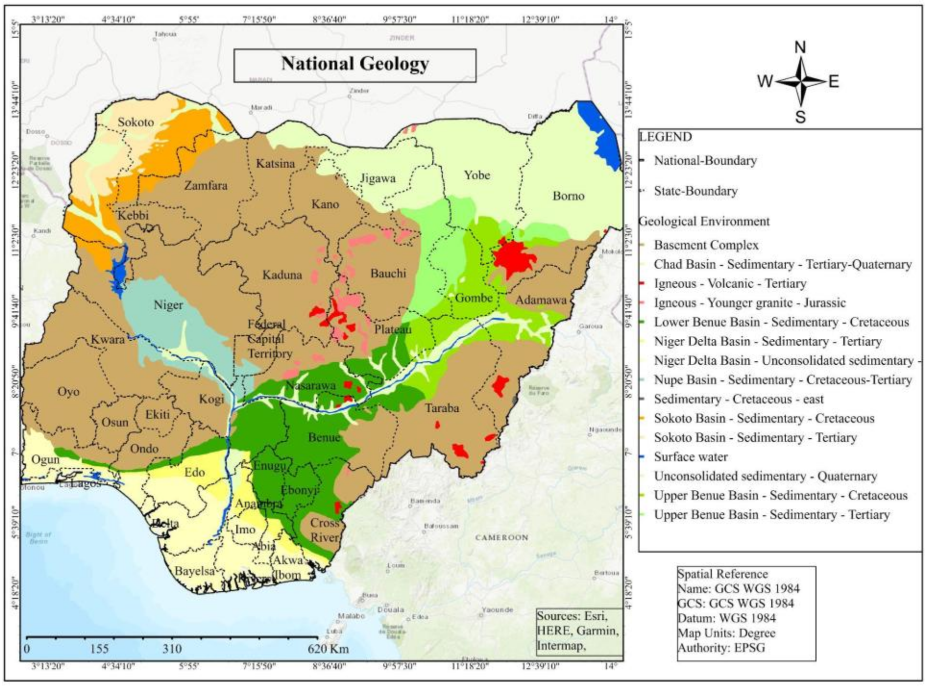


*Figure 17: National Geology (source: BGS, 2019-2021)*

## Groundwater Hydrogeology

Aquifer productivity serves as a proxy for the presence of porous, permeable formations that may be suitable for hydrogen storage, though storage targets are typically deeper than freshwater aquifers. Groundwater hydrogeology data were sourced from the British Geological Survey Africa Groundwater Atlas (BGS, 2019–2021).

Areas with very low aquifer productivity were assigned the highest suitability (score 5) as these are less likely to contain productive freshwater resources that would compete with storage operations. Low productivity areas were assigned high suitability (score 4). Moderate productivity areas received moderate suitability (score 3). High and very high productivity areas were assigned lower suitability scores (2 and 1 respectively) due to potential competition with groundwater resources (Figure 16 and Table 1).

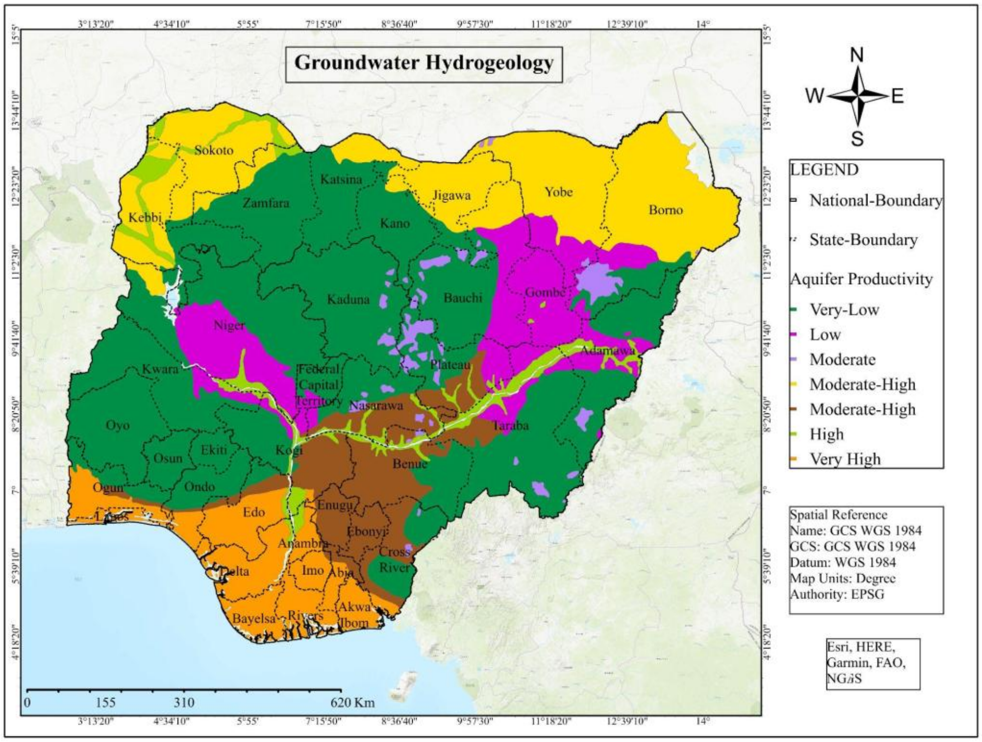


*Figure 18: Groundwater Hydrogeology (source: BGS, 2019-2021)*

### *3.6.2 Structural and Safety Criteria*

**Lineaments (Fault Density)**

Faults and fractures represent potential leakage pathways that compromise storage integrity. High fault density increases the probability of intersecting features that connect the storage formation to overlying aquifers or the surface (Miocic *et al.,* 2023). Fault reactivation potential under cyclic pressure loading during hydrogen injection and withdrawal requires assessment, as pressure cycling can alter effective stress on fault planes (Edlmann *et al.,* 2021). Fault density was calculated from lineaments extracted from the digital elevation model using the line density tool in ArcGIS Pro.

Areas with very low fault density were assigned the highest suitability (score 5). Low fault density areas were assigned high suitability (score 4). Moderate fault density areas received moderate suitability (score 3). High fault density areas were assigned low suitability (score 2). Very high fault density areas were assigned the lowest suitability (score 1). See, Figure 17 and Table 1.


*Figure 19: Lineaments (source: WFP 2012)*

### *3.6.3 Infrastructure and Accessibility Criteria*

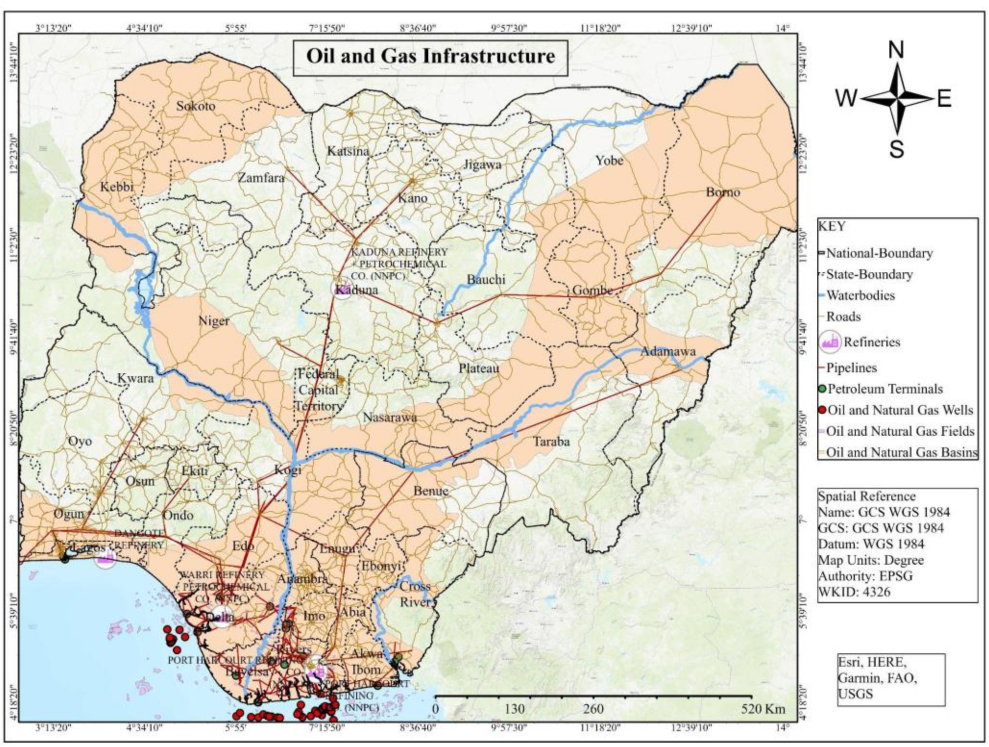


*Figure 20: Oil and gas infrastructure (sources: OGIM; O'Brien et al., 2025, World Bank, 2023)*

**Proximity to Pipelines**

Proximity to existing pipeline infrastructure significantly influences the economic viability of UHS projects, as hydrogen transport via pipeline represents the most cost-effective option for large-scale gas movement over land (IEA, 2023). Existing natural gas pipelines may be repurposed for hydrogen transport with appropriate modifications, though hydrogen embrittlement concerns require assessment of pipeline metallurgy and operating pressures (Melaina *et al.,* 2013). Pipeline data were obtained from the Oil and Gas Infrastructure Mapping (OGIM) database (O'Brien *et al.,* 2025).

Areas within 5 km of existing pipelines were assigned the highest suitability (score 5). Areas 5–10 km from pipelines were assigned high suitability (score 4). Areas 10–20 km from pipelines received moderate suitability (score 3). Areas 20–50 km from pipelines were assigned low suitability (score 2). Areas beyond 50 km received the lowest suitability (score 1) (Table 1 and Figure 18).

**Proximity to Energy Demand Centers**

Storage sites located near hydrogen demand centres minimise transportation costs and energy losses associated with gas movement (Caglayan *et al.,* 2020). Major demand centres include industrial clusters requiring hydrogen as feedstock (fertiliser production, methanol synthesis, steel manufacturing) and power generation facilities capable of hydrogen co-firing or conversion (IEA, 2023). Proximity also enables faster response times for load-following storage services (Heinemann *et al.,* 2021). Energy hub data, including refineries and petroleum terminals, were obtained from the OGIM database (O'Brien *et al.,* 2025).

Areas within 20 km of energy hubs were assigned the highest suitability (score 5). Areas 20–50 km from energy hubs were assigned high suitability (score 4). Areas 50–100 km from energy hubs received moderate suitability (score 3). Areas 100–200 km from energy hubs were assigned low suitability (score 2). Areas beyond 200 km received the lowest suitability (score 1) (Figure 18 and Table 1).

**Proximity to Roads**

Proximity to road networks enables movement of drilling rigs, completion equipment, and maintenance personnel during construction and operations (Goodman *et al.,* 2020). Road data were obtained from the Global Road Inventory Project (Meijer *et al.,* 2018) and the World Bank Nigeria Roads dataset (World Bank, 2023).

Areas within 2 km of roads were assigned the highest suitability (score 5). Areas 2-5 km from roads were assigned high suitability (score 4). Areas 5–10 km from roads received moderate suitability (score 3). Areas 10–20 km from roads were assigned low suitability (score 2). Areas beyond 20 km received the lowest suitability (score 1) (Table 2 and Figure 18).

**Slope**

Terrain suitability, represented by slope, affects facility siting and earthworks costs. Flat to gently sloping terrain facilitates facility siting, while steep slopes present construction challenges and may indicate unstable ground conditions (Miocic *et al.,* 2023). Slope was derived from the digital elevation model (WFP, 2012).

Flat terrain (0.001-0.57 degrees) was assigned the highest suitability (score 5). Slightly inclined terrain (0.57-2.86 degrees) received high suitability (score 4). Inclined terrain (2.86-8.53 degrees) received moderate suitability (score 3). Steep terrain (8.53-30.96 degrees) received low suitability (score 2). Very steep terrain (30.96-85.52 degrees) received the lowest suitability (score 1) *(Table 2 and Figure 19).*

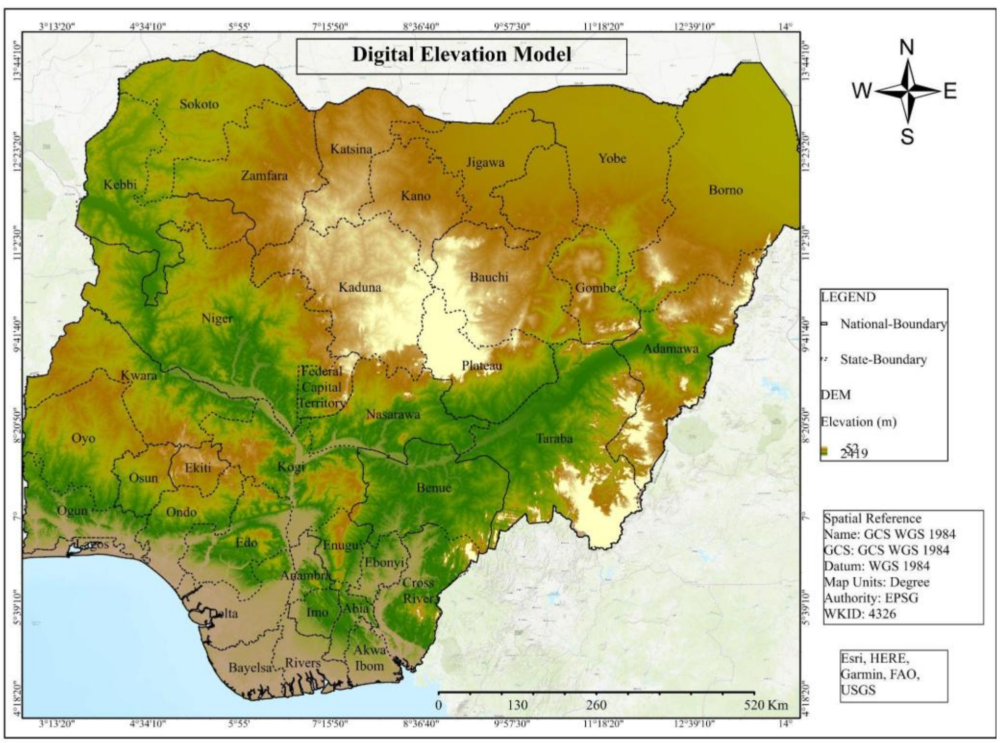


*Figure 21: Digital elevation model (source: WFP, 2012)*

*3.6.4 Environmental*

**Land Use and Land Cover Compatibility**

Land use compatibility affects land acquisition costs, community opposition, and permitting requirements (Edlmann *et al.,* 2021). Land use and land cover data were derived from ESRI Sentinel-2 classification (Karra *et al.,* 2021).

Bare ground was assigned the highest suitability (score 5) due to minimal land-use conflict. Rangeland was assigned high suitability (score 4) as it allows surface facilities with limited agricultural disruption. Trees were assigned moderate suitability (score 3). Crops were assigned low suitability (score 2) due to agricultural land use. Built areas were assigned the lowest suitability (score 1) due to dense population, infrastructure, and permitting challenges *(Table 1 and Figure 20).*

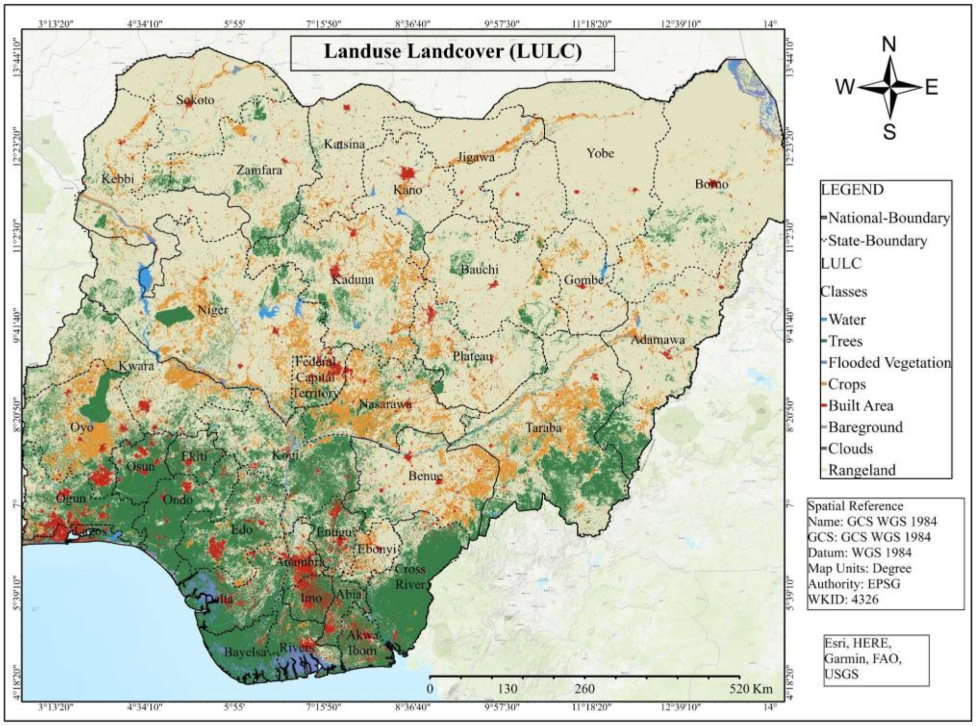


*Figure 22: Land Use Land Cover (LULC) (source: ESRI Sentinel-2 classification; Karra et al., 2021)*

**Protected and Conservation Areas**

Protected areas were treated as hard exclusion constraints rather than weighted criteria. Areas designated as national parks, game reserves, forest reserves, wetlands, and other protected areas were excluded from suitability analysis. A Binary mask was applied to remove all protected zones from the final suitability map, ensuring that no development is permitted within environmentally sensitive areas. The protected areas layer was converted to a binary mask, with protected areas assigned a value of 0 (excluded) and non-protected areas assigned a value of 1 (retained for analysis). This binary mask was multiplied with the weighted sum of the nine criteria to produce the final suitability map, ensuring that protected areas were entirely removed from consideration. Protected area data were sourced from the World Database on Protected Areas (UNEP-WCMC & IUCN, 2026). Water bodies and flooded vegetation, as identified in the land use/land cover classification, were also excluded from analysis due to physical constraints on infrastructure development. *See, Figure 21 and Table 1.*

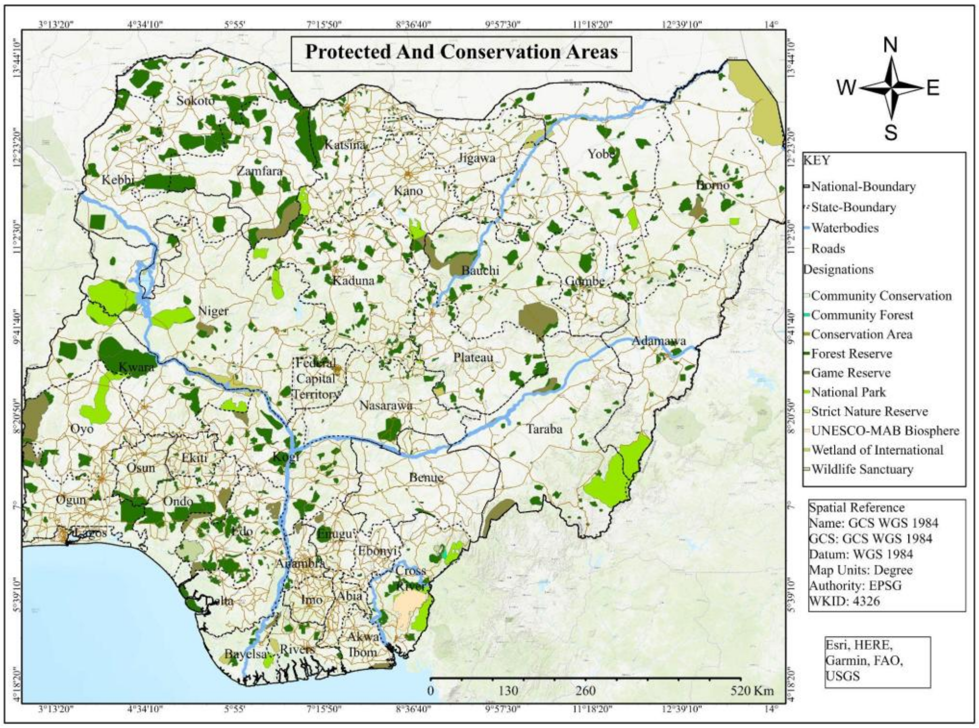


*Figure 23: Protected and Conservation Areas (source: UNEP-WCMC and IUCN, 2026*

*Table 1: Reclassification Table.*

| Criteria Category | Data | Required Layer | Classifications | Importance/Weight | Suitability Score |
|---|---|---|---|---|---|
| **Geological and Subsurface Criteria** | Geology | Geological Environment | Niger Delta Basin - Sedimentary - Tertiary | Very High/ 16 | 5 |
| | | | Lower Benue Basin - Sedimentary – Cretaceous | | 5 |
| | | | Upper Benue Basin - Sedimentary – Cretaceous | | 5 |
| | | | Upper Benue Basin - Sedimentary - Tertiary | | 5 |
| | | | Sokoto Basin - Sedimentary - Cretaceous | | 5 |
| | | | Sokoto Basin - Sedimentary - Tertiary | | 5 |
| | | | Nupe Basin - Sedimentary - Cretaceous-Tertiary | | 4 |
| | | | Sedimentary - Cretaceous - east | | 4 |
| | | | Chad Basin - Sedimentary - Tertiary-Quaternary | | 4 |
| | | | Niger Delta Basin - Unconsolidated sedimentary - Tertiary-Quaternary | | 4 |
| | | | Unconsolidated sedimentary - Quaternary | | 3 |
| | | | Igneous - Volcanic – Tertiary | | 2 |

| | | | | | |
|---|---|---|---|---|---|
| | | | Igneous - Younger granite – Jurassic | | 1 |
| | | | Basement Complex | | 1 |
| | Lithology | Rock Types | Siliciclastic sedimentary rocks | Very High/ 12 | 5 |
| | | | Unconsolidated sediments | | 4 |
| | | | Metamorphics | | 3 |
| | | | Basic volcanic rocks | | 2 |
| | | | Intermediate volcanic rocks | | 2 |
| | | | Acid plutonic rocks | | 1 |
| | Groundwater Hydrogeology | Aquifer Productivity | Very Low Productivity | Very High/ 15 | 5 |
| | | | Low Productivity | | 4 |
| | | | Moderate Productivity | | 3 |
| | | | High Productivity | | 2 |
| | | | Very High Productivity | | 1 |
| **Structural & Safety** | Lineaments | Fault Density | 0.1 - 429.2 (Very low) | Very High/ 15 | 5 |

| | | | | | |
|---|---|---|---|---|---|
| **Criteria** | | | 429.3 - 1,019.4 (Low) | | 4 |
| | | | 1,019.5 - 1,573.8 (Moderate) | | 3 |
| | | | 1,573.9 - 2,128.2 (High) | | 2 |
| | | | 2,128.3 - 4,560.5 (Very high) | | 1 |
| **Infrastructure & Accessibility Criteria** | Pipelines | Pipeline network | <5 km | High/ 11 | 5 |
| | | | 5 – 10 km | | 4 |
| | | | 10 – 20 km | | 3 |
| | | | 20 – 50 km | | 2 |
| | | | >50 km | | 1 |
| | Energy Hubs | Refinery & Petroleum Terminal Locations Point dataset | <20 km | High/ 9 | 5 |
| | | | 20 – 50 km | | 4 |
| | | | 50 – 100 km | | 3 |
| | | | 100 – 200 km | | 2 |
| | | | >200 km | | 1 |

| | | | | | |
|---|---|---|---|---|---|
| | Road Networks | Road shapefile | <2 km | Medium/8 | 5 |
| | | | 2 – 5 km | | 4 |
| | | | 5 – 10 km | | 3 |
| | | | 10 – 20 km | | 2 |
| | | | >20 km | | 1 |
| | Dem | Slope | 0.001 - 0.57 (Flat) | Low/ 5 | 5 |
| | | | 0.571 - 2.86 (Slightly Inclined) | | 4 |
| | | | 2.861 - 8.53 (Inclined) | | 3 |
| | | | 8.531 - 30.96 (Steep) | | 2 |
| | | | 30.961 - 85.515 (Very Steep) | | 1 |
| **Environmental & Landuse Criteria** | Protected Areas | WDPA / National Parks/Game Reserves, Forest Reserve, Wetlands | Protected Areas | Very High (Constraint) | 0 |
| | | | Non-Protected Areas | | 1 |
| | ESRI Land Cover | LULC raster | Bareground | High/ 9 | 5 |
| | | | Rangeland | | 4 |

| | | | | | |
|---|---|---|---|---|---|
| | | | Trees | | 3 |
| | | | Crops | | 2 |
| | | | Built Area | | 1 |

### 3.7 Criteria Weighting Using Analytical Heierachy Process (AHP)

AHP was adopted to assign weights to nine evaluation criteria: geology/geological environment, groundwater hydrogeology, lithology, lineaments, pipelines, refineries/petroleum terminals, roads, slope, and LULC based on level of influence, to the criteria. The process involved the use of a pairwise comparison matrix. Protected areas were excluded from the weighting process because they were treated as hard constraints. The pairwise comparison matrix yielded a consistency ratio of 0.012, which is below the acceptable threshold of 0.10, indicating that the judgments were logically consistent.

*Table 2:Saaty's Fundamental Scale for Pairwise Comparisons*

| **Intensity of importance** | **Definition** | **Explanation** |
|---|---|---|
| 1 | Equal importance | Two elements contribute equally to the objective |
| 3 | Moderate importance | Experience and judgment slightly favor one element over another |
| 5 | Strong importance | Experience and judgment strongly favors one element over another |
| 7 | Very strong importance | One element is favored very strongly over another. Its dominance is demonstrated in practice. |
| 9 | Extreme importance | The evidence favoring one element over another is of the highest possible order of affirmation. |
| Intensities of 2, 4, 6 & 8 can be used to express intermediate values. Intensities of 1.1, 1.2, 1.3 e.t.c can be used for elements that are very close in importance. | | |

*Citation;* (Mat *et al.,* 1987) and (Mu & Making, 2017)

Using the above scale, weights of influence were assigned to each criterion against others, and were further normalized to determine the priority weights. A sensitivity analysis called consistency ratio (C.R) was carried out to determine the level of inconsistency in the weighting matrix. The consistency Ratio was calculated and a value of 0.05 was obtained using the formula;

$$CR = \frac{CI}{RI}$$

$$CI = \frac{(\lambda max - n)}{(n-1)}$$

$$\lambda max = average\ of\ \sum_{i=1}^{n} \frac{weighted\ sum\ vector}{priority\ weight})$$

Where CI is consistency index, RI is random matrix index (1.45), and n is number of criterion.

According to Saaty, when the value of consistency ratio is less than 0.10, then the judgement matrix is reasonably consistent and accurate for the AHP (Saaty, 2012).

**WEIGHTING MATRIX/CONSISTENCY RATIO: EXCEL DOCUMENT**

The pairwise comparison matrix was carried out using knowledge and information from global, regional, and national literature to determine their influences. Below is the datasheet for the Pairwise comparison matrix and weighting datasheet can be found in Appendix II.

### 3.8 Software and Analytical Tools

#### *3.8.1 GIS Software Used*

All spatial analysis and mapping operations were performed using ESRI's ArcGIS Pro version 3.5 (PDF Page 9). The software's Spatial Analyst extension was used for weighted sum operations, Euclidean distance calculations, line density analysis, and region grouping algorithms. ArcGIS Pro is an industry standard for GIS-MCDA applications in energy siting studies (Olatunji *et al.,* 2024).

#### *3.8.2 Supporting Analytical Tools*

Microsoft Excel was used for all non-spatial calculations, including pairwise comparison matrix construction, eigenvector calculations for AHP weights, and consistency ratio determination (Derakhshani *et al.,* 2024). The AHP weighting matrix and consistency ratio calculations were documented in a Google Sheets workbook, with the link provided in the appendix. Excel's built-in matrix functions provide transparent, auditable calculations essential for methodological rigour (Awasthi *et al.,* 2024).

*3.8.3 Suitability Analysis*

To obtain the suitability models, the weighted sum tool was used to add the reclassified models of the 9 criteria considering their weights. The weighted sum tool was adopted and prioritized because after this weighted sum, the constraint layer was added.

The product of this initial nine criteria suitability analysis was multiplied by the binary mask result of the protected areas layer to produce the final suitability map.

**Suitability Maps**

The result of the weighted sum of 9 criteria maps is the initial suitability model or map, while the result after applying environmental constraints is the final suitability map showing the suitability levels of different areas in the study area. The suitability levels are classified into 5 ranges from unsuitable areas to very high suitability areas.

## 3.9 Methodological Limitations

*3.9.1 Data Availability and Resolution Limits*

This study relied exclusively on secondary data from global datasets, government agencies, and published sources. No primary field data collection or laboratory analysis was undertaken. Data coverage is uneven across Nigeria's sedimentary basins, with better availability in the Niger Delta than inland basins. The 100-metre raster resolution, appropriate for regional assessment, cannot resolve site-specific features required for engineering design. Results therefore indicate regional prospects, not proven storage locations (Heinemann *et al.,* 2021). Several criteria identified in the UHS literature as important including porosity, permeability, caprock integrity, depth, pressure, and temperature were excluded from this study due to lack of publicly available, spatially continuous data covering Nigeria's sedimentary basins. Their absence is a limitation of regional-scale screening using secondary data and does not diminish their technical relevance for site-specific evaluation (Heinemann *et al.*, 2021; Miocic *et al.*, 2023).

*3.9.2 Uncertainty in Subsurface Interpretation*

Subsurface characterization inherently involves interpretive uncertainty. The global datasets used represent regional-scale generalizations that may mask local variations. Fault density derived from lineament analysis depends on DEM resolution and extraction parameters. The data quality classification (Tier 1-3) acknowledged these uncertainties but could not eliminate them entirely.

*3.9.3 Scope of Applicability of Results*

This assessment provides a first-order regional screening intended to guide future investigation, not to support final investment decisions (Tarkowski & Uliasz-Misiak, 2022). Results identify where detailed site characterization should occur, not where storage should definitively be developed. The study did not consider dynamic hydrogen behaviour, geochemical reactions, or detailed economic analysis. Findings are valid within current technical understanding; evolving knowledge may alter suitability assessments.

## 4.0 Results and Discussion

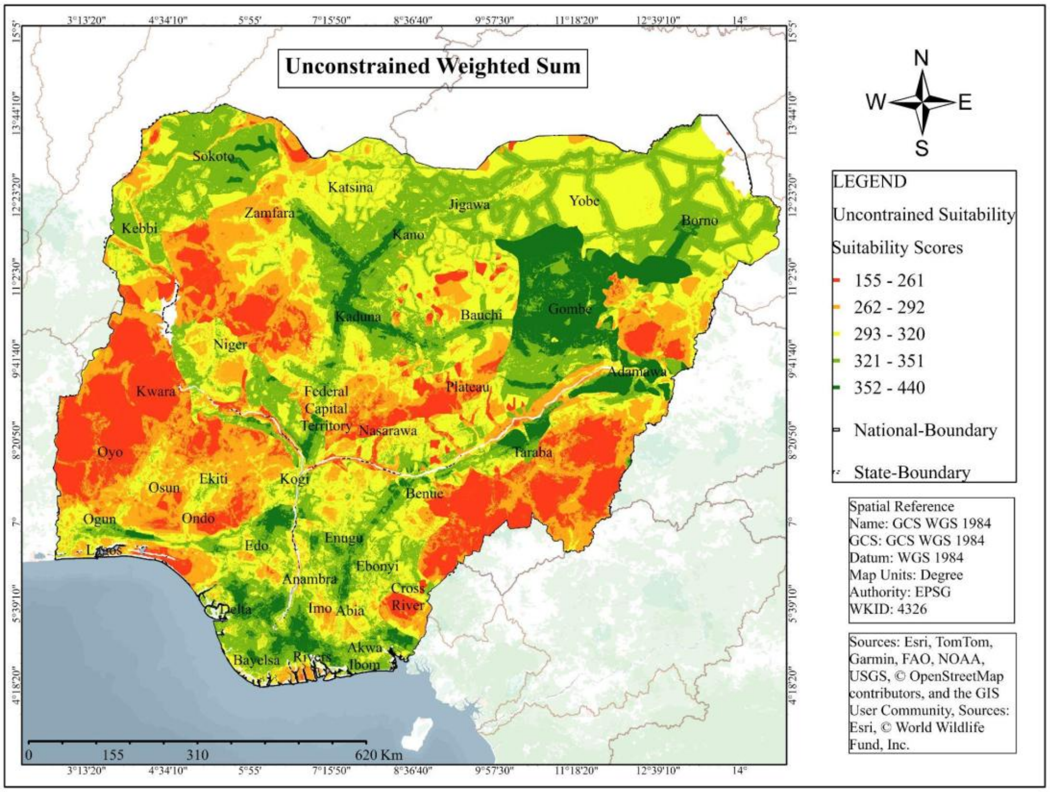


*Figure 24: Unconstrained weighted sum*

### 4.1 Unconstrained Weighted Sum

Figure 22 presents the unconstrained weighted sum of nine criteria before the application of the protected areas exclusion mask. The weighted sum was calculated using the AHP-derived weights (Section 3.5) applied to reclassified suitability scores ranging from 1 (least suitable) to 5 (most suitable). This output shows the theoretical maximum suitability distribution across Nigeria's sedimentary basins without constraints, environmental or legal.

The Niger Delta basin shows the highest concentration of scores (4 and 5), with contiguous suitable areas including Rivers, Bayelsa, Delta, and Akwa Ibom States. This pattern shows the basin's favourable combination of sedimentary geology (Section 3.2.1), dense pipeline infrastructure, multiple energy hubs (refineries and terminals), and flat terrain (low slope values). The Anambra Basin (Enugu State) and sections of the Benue Trough (Gombe, Adamawa, Taraba States) also show clusters of high scores, though with greater fragmentation. The Chad Basin (Borno State) displays isolated high-score areas, consistent with its less developed infrastructure network and lower density of energy facilities.

These findings align with the screening methodology of Wei *et al*. (2025), who developed a comprehensive scoring system for UHS in depleted oil and gas reservoirs using a 1 to 5 scale identical to the one employed in this study. Their framework established criteria covering storage efficiency, risk minimization, and economic impacts, with disqualifying thresholds defined based on experimental and simulation data. The dominance of the Niger Delta in Figure 20 corresponds to Wei *et al*.'s (2025) observation that basins with proven reservoir-seal systems and existing infrastructure receive the highest suitability scores in unconstrained assessments.

The application of AHP for criteria weighting in this study follows the approach of Ridolfi *et al*. (2024), who utilized the Analytical Hierarchy Process with the Delphi technique to weight 27 screening parameters for Italian hydrocarbon reservoirs. Their analysis, tested on a confidential dataset of 48 production sites, highlighted fault description, reservoir architecture, and well number as the highest-weighted parameters. The consistency ratio achieved in their study (below 0.10) aligns with the CR = 0.012 obtained in Section 3.5, confirming the reliability of both weighting procedures.

The concentration of high scores in Nigeria's southern basins is consistent with the findings of Agbaje *et al*. (2025), who conducted a comparative analysis of geological storage solutions for Nigeria, identifying substantial hydrogen storage opportunities across the Niger Delta, Benue Trough, and Sokoto Basin. Their study emphasised the potential advantages of repurposing the country's extensive oil and gas infrastructure for energy storage purposes, an advantage shown in the high infrastructure proximity scores assigned to the Niger Delta in this study.

Lower scores (1-2) are observed in basement complex terrains (Section 3.1.1) and areas with sparse infrastructure. This pattern is consistent with the AHP weighting (Section 3.5), where geological environment (sedimentary basins scored highest) and infrastructure proximity (pipelines, energy hubs, roads) received the highest weights.

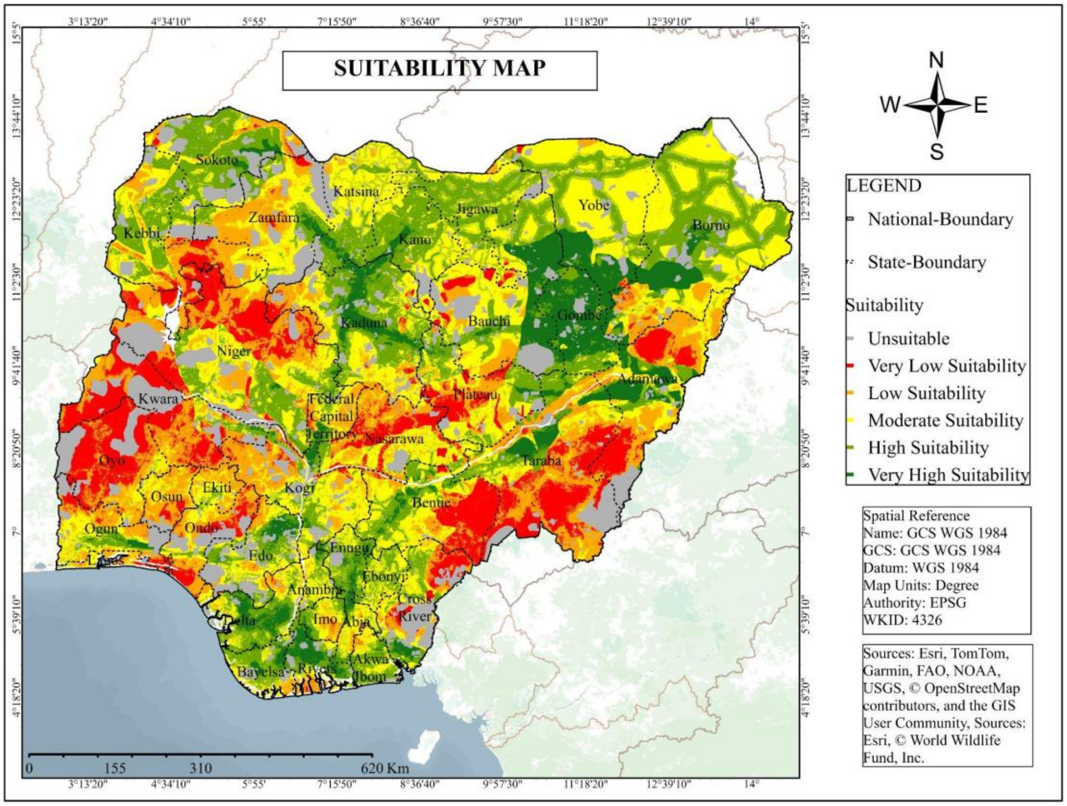


*Figure 25: Suitability map*

## 4.2 Constrained Suitability Map

Figure 23 presents the final suitability map after applying the protected areas binary mask to the unconstrained weighted sum shown in Figure 22. Protected areas including national parks, game reserves, forest reserves, and wetlands were assigned a value of 0 (excluded), while non-protected areas retained their weighted sum scores (1 to 5). This transformation removes environmentally sensitive and legally restricted lands from further consideration, producing a realistic suitability surface for UHS development.

Comparison between Figure 22 and Figure 23 reveals the spatial impact of environmental constraints. The Niger Delta basin, while still containing the largest contiguous areas of High and Very High suitability, shows reduction in certain zones where protected mangroves and forest reserves occur. The inland basins (Benue Trough, Anambra, and Chad) experience less exclusion overall, as protected area coverage is lower in these regions compared to the coastal zone. This pattern indicates that environmental constraints unequally affect the southern basins, though sufficient suitable area remains for further prioritisation.

The exclusion of protected areas as hard constraints follows established practice in energy infrastructure siting. Tarkowski and Czapowski (2023) applied identical binary masking to national parks and nature reserves in their GIS-based suitability mapping for Polish salt caverns, noting that such exclusions are legally mandatory and non-negotiable for commercial development. Their study confirmed that excluding protected areas does not eliminate all suitable sites but rather focuses attention on remaining viable zones.

The moderate to high scores (3-5) retained after masking are concentrated in three primary regions. First, the Niger Delta and its adjoining basins (Anambra, Lower Benue) form a continuous belt of suitability from Delta State through Rivers, Akwa Ibom, and into Enugu State. Second, the Upper Benue Trough (Gombe, Adamawa, Taraba States) shows isolated patches of High suitability. Third, the Chad Basin around Maiduguri shows pockets of Moderate to High suitability, though with lower scores than the southern basins.

The reduction in suitable area due to protected areas is quantified in Table 3, which presents the total area per suitability class across Nigeria. The proportion of Very High suitability (8.0% of total area) is sufficiently large to support multiple storage development scenarios, while the High suitability category (23.1%) provides additional capacity for expansion. The total combined area of High and Very High suitability in total is 277,192.41 km², or 31.1% of Nigeria.

The constrained suitability map *(Figure 23)* forms the basis for candidate site identification. The High and Very High zones shown in this figure are extracted and presented at higher resolution in *Figure 23*, with their areas disaggregated by State and Local Government Area in *Table 4*.

*Table 3: Suitability Areas in Nigeria*

| SUITABILITY AREAS IN NIGERIA | | | | |
|---|---|---|---|---|
| S/N | Class Value | Suitability Class | Area ($Km^2$) | Percentage Area (%) |
| 1 | 0 | Unsuitable | 112984.53 | 12.7 |
| 2 | 1 | Very Low suitability | 104762.44 | 11.7 |
| 3 | 2 | Low suitability | 163511.68 | 18.3 |
| 4 | 3 | Moderate suitability | 233320.83 | 26.2 |
| 5 | 4 | High suitability | 206307.17 | 23.1 |
| 6 | 5 | Very High suitability | 70885.24 | 8.0 |
| Total | | | 891771.89 | 100 |

**4.3 Suitability Areas in Nigeria**

Table 3 quantifies the total area and percentage distribution of each suitability class across Nigeria following the application of the protected areas mask. The classification ranges from 0 (Unsuitable, excluded areas) to 5 (Very High suitability). The total land area assessed is 891,771.89 km², representing Nigeria's terrestrial extent after excluding water bodies and areas with persistent cloud cover from the land cover classification.

The Unsuitable class (Class 0) covers 112,984.53 km² (12.7% of total area). This category includes all protected areas national parks, game reserves, forest reserves, and wetlands that were removed as hard constraints. The proportion of land under protection in Nigeria (approximately 12-13%) is comparable to other West African nations and reflects the country's commitment to biodiversity conservation under international frameworks.

The Very Low suitability class (Class 1) covers 104,762.44 km² (11.7%). These areas are primarily located in the Basement Complex terrains (Section 3.1.1) and regions with extremely sparse infrastructure. Low suitability (Class 2) covers 163,511.68 km² (18.3%), representing areas with multiple limiting factors such as poor lithology, high fault density, or distance from infrastructure.

Moderate suitability (Class 3) occupies the largest proportion at 233,320.83 km² (26.2%). This class represents areas that are technically acceptable for UHS but require one or more mitigating measures for example, adequate geology but distant from pipelines, or good infrastructure access but less favourable lithology. Ridolfi *et al*. (2024) observed a similar distribution pattern in their screening of Italian hydrocarbon reservoirs, where the majority of sites fell into moderate suitability categories, with only a minority achieving high or very high scores.

High suitability (Class 4) covers 206,307.17 km² (23.1%). Very High suitability (Class 5) covers 70,885.24 km² (8.0%). The combined High and Very High area totals 277,192.41 km², representing 31.1% of Nigeria's total land area. This proportion is substantial when compared to similar regional screening studies. Tarkowski and Czapowski (2023) reported that High and Very High suitability areas for salt cavern storage in Poland constituted approximately 15-20% of the studied sedimentary basin area, suggesting that Nigeria's potential, as assessed in this study, is relatively favourable.

The dominance of Moderate suitability (26.2%) followed by High suitability (23.1%) indicates that while Nigeria contains extensive areas favourable for UHS, the truly optimal sites (Very High, 8.0%) are limited. This distribution pattern is consistent with the findings of Wei *et al*. (2025), who observed that in regional screening of depleted reservoirs, the highest suitability class typically comprises less than 10% of assessed area, with the majority falling into moderate categories requiring further site-specific investigation.

The High and Very High suitability classes (Classes 4 and 5) are extracted for detailed spatial presentation in Figure 22 and area disaggregation by State and Local Government Area in *Table 4*.

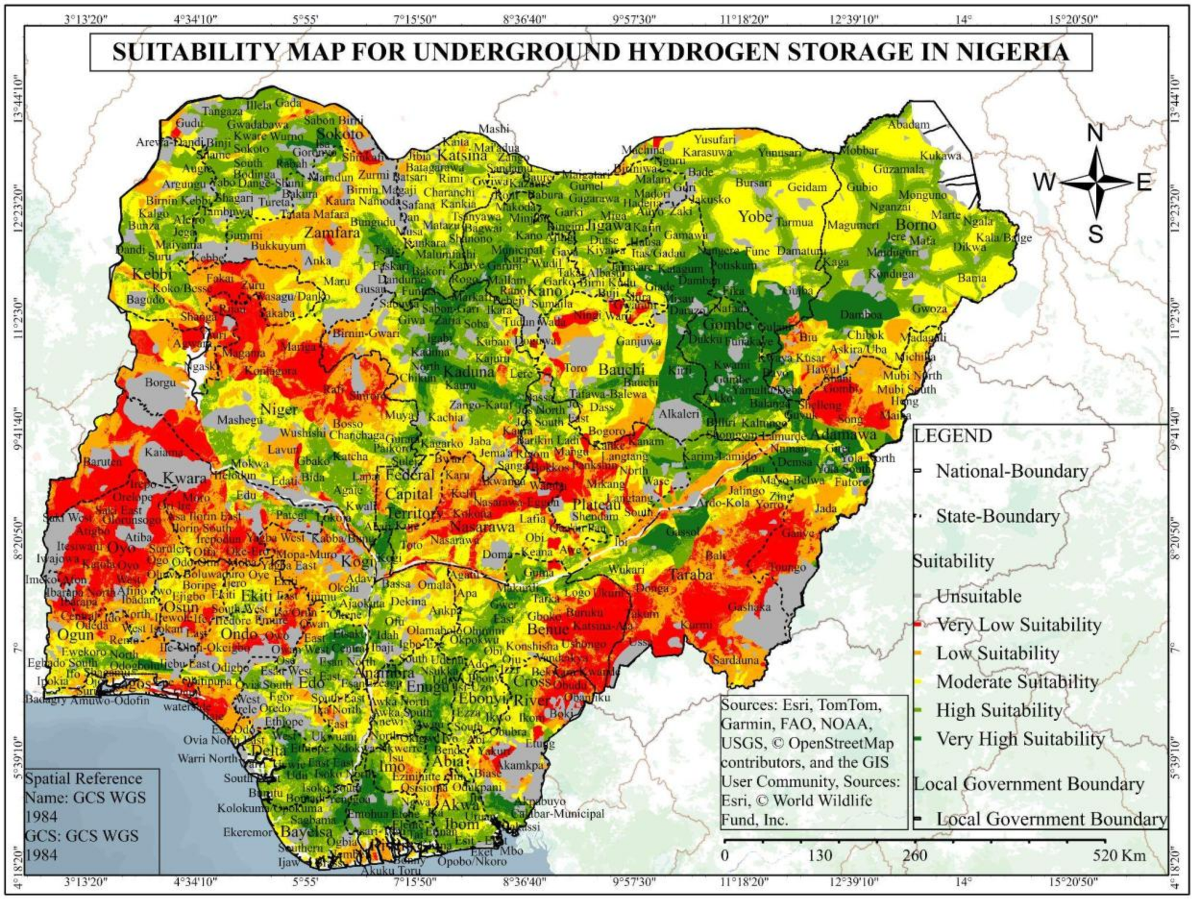


*Figure 26: Suitability maps for UHS in Nigeria*

### 4.4 Suitability Maps for UHS in Nigeria

Figure 24 presents the extracted High (Class 4) and Very High (Class 5) suitability zones from the constrained suitability map (Figure 4.2), shown at higher resolution with State and Local Government Area boundaries overlaid. This visualization isolates only the areas suitable for further consideration, removing Unsuitable and Low to Moderate classes for clarity. The map serves as the primary tool for identifying specific regions where UHS development should be prioritised.

The spatial distribution of High and Very High zones reveals three distinct clusters. The largest and most contiguous cluster occupies the Niger Delta basin and extends northward into the Anambra Basin and Lower Benue Trough. This cluster spans Rivers, Bayelsa, Delta, Akwa Ibom, and Enugu States, with High suitability zones forming a nearly continuous belt from the Atlantic coast inland to approximately latitude 7°N. The continuity of this cluster shows the region's consistently favourable geology (sedimentary basins with thick reservoir sequences), dense infrastructure (pipelines, refineries, roads), and suitable terrain (low slope values).

The second cluster is located in the Upper Benue Trough, spanning Gombe, Adamawa, and Taraba States. This cluster is more fragmented than the southern cluster, with High suitability zones appearing as discrete patches rather than a continuous belt. The fragmentation reflects the region's less developed infrastructure network and more variable geological conditions. However, the presence of High suitability zones confirms that the Benue Trough contains formations suitable for UHS, consistent with the findings of Agbaje *et al*. (2025), who identified substantial hydrogen storage opportunities across the Benue Trough.

The third cluster is located in the Chad Basin (Bornu State), centred around Maiduguri and extending to Konduga and Damboa Local Government Areas. This cluster is the smallest and least continuous of the three, with only isolated patches of High suitability and no Very High zones. The lower scores are a result of the basin's distance from major pipeline infrastructure and energy hubs, though the geological environment remains favourable. Tarkowski and Czapowski (2023) noted that frontier basins with favourable geology but limited infrastructure should be classified as secondary prospects, appropriate for longer-term development once infrastructure expands.

A fourth, smaller cluster appears in Kaduna State (North-West region), with High suitability zones in Kaduna North, Kaduna South, Zaria, Kauru, Kudan, Sabon Gari, and Markafi Local Government Areas. This cluster is unexpected given the region's Basement Complex geology (Section 3.1.1). The High suitability scores here are driven primarily by infrastructure proximity (roads, pipelines, and proximity to Kaduna Refinery) rather than geological factors. This pattern shows the influence of infrastructure weighting in the AHP and suggests that even areas without optimal geology may become suitable if infrastructure access is sufficiently favourable. Wei *et al*. (2025) noted that infrastructure proximity can compensate for moderate geological suitability in regional screening, though site-specific confirmation would be required.

The Local Government Areas identified in Figure 23 and listed in Table 4 represent the smallest administrative units for which suitability data are available. This resolution enables direct communication of results to local planning authorities and supports targeted field investigations. The area of High and Very High suitability within each State and LGA is quantified in *Table 4*.

*Table 4: Area of Very High and High Suitability Sites*

| AREAS OF VERY HIGH AND HIGH SUITABILITY SITES | | |
|---|---|---|
| **S/N** | **STATES** | **LOCAL GOVERNMENT AREAS** |
| 1 | Borno | Damboa, Maiduguri, Konduga |
| 2 | Gombe | Dukku, Kwami, Gombe, Akko, Balanga |
| 3 | Adamawa | Demsa, Yola South |
| 4 | Taraba | Gassol, Lau |
| 5 | Enugu | Nkanu, Udi, Awgu, Enugu South, Igbo Etiti |
| 6 | Rivers | Ahouda West, Ahouda East, Ikwerre, Emohua, Gokana, Tai, Oyigbo, Ibia Akpor |
| 7 | Bayelsa | Yenogoa |
| 8 | Delta | Isoko North, Isoko South, Ughelli North, Ughelli South, Udu, Okpe, Ndokwa west, Uvwie, Ethiope West, Ethiope East |
| 9 | AkwaIbom | Ibesikpo, Ikot Abasi, Onna, Ukanafun |
| 10 | Edo | Etsako West, Etsako East, Etsako Central, Esan North, Esan East, Esan West, Esan Central |
| 11 | Kaduna | Kaduna North, Kaduna South, Zaria, Kauru, Kudan, Sabon Gari, Markafi |

## 4.5 Area of Very High and High Suitability Sites

Table 4 presents the States and Local Government Areas (LGAs) that contain High (Class 4) and Very High (Class 5) suitability zones, as extracted from Figure 23. A total of eleven States across four geopolitical regions are represented, with the number of LGAs per State ranging from one (Bayelsa) to ten (Delta).

The South-South region is the most heavily represented, with five States: Rivers (eight LGAs), Bayelsa (one LGA), Delta (ten LGAs), Akwa Ibom (four LGAs), and Edo (seven LGAs). This concentration reflects the Niger Delta basin's favourable combination of sedimentary geology, dense infrastructure, and suitable terrain. The ten LGAs identified in Delta State including Isoko North and South, Ughelli North and South, Udu, Okpe, Ndokwa West, Uvwie, and Ethiope West and East form a contiguous block of High and Very High suitability extending from the coast inland. This area corresponds to the core of Nigeria's hydrocarbon industry, with existing pipelines, refineries, and energy hubs already in place. Ridolfi *et al*. (2024) observed a similar pattern in their screening of Italian hydrocarbon reservoirs, where the highest suitability sites were concentrated in regions with existing production infrastructure, confirming that infrastructure availability is a primary determinant of UHS suitability.

The South-East region is represented by Enugu State (five LGAs: Nkanu, Udi, Awgu, Enugu South, and Igbo Etiti). These LGAs lie within the Anambra Basin, where the Ajali Sandstone formation provides potential reservoir quality. The presence of High suitability zones in Enugu, despite the absence of major pipeline infrastructure, indicates that geological factors can compensate for moderate infrastructure access. Agbaje *et al*. (2025) noted that the Anambra Basin's Cretaceous sandstones offer storage potential that warrants further investigation, a finding supported by the classification of Enugu LGAs as High suitability in this study.

The North-East region is represented by four States: Borno (three LGAs: Damboa, Maiduguri, Konduga), Gombe (five LGAs: Dukku, Kwami, Gombe, Akko, Balanga), Adamawa (two LGAs: Demsa, Yola South), and Taraba (two LGAs: Gassol, Lau). These LGAs lie within the Chad Basin (Borno) and Upper Benue Trough (Gombe, Adamawa, Taraba). The presence of High suitability zones in this region confirms the findings of Agbaje *et al*. (2025), who identified storage opportunities across the Benue Trough and Chad Basin. However, the fragmentation of suitable zones (multiple LGAs but smaller contiguous areas) suggests that these basins are secondary prospects compared to the Niger Delta.

The North-West region is represented by Kaduna State (seven LGAs: Kaduna North, Kaduna South, Zaria, Kauru, Kudan, Sabon Gari, Markafi). This cluster is notable because it lies outside the main sedimentary basin areas, instead occupying the Basement Complex (Section 3.1.1). The High suitability classification here is driven primarily by infrastructure proximity Kaduna Refinery, road networks, and pipeline connections rather than geological factors. This pattern illustrates that the AHP weighting (Section 3.5) assigned sufficient importance to infrastructure criteria to elevate areas with favourable infrastructure access even where geology is suboptimal. Wei *et al*. (2025) noted that regional screening should not prematurely exclude areas where infrastructure could compensate for moderate geological suitability, and the Kaduna cluster exemplifies this principle.

The LGAs identified in Table 4 represent the primary targets for subsequent site-specific investigation. The Niger Delta LGAs (Rivers, Delta, Bayelsa, Akwa Ibom, Edo) should receive highest priority due to their contiguous High and Very High suitability zones, existing infrastructure, and proven reservoir-seal systems. The Anambra Basin LGAs (Enugu) represent secondary priority, requiring additional geological characterisation but offering substantial potential. The Upper Benue Trough and Chad Basin LGAs (Gombe, Adamawa, Taraba, Borno) represent longer-term prospects, suitable for development after infrastructure expands or if regional hydrogen demand materialises. The Kaduna LGAs represent a special case where infrastructure access compensates for geology, warranting further investigation to confirm subsurface conditions.

## 5.0 Conclusion and Recommendations

### 5.1 Summary of Findings

This study aimed to identify, evaluate, and prioritise suitable geological formations for underground hydrogen storage in Nigeria using a GIS-based Multi-Criteria Decision Analysis framework integrated with the Analytical Hierarchy Process. The research was motivated by the absence of systematic UHS site assessment in Nigeria despite growing national interest in hydrogen as an energy carrier and the existence of extensive sedimentary basins with proven reservoir-seal systems.

Nine criteria were identified and defined across geological (geology, lithology, hydrogeology), structural (fault density), infrastructure (pipelines, energy hubs, roads, slope), and environmental (land use/land cover) categories. The AHP weighting procedure yielded a consistency ratio of 0.012, well below the 0.10 threshold, confirming the reliability of the derived weights.

The constrained suitability map (Figure 22) showed that after excluding protected areas (12.7% of Nigeria's land area), High and Very High suitability zones occupy 206,307.17 km² (23.1%) and 70,885.24 km² (8.0%) respectively. The combined area of 277,192.41 km² (31.1% of Nigeria) exceeds the land area of the United Kingdom (243,610 km²), indicating substantial storage potential.

The prioritised zones (Table 4) identified eleven States across four geopolitical regions. The South-South region (Rivers, Bayelsa, Delta, Akwa Ibom, Edo) contains the largest contiguous High and Very High suitability areas, concentrated in the Niger Delta basin. The South-East (Enugu) represents secondary priority in the Anambra Basin. The North-East (Borno, Gombe, Adamawa, Taraba) and North-West (Kaduna) represent longer-term prospects requiring additional characterisation or infrastructure development.

### 5.2 Contribution to Knowledge

1. This study provides the first criteria-based identification and prioritisation of underground hydrogen storage sites in Nigeria, filling a critical gap in the literature and providing the foundational evidence base for hydrogen storage planning.
2. The study demonstrates that global datasets can be successfully integrated to conduct regional-scale UHS assessment in data-limited settings, providing a replicable template for similar assessments in other West African countries.
3. Table 4 identifies specific administrative units where UHS development should be prioritised, enabling targeted field investigations and informed investment decisions.
4. The study quantitatively confirms that the Niger Delta basin offers the largest contiguous High and Very High suitability zones, supporting prioritisation of UHS investigations in this basin.

5. The Anambra Basin (Enugu State) and Upper Benue Trough (Gombe, Adamawa, Taraba States) are identified as secondary prospects requiring additional characterisation, providing baseline data for future exploration.
6. The consistency ratio of 0.012 confirms that literature-informed pairwise comparisons can produce reliable weights without formal expert consultation, offering a practical approach for data-limited regions.

### 5.3 Recommendations

1. A national hydrogen storage framework should be established, addressing permitting procedures, safety regulations, and monitoring requirements, with the suitability zones identified in this study serving as the spatial evidence base for policy development.
2. The Niger Delta should be prioritised for pioneer storage projects, specifically in Rivers, Delta, Bayelsa, Akwa Ibom, and Edo States, where existing hydrocarbon infrastructure can reduce capital requirements and accelerate deployment timelines.
3. Underground hydrogen storage should be integrated into Nigeria's Energy Transition Plan, with storage targets and timelines established alongside hydrogen production goals, informed by the 277,192.41 km$^2$ of High and Very High suitability identified in this study.
4. Site-specific characterisation should be conducted in priority Local Government Areas (Table 4.), including seismic surveys, well testing, and core analysis, to confirm suitability at engineering scale before investment decisions are made.
5. Excluded criteria including porosity, permeability, caprock integrity, depth, pressure, and temperature should be added as spatially continuous datasets become available for Nigerian sedimentary basins.

### 5.4 Limitations of the Study

This study relied exclusively on secondary data from global datasets. No primary field data collection or laboratory analysis was undertaken. Data coverage is uneven across Nigeria's sedimentary basins, with better availability in the Niger Delta than inland basins. The 100-metre raster resolution, appropriate for regional assessment, cannot resolve site-specific features required for engineering design. Several criteria identified in the UHS literature as important including porosity, permeability, caprock integrity, depth, pressure, and temperature were excluded due to lack of publicly available, spatially continuous data. Their absence is a limitation of regional-scale screening using secondary data and does not diminish their technical relevance for site-specific evaluation (Heinemann *et al*., 2021; Miocic *et al*., 2023).

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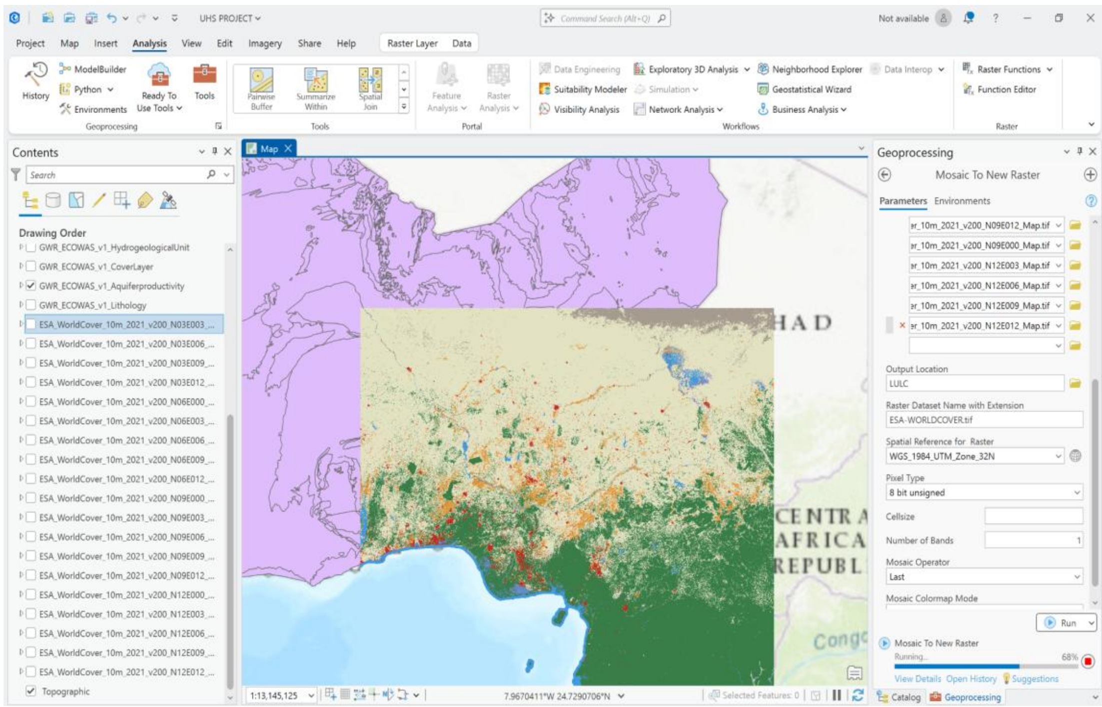
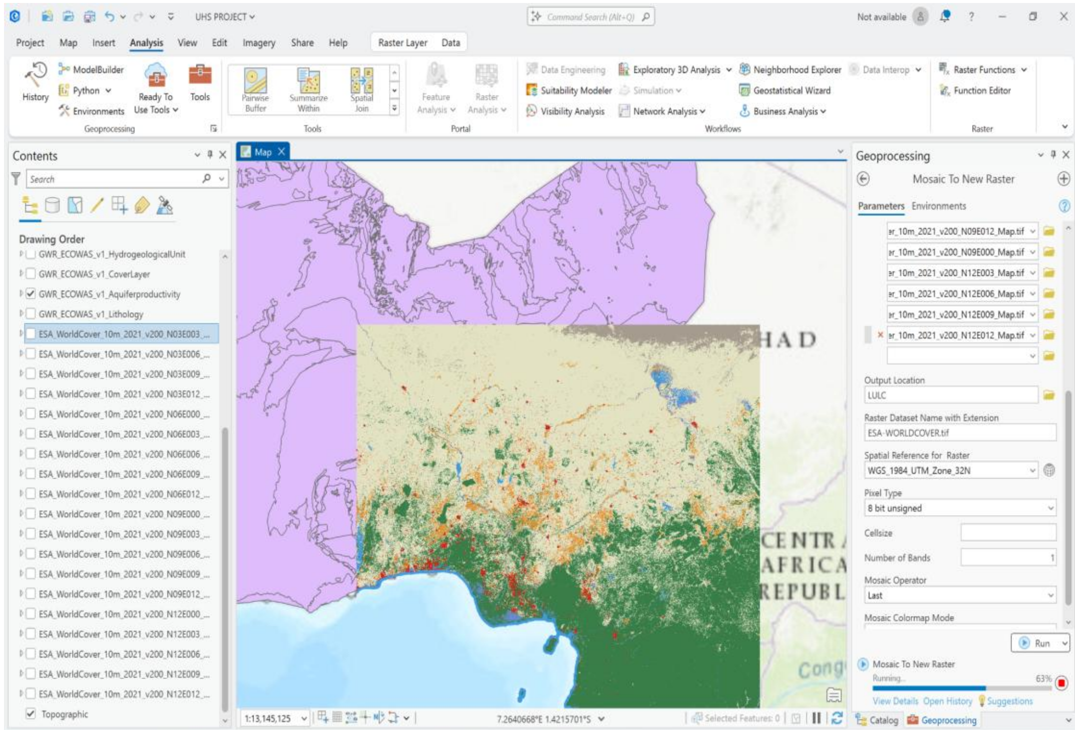
# Appendices

## Appendix I

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GWR_ECOWAS_v1_Aquiferproductivity
GWR_ECOWAS_v1_Lithology
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Geoprocessing
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LULC
Raster Dataset Name with Extension
ESA-WORLDCOVER.tif
Spatial Reference for Raster
WGS_1984_UTM_Zone_32N
Pixel Type
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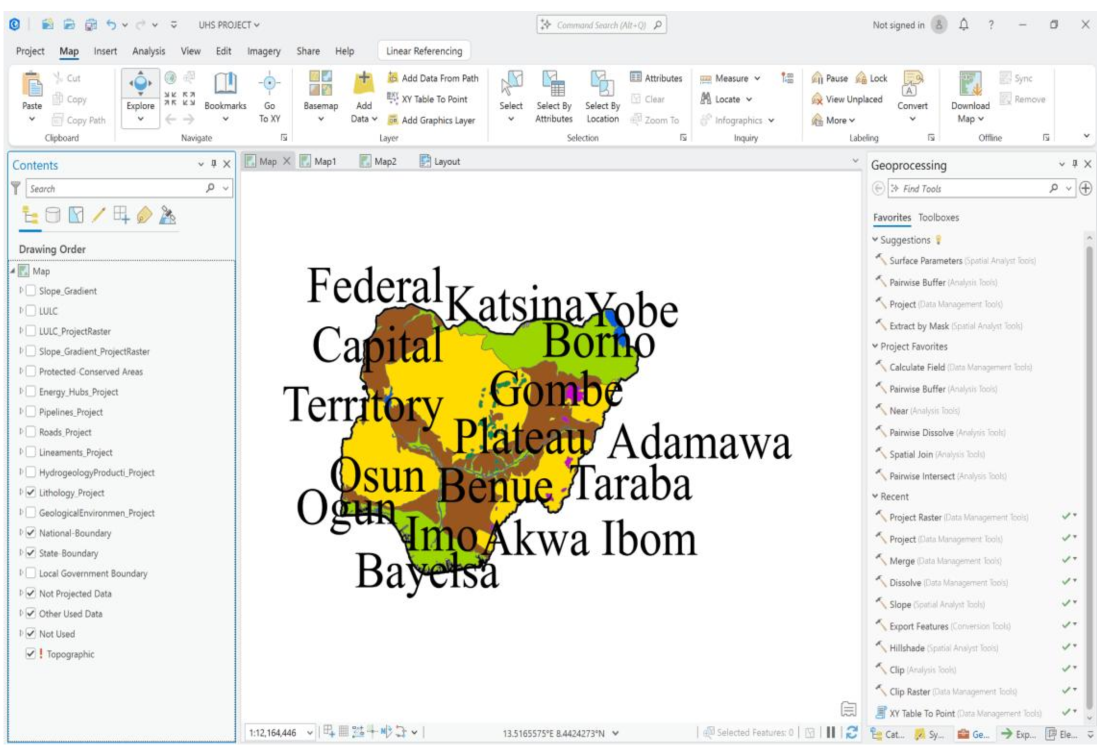
Cellsize
Number of Bands
Mosaic Operator
Last
Mosaic Colormap Mode
Run
Mosaic To New Raster
Running...
63%
1:13,145,125


UHS PROJECT
Project
Map
Insert
Analysis
View
Edit
Imagery
Share
Help
Linear Referencing
Contents
Drawing Order
Map
Slope_Gradient
LULC
LULC_ProjectRaster
Slope_Gradient_ProjectRaster
Protected-Conserved Areas
Energy_Hubs_Project
Pipelines_Project
Roads_Project
Lineaments_Project
HydrogeologyProducti_Project
Lithology_Project
GeologicalEnvironmen_Project
National-Boundary
State-Boundary
Local Government Boundary
Not Projected Data
Other Used Data
Not Used
Topographic
Federal
Capital
Territory
Katsina
Yobe
Borno
Gombe
Plateau
Adamawa
Osun
Ogun
Benue
Taraba
Imo
Akwa Ibom
Bayelsa
Geoprocessing
Favorites
Toolboxes
Suggestions
Project Favorites
Recent
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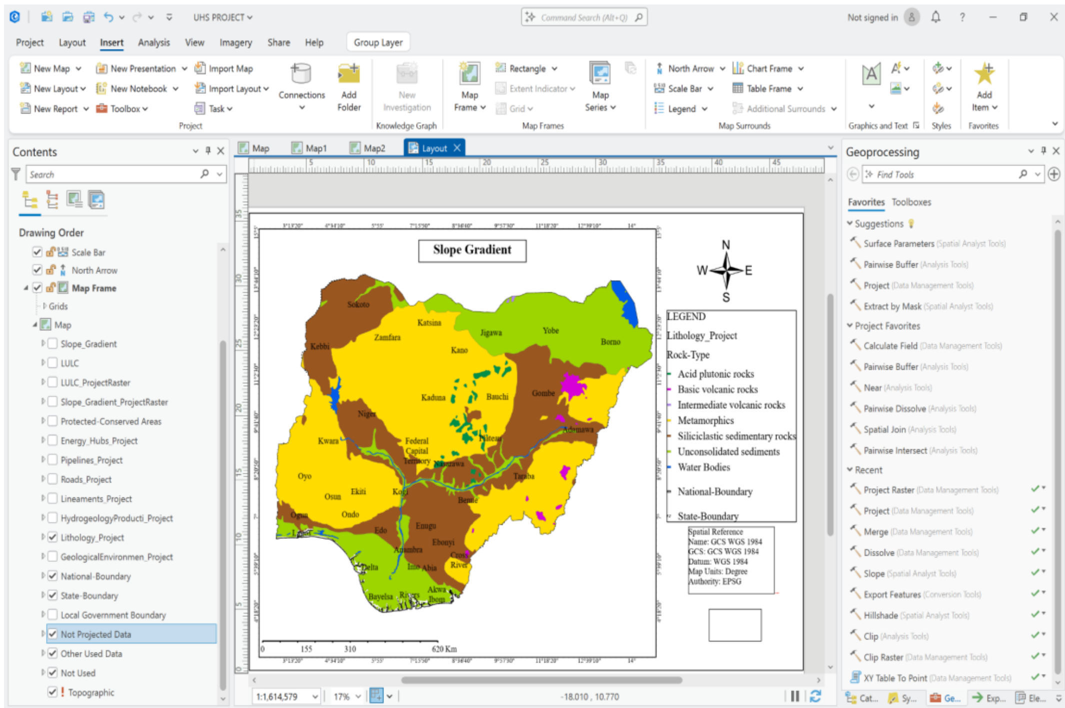
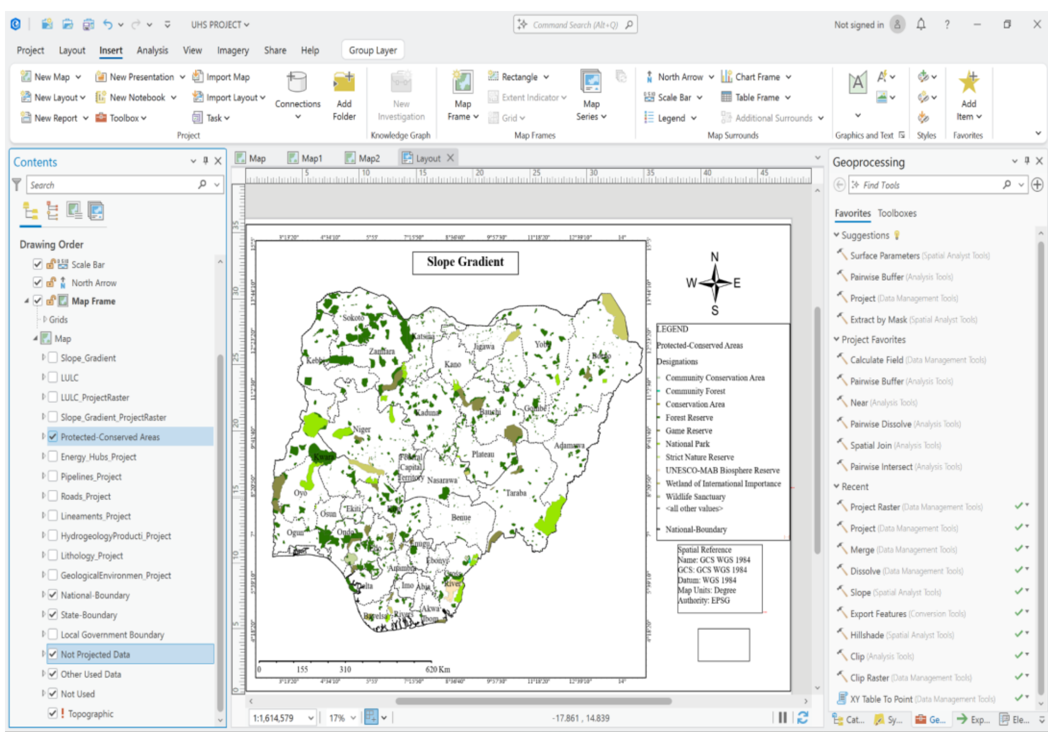
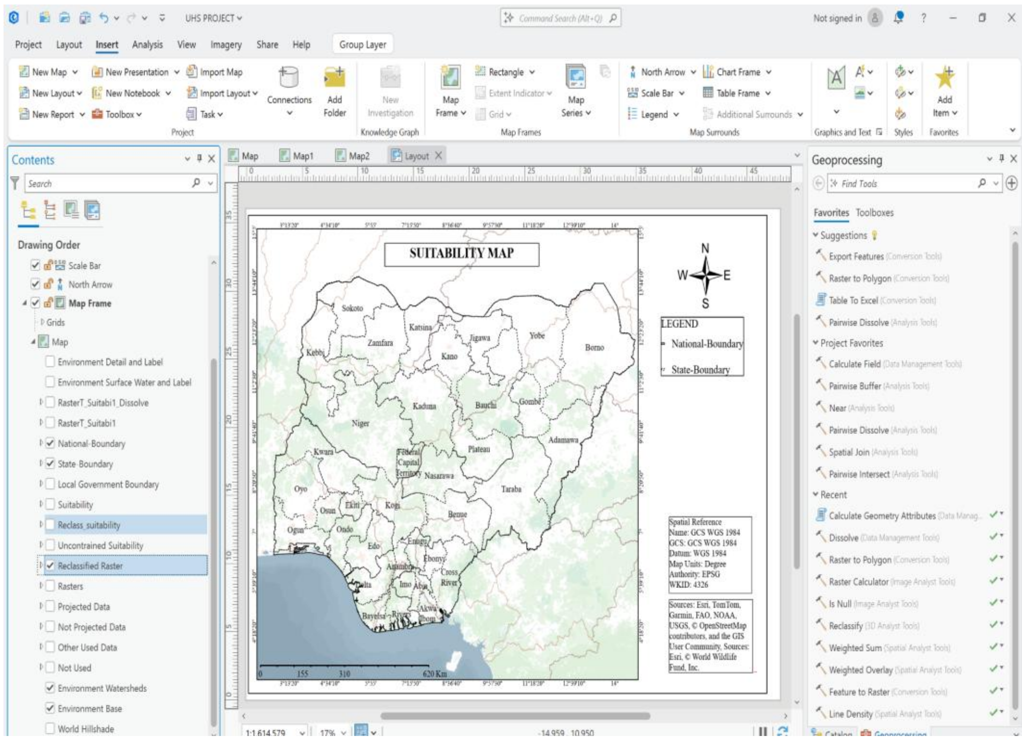

Slope Gradient
LEGEND
Lithology_Project
Rock-Type
Acid plutonic rocks
Basic volcanic rocks
Intermediate volcanic rocks
Metamorphics
Siliciclastic sedimentary rocks
Unconsolidated sediments
Water Bodies
National-Boundary
State-Boundary
Spatial Reference
Name: GCS WGS 1984
GCS: GCS WGS 1984
Datum: WGS 1984
Map Units: Degree
Authority: EPSG
0 155 310 620 Km
Contents
Drawing Order
Geoprocessing


Slope Gradient
LEGEND
Protected-Conserved Areas
Designations
Community Conservation Area
Community Forest
Conservation Area
Forest Reserve
Game Reserve
National Park
Strict Nature Reserve
UNESCO-MAB Biosphere Reserve
Wetland of International Importance
Wildlife Sanctuary
<all other values>
National-Boundary
Spatial Reference
Name: GCS WGS 1984
GCS: GCS WGS 1984
Datum: WGS 1984
Map Units: Degree
Authority: EPSG
0 155 310 620 Km
Contents
Drawing Order
Geoprocessing


SUITABILITY MAP
LEGEND
National-Boundary
State-Boundary
Spatial Reference
Name: GCS WGS 1984
GCS: GCS WGS 1984
Datum: WGS 1984
Map Units: Degree
Authority: EPSG
WKID: 4326
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community, Sources: Esri, © World Wildlife Fund, Inc.
Contents
Drawing Order
Geoprocessing
UHS PROJECT

# Appendix II

## WEIGHTING MATRIX/ CONSISTENCY RATIO EXCEL DOCUMENT

| | PAIRWISE COMPARISON TABLE | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | **Geology** | **Hydrogeology** | **Lithology** | **Fault Density** | **Distance to Pipeline** | **Distance to Energyhubs** | **Distance to Roads** | **Slope** | **LULC** |
| **Geology** | 1.00 | 1.00 | 1.00 | 1.00 | 2.00 | 2.00 | 2.00 | 3.00 | 2.00 |
| **Hydrogeology** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 2.00 | 2.00 | 3.00 | 2.00 |
| **Lithology** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 2.00 | 2.00 | 1.00 |
| **Fault Density** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 2.00 | 2.00 | 3.00 | 2.00 |
| **Distance to Pipeline** | 0.50 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 2.00 | 2.00 | 1.00 |
| **Distance to Energyhubs** | 0.50 | 0.50 | 1.00 | 0.50 | 1.00 | 1.00 | 1.00 | 2.00 | 1.00 |
| **Distance to Roads** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 1.00 | 1.00 | 2.00 | 1.00 |
| **Slope** | 0.33 | 0.33 | 0.50 | 0.33 | 0.50 | 0.50 | 0.50 | 1.00 | 0.5 |
| **LULC** | 0.50 | 0.50 | 1.00 | 0.50 | 1.00 | 1.00 | 1.00 | 2.00 | 1.00 |
| **SUM** | 6.33 | 6.83 | 8.00 | 6.83 | 9.00 | 11.50 | 13.50 | 20.00 | 11.50 |

| NORMALIZATION TABLE | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Geology** | **Hydrogeology** | **Lithology** | **Fault Density** | **Distance to Pipeline** | **Distance to Energyhubs** | **Distance to Roads** | **Slope** | **LULC** | **SUM** | **PRIORITY (MEAN)** | **WEIGHT %** |
| **Geology** | 0.16 | 0.15 | 0.13 | 0.15 | 0.22 | 0.17 | 0.15 | 0.15 | 0.17 | 1.44 | 0.16 | 16.04 |
| **Hydrogeology** | 0.16 | 0.15 | 0.13 | 0.15 | 0.11 | 0.17 | 0.15 | 0.15 | 0.17 | 1.33 | 0.15 | 14.81 |
| **Lithology** | 0.16 | 0.15 | 0.13 | 0.15 | 0.11 | 0.09 | 0.15 | 0.10 | 0.09 | 1.11 | 0.12 | 12.32 |
| **Fault Density** | 0.16 | 0.15 | 0.13 | 0.15 | 0.11 | 0.17 | 0.15 | 0.15 | 0.17 | 1.33 | 0.15 | 14.81 |
| **Distance to Pipeline** | 0.08 | 0.15 | 0.13 | 0.15 | 0.11 | 0.09 | 0.15 | 0.10 | 0.09 | 1.03 | 0.11 | 11.44 |
| **Distance to Energyhubs** | 0.08 | 0.07 | 0.13 | 0.07 | 0.11 | 0.09 | 0.07 | 0.10 | 0.09 | 0.81 | 0.09 | 8.99 |
| **Distance to Roads** | 0.08 | 0.07 | 0.06 | 0.07 | 0.06 | 0.09 | 0.07 | 0.10 | 0.09 | 0.69 | 0.08 | 7.68 |
| **Slope** | 0.05 | 0.05 | 0.06 | 0.05 | 0.06 | 0.04 | 0.04 | 0.05 | 0.04 | 0.44 | 0.05 | 4.91 |
| **LULC** | 0.08 | 0.07 | 0.13 | 0.07 | 0.11 | 0.09 | 0.07 | 0.10 | 0.09 | 0.81 | 0.09 | 8.99 |
| **SUM** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 9.00 | 1.00 | 100.00 |

| Weighted Sum Vector | | | | | | | | | | | | | CONSISTENCY RATIO (CR) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Geology** | **Hydrogeology** | **Lithology** | **Fault Density** | **Distance to Pipeline** | **Distance to Energy hubs** | **Distance to Roads** | **Slope** | **LULC** | **WSV** | **PRIORITY (MEAN)** | **WSV / PRIORITY** | **λ Max** | **CONSISTENCY INDEX (CI)** | **RANDOM MATRIX INDEX (RI)** | **CR** |
| **Geology** | 0.16 | 0.15 | 0.12 | 0.15 | 0.23 | 0.18 | 0.15 | 0.15 | 0.18 | 1.47 | 0.16 | 9.11 | 9.11 | 0.01 | 1.45 | 0.01 |
| **Hydrogeology** | 0.16 | 0.15 | 0.12 | 0.15 | 0.11 | 0.18 | 0.15 | 0.15 | 0.18 | 1.35 | 0.15 | 9.11 | | | | |
| **Lithology** | 0.16 | 0.15 | 0.12 | 0.15 | 0.11 | 0.09 | 0.15 | 0.10 | 0.09 | 1.13 | 0.12 | 9.11 | | | | |
| **Fault Density** | 0.16 | 0.15 | 0.12 | 0.15 | 0.11 | 0.18 | 0.15 | 0.15 | 0.18 | 1.35 | 0.15 | 9.11 | | | | |
| **Distance to Pipeline** | 0.08 | 0.11 | 0.12 | 0.15 | 0.11 | 0.09 | 0.15 | 0.10 | 0.09 | 1.01 | 0.11 | 9.11 | | | | |
| **Distance to Energyhubs** | 0.08 | 0.07 | 0.12 | 0.07 | 0.11 | 0.09 | 0.08 | 0.10 | 0.09 | 0.82 | 0.09 | 9.11 | | | | |
| **Distance to Roads** | 0.08 | 0.07 | 0.06 | 0.07 | 0.06 | 0.09 | 0.08 | 0.10 | 0.09 | 0.70 | 0.08 | 9.11 | | | | |
| **Slope** | 0.05 | 0.05 | 0.06 | 0.05 | 0.06 | 0.04 | 0.04 | 0.05 | 0.04 | 0.45 | 0.05 | 9.11 | | | | |
| **LULC** | 0.08 | 0.07 | 0.12 | 0.07 | 0.11 | 0.09 | 0.08 | 0.10 | 0.09 | 0.82 | 0.09 | 9.11 | | | | |
| **SUM** | | | | | | | | | | 9.11 | | 81.99 | | | | |